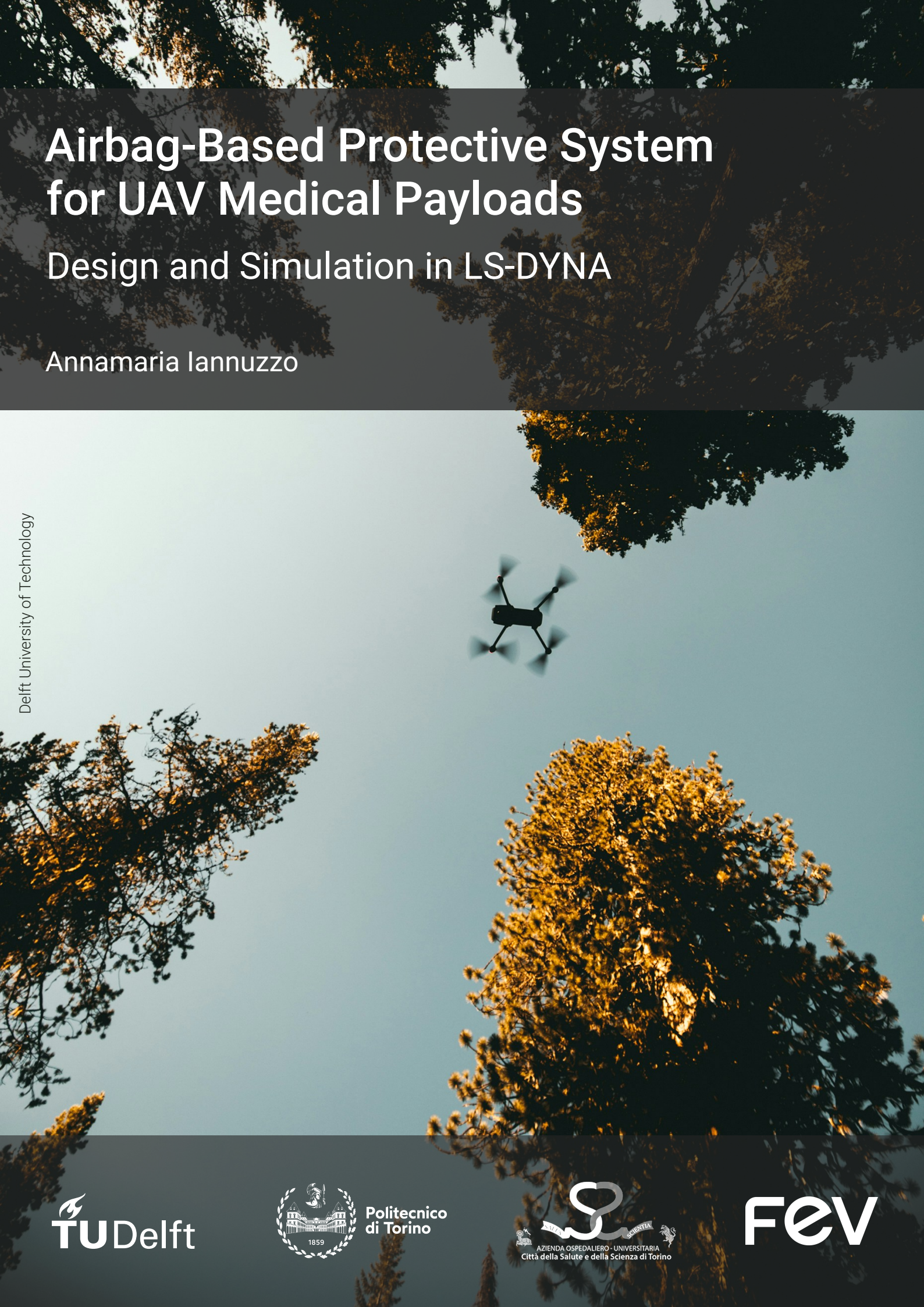
Airbag-Based Protective System for UAV Medical Payloads

Design and Simulation in LS-DYNA

Annamaria Iannuzzo

Delft University of Technology



TU Delft

Politecnico di Torino

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Città della Salute e della Scienza di Torino

FEV

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Design and Simulation in LS-DYNA

by

Annamaria Iannuzzo

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Preface

A preface...

Annamaria Iannuzzo
Delft, November 2026

Summary

In Turin (north-west Italy), a research line between *Politecnico di Torino*, the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex and *FEV Italia S.r.l.* has studied since 2020 how drones could support the ambulance-based transport of medical supplies between hospitals. If a drone fails in flight, however, its payload may detach and fall onto the city, where the impact could, among other consequences, destroy urgently needed medical goods and disperse potentially hazardous substances. No device currently exists to protect such a payload, nor a method to design one for a given mission. An airbag could fulfil this role, but sizing it requires two quantities: the impact velocity of the payload and the largest acceleration it can withstand without damage.

This thesis addresses the problem on two levels. An outer design loop formalises a safety-first principle, in which the payload and the drone are treated as separate entities and the protection of the payload is designed at the start rather than added at the end, so that it supplies the drone design with inputs such as airbag mass, volume, inflator type and inflation time. Within this loop, an inner design loop forms the workflow delivered by the thesis: it establishes the two quantities above and returns the lightest airbag that keeps the payload below its allowable acceleration. Its first five steps, from mission analysis to design optimisation, were demonstrated on a reference delivery mission between two hospitals in Turin, flown largely above the river Po and carrying 5 ml borosilicate glass phials, while the remaining three steps depend on the geometry of the container and the drone and are left to future work.

The first step of the workflow, a mission analysis, established the impact conditions: a Monte Carlo simulation of the fall along the whole route gave, at its 98.5th percentile, a design impact of 7.5 m/s horizontally and 25 m/s vertically. The next two steps characterised the payload: compression tests on 76 phials from two producers showed that nominally identical phials behaved differently, so only the batch with the smaller scatter was retained, and its 5th-percentile failure load of 98.7 N became the target for a finite element model of the same test. With a glass tensile strength of 60 MPa, the model matched this load within 2% and reproduced the two most common failure modes of that batch. In the fourth step, the calibrated phial was placed in a simulated sled test, in which an assumed container, lined with a foam sponge and a plastic tray, was subjected to an acceleration pulse emulating the impact at 26.1 m/s. Loaded perpendicular to its axis, the most critical of the directions tested, the phial withstood a peak of 225 g and cracked at 230 g, so 225 g was taken as the allowable acceleration. The final step brought both strands together: an analytical airbag impact model, extended to account for horizontal velocity, fabric porosity and bursting, was embedded in a differential evolution optimiser. Given a 1 kg payload, both velocity components and an acceleration limit of 187.5 g (the allowable divided by a safety factor of 1.2), the optimiser selected the same design for every landing-speed limit between 0.5 and 2.0 m/s: a single-ply cylinder of silicone-coated Nylon 6,6 inflated with helium, 0.522 to 0.774 m in diameter depending on that limit, with fabric and gas together below 0.4 kg.

The workflow thus sizes the inflated airbag for a specific payload and impact scenario. Its outputs point to a stored-gas helium inflator, since helium cannot be produced pyrotechnically, although the feasibility and mass of the pressurised gas containers, which the optimisation does not include, remain to be verified. The design is only as reliable as its inputs: a single phial geometry and batch were modelled, every optimum lies on at least one bound of the search space, and the airbag model has not yet been checked against physical tests. Once the capsule geometry is fixed, the optimised airbag should be validated through simulations of the real assembly and drop tests, and the workflow could be extended to a reliability-based design, combining the scatter in glass strength with that of the impact velocities, or to an airbag working in tandem with a parachute.

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Nomenclature

Abbreviations

Abbreviation	Definition
ABS	Acrylonitrile butadiene styrene
A.O.U.	*Azienda Ospedaliero-Universitaria* (University Hospital Trust)
ADR	*Accord relatif au transport international des marchandises Dangereuses par Route* (European Agreement concerning the International Carriage of Dangerous Goods by Road)
ASL	*Azienda Sanitaria Locale* (Local Health Authority)
ASST	*Azienda Socio-Sanitaria Territoriale* (Territorial Social and Health Authority)
ATD	Anthropomorphic test device
CAD	Computer-aided design
CFL	Courant-Friedrichs-Lewy (stability condition)
ConOps	Concept of Operations
CTO	*Centro Traumatologico Ortopedico* (Orthopaedic Trauma Centre, Turin)
DE	Differential evolution
DIMEAS	*Dipartimento di Ingegneria Meccanica e Aerospaziale* (Department of Mechanical and Aerospace Engineering, Politecnico di Torino)
DoF	Degree(s) of freedom
DTA	*Distretto Tecnologico Aerospaziale* (Apulian Aerospace Technology Cluster)
EASA	European Union Aviation Safety Agency
ENAC	*Ente Nazionale per l'Aviazione Civile* (Italian Civil Aviation Authority)
ENAV	*Ente Nazionale di Assistenza al Volo* (Italian Air Navigation Service Provider)
EPSG	European Petroleum Survey Group (geodetic parameter registry)
EU	European Union
eVTOL	Electric vertical take-off and landing
FE	Finite element
FEM	Finite Element Method
H2020	Horizon 2020 (EU research and innovation programme)
HPC	High-performance computing
ICAO	International Civil Aviation Organization
ISA	International Standard Atmosphere
ISO	International Organization for Standardization
JARUS	Joint Authorities for Rulemaking on Unmanned Systems
LEA	*Livelli Essenziali di Assistenza* (Essential Levels of Care)
MTOM	Maximum take-off mass
OPC	One-point cut (ampoule breaking system)
OSM	OpenStreetMap
PETG	Polyethylene terephthalate glycol-modified
PIC4SeR	PoliTo Interdepartmental Centre for Service Robotics
PP	Polypropylene
RRT	Rapidly-exploring Random Tree (RRT*: asymptotically optimal variant)
SIS 118	*Società Italiana Sistema 118* (Italian Society of the 118 Emergency Medical System)
SORA	Specific Operations Risk Assessment
SSN	*Servizio Sanitario Nazionale* (Italian National Health Service)
SSR	*Servizio Sanitario Regionale* (Regional Health Service)

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Abbreviation	Definition
UAV	Unmanned aerial vehicle
UN3373	UN number for Biological Substance, Category B (transport packaging standard)
UTM	Universal Transverse Mercator (map projection)
VPN	Virtual private network
VTOL	Vertical take-off and landing
WGS84	World Geodetic System 1984
XML	Extensible Markup Language

Symbols

Symbol	Definition	Unit
Latin symbols		
A_{or}	Area of the discrete vent orifice	[m^2]
A_t	Airbag–ground footprint area at time t	[m^2]
A_{vent}	Effective vent area	[m^2]
$A_{\text{vent},0}$	Initial vent area	[m^2]
a	Payload acceleration	[m/s^2]
a_{allow}	Allowable payload acceleration	[m/s^2]
a_{lim}	Design acceleration limit, a_{allow}/SF	[m/s^2]
$a_{\text{peak},n}$	Peak payload acceleration along $\mathbf{n}$, $\max_t F_n/M$	[m/s^2]
C_{D}	Aerodynamic drag coefficient of the payload	[–]
C_{dis}	Discharge coefficient of the vent	[–]
CR	Crossover probability of the differential evolution algorithm	[–]
$\mathcal{C}$	Constraint set of the optimisation	[–]
c	Drag constant (Step 0); acoustic wave speed in the material (Step 3)	[kg/m]; [m/s]
D_0	Initial airbag diameter (characteristic dimension)	[m]
$D_{0,\text{min}}$	Minimum airbag diameter, stroke needed to stop the payload at a_{lim}	[m]
D_t	Airbag diameter at time t	[m]
d_{or}	Diameter of the discrete vent orifice	[m]
E	Young's modulus	[GPa]
F	Mutation factor of the differential evolution algorithm	[–]
F_d	Aerodynamic drag force on the payload	[N]
F_{fail}	Phial failure force under axial compression	[N]
F_n	Normal ground reaction force on the airbag	[N]
FT_{fail}	Phial brittle failure threshold	[GPa]
f	Oracle function, $\mathbf{y} = f(\mathbf{x})$	[–]
$f(\mathbf{x})$	Objective function of the optimisation	[kg]
f_1	First natural frequency of the phial	[Hz]
f_t, f_c	Tensile and compressive strength (Rankine criterion)	[GPa]
G	Maximum number of generations of the differential evolution algorithm (Step 4); shear modulus (Step 3)	[–]; [GPa]
g	Gravitational acceleration	[m/s^2]
h	Stroke, i.e. payload travel while compressing the airbag	[m]
$\mathcal{J}$	Objective function of the optimisation wrapper, $\mathcal{J}(\mathbf{y})$	[–]
j_{rand}	Randomly chosen variable index always taken from the mutant $\mathbf{v}$	[–]
K	Bulk modulus	[GPa]
k	Airbag shape coefficient (sphere: 0.5; cylinder: 1)	[–]
K_{FEM}, K_{PETG}, K_{eq}	Stiffness of the FE phial model, of the PETG adaptor, and equivalent series stiffness	[N/mm]
L_0	Initial airbag length (cylindrical airbag)	[m]

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Symbol	Definition	Unit
L_e	Characteristic length of the critical element	[mm]
L_t	Contact patch length (cylindrical airbag)	[m]
M	Payload mass; elastic modulus (Step 3)	[kg]; [GPa]
m_{bag}	Fabric mass of the airbag	[kg]
m_{gas}	Gas mass inside the airbag	[kg]
$m_{\text{gas},0}$	Initial gas charge	[kg]
m_{tot}	Airbag system mass, $m_{\text{bag}} + m_{\text{gas},0}$ (payload excluded)	[kg]
$\dot{m}$	Vented gas mass flow rate	[kg/s]
N	Membrane force per unit width, $k\,\Delta P\,R$ (oracle); population size of the differential evolution algorithm (wrapper)	[N/m]; [–]
N_{ult}	Ultimate membrane force per unit width, $100\,n_{\text{ply}}\,N'_{\text{ult}}$	[N/m]
N'_{ult}	Supplier fabric strength per unit width (one ply)	[N/cm]
$\mathbf{n}$	Unit vector along the impact (stroke) direction	[–]
n_{gas}	Number of inflation gas options	[–]
n_{mat}	Number of fabric options	[–]
n_{ply}	Number of fabric plies	[–]
n_{seed}	Number of independent differential evolution runs (seeds) per combination	[–]
n_{shape}	Number of airbag shape options	[–]
n_{ures}	Number of residual speed caps $u_{\text{res,max}}$	[–]
n_{var}	Number of numerical design variables	[–]
O	Optimisation wrapper	[–]
P_0	Initial inflation pressure	[Pa]
$P_{0,\text{max}}$	Upper bound of the inflation pressure	[Pa]
P_{amb}	Ambient (atmospheric) pressure	[Pa]
P_{bag}	Airbag internal pressure	[Pa]
P_{burst}	Airbag burst pressure	[Pa]
P_{open}	Vent opening threshold pressure	[Pa]
P_{peak}	Peak airbag internal pressure	[Pa]
P_{up}, P_{down}	Upstream and downstream pressure across the vent	[Pa]
PEN	Penalty weight on constraint violations	[–]
R	Specific gas constant	[J/(kg K)]
R_0	Initial airbag radius	[m]
r_j	Random number in $[0, 1]$ drawn for variable j in the crossover	[–]
S	Payload reference area for aerodynamic drag	[m^2]
S_{surface}	Undeformed fabric surface area of the airbag	[m^2]
SF	Safety factor on the allowable acceleration	[–]
SF_{burst}	Design factor on the allowable membrane force	[–]
T	Gas temperature	[K]
T_0	Initial gas temperature	[K]
T_1	Period of the first elastic mode of the phial	[s]
T_{amb}	Ambient temperature	[K]
t	Time	[s]
t_0	Initial time (onset of impact)	[s]
t_{fall}	Free-fall time	[s]
t_{max}	Maximum simulated duration of the impact event	[s]
t_p	Plateau duration of the acceleration pulse	[s]
t_r	Ramp duration of the acceleration pulse	[s]
t_{stop}	Time of the stopping event (burst, full stroke or rebound)	[s]
$\mathbf{u}$	Challenger (trial) design of the differential evolution algorithm, with components u_j	[–]
u_0	Initial impact velocity of the 1-DoF model (Ch. 2)	[m/s]

continued on next page

Symbol	Definition	Unit
$\mathbf{u}_0$	Initial impact velocity vector, (u_{x0}, u_{y0})	[m/s]
u_t	Payload velocity at time t	[m/s]
u_{land}	Payload landing speed	[m/s]
u_n	Payload velocity along $\mathbf{n}$	[m/s]
u_{n0}	Initial impact velocity along $\mathbf{n}$	[m/s]
$u_{\text{reb},y}$	Vertical rebound velocity	[m/s]
$u_{\text{res,max}}$	Maximum allowable residual velocity	[m/s]
u_x	Horizontal payload velocity	[m/s]
u_{x0}	Initial horizontal impact velocity	[m/s]
u_y	Vertical payload velocity (positive downward)	[m/s]
u_{y0}	Initial vertical impact velocity (positive downward)	[m/s]
V_0	Initial airbag volume	[m^3]
V_t	Airbag volume at time t	[m^3]
V_w	Wind speed	[m/s]
$\mathbf{v}$	Mutant design of the differential evolution algorithm, with components v_j	[–]
v_i	Graded constraint violation in the objective (0 = satisfied)	[–]
$\mathbf{x}$	Design variable vector	[–]
$\mathbf{x}_{\text{best}}$	Best design of the current differential evolution population	[–]
$\mathbf{y}$	System response computed by the oracle	[as variable]
y_{pen}	Vertical penetration of the payload, measured from the point of first contact	[m]
Greek symbols		
α	Scale parameter of the Weibull distribution	[N]
β	Shape parameter of the Weibull distribution	[–]
Γ	Terminal velocity of the payload	[m/s]
γ	Specific heat ratio	[–]
δ_{FEM}, δ_{PETG}, δ_{eq}	Displacement of the FE phial model, of the PETG adaptor, and of the series system	[mm]
Δm	Gas mass vented in one timestep	[kg]
ΔP	Gauge pressure inside the airbag, $P_{\text{bag}} - P_{\text{amb}}$	[Pa]
Δt	Timestep	[s]
Δt_{crit}	Critical (stable) timestep of the explicit analysis	[s]
η_{seam}	Seam efficiency (fraction of fabric strength retained at seams)	[–]
θ	Impact angle, measured from the vertical	[°]
λ	Pressure ratio across the vent, $P_{\text{down}}/P_{\text{up}}$	[–]
μ, σ	Mean and standard deviation of a Gaussian distribution	[as variable]
ρ	Air density; material density (Step 3)	[kg/m^3]
ρ_t	Gas density at time t	[kg/m^3]
σ_1, σ_2	Principal stresses	[GPa]
φ_{vent}	Vent open-area fraction	[–]
ψ_w	Wind direction	[°]
ω_1	Fundamental angular frequency of the phial	[rad/s]

1 Introduction

1.1. Motivation

Unmanned aerial vehicles (UAVs) are increasingly used to carry medical goods, from blood samples to life-saving drugs [1], [2]. In Turin, a research line between *Politecnico di Torino*, the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex and *FEV Italia S.r.l.* has studied since 2020 the possibility of flanking the current ambulance-based transport of medical supplies with drone-based missions, both at an inter- and intra-hospital level (Section 1.2.2). In such missions, a failure of the drone may cause its payload to detach and fall onto the city. The medical payload, typically drugs stored in glass phials, would then break on impact, resulting in the loss of urgently needed medical goods and in the dispersion of potentially hazardous substances into the environment [3], [4].

The previous research has taken a legislative, logistic, market- and mission-evaluation approach, and it has delineated a scenario in which the approval of the Italian Civil Aviation Authority (ENAC, *Ente Nazionale per l'Aviazione Civile*) could potentially be obtained for this type of operation. However, certifications are unavoidably specific to the mission type, the trajectory and the vehicle, and no device currently exists to protect a medical payload falling from a drone (Section 2.1), nor a method to design one for a given mission. To move forward, the research line needs a more concrete approach, rooted in *design*.

Therefore, the overall objective of the present thesis is to propose a *design workflow* for an airbag-like device that protects the free-falling medical payload in the event of a drone failure, and to demonstrate it through testing and simulation. If adopted in the preliminary design phase of a drone-based delivery mission, this workflow yields an optimised airbag-type device tailored to a specific failure-and-drop application, which brings the impact accelerations within the allowable limits of the payload container and of the medical payload being transported. In this thesis, the workflow is demonstrated on 5 ml borosilicate glass phials carried along a trajectory over the Po river in Turin, from the mission analysis to a preliminary sizing of the airbag. The successful completion of this thesis is therefore achieved by:

1. Proposing an application-specific design workflow that fits within the broader context of the ongoing research effort;
2. Testing the workflow, both experimentally and through simulation, to realistically inform the design of the device.

Overall, this MSc thesis is a multidisciplinary project, in which the techniques of Structural Engineering are put to the service of the medical field. The project adopts a *safety-first* approach, in which the protection of the medical cargo is prioritised over the integrity of the drone carrying it. Distinguishing between the safety of the UAV and that of its payload may facilitate the certification process of the UAV, as well as the modular implementation of the safety device on a variety of UAVs serving in different, specific delivery missions. This design choice is justified in Sections 2.1 and 2.2, on the basis of a regional market analysis and of an ongoing paradigm shift in the design process of biomedical devices.

The rest of this chapter frames the project within the Italian healthcare context (Section 1.2.1), summarises the prior research and positions the present thesis within it (Section 1.2.2), and introduces the stakeholders (Section 1.2.3).

1.2. Project framing

1.2.1. Care centralisation and territorial inequity in the Italian healthcare system

The idea of transitioning from road- to air-transport in healthcare delivery systems stems from a growing consensus within the Italian medical field, consolidated during the COVID-19 pandemic (2020-2022). According to this view, the current ambulance-based logistics of the National Health Service (described in detail[1] in [5]) can no longer keep up with the needs of the healthcare system and its ongoing transitions. The two main competing factors are:

1. **Care centralisation**: Abrescia et al. [6] describe care centralisation as the ongoing process of concentrating medical expertise and services in key geographical areas, e.g. provincial and/or regional capitals, to meet the demands of high-volume care. The downside of care centralisation is that at a regional level, it induces "distance-related inequity, particularly for rural and peripheral populations", especially heightened in emergency scenarios.
2. **Inhomogeneous distribution of emergency services**: Garattini et al. [7] affirm that, in contrast with the centralisation manoeuvre described above, emergency resources are still inhomogeneously distributed over the regional territories, with services overloaded in densely populated areas and underused in low-density ones, resulting in inefficient overall utilisation.

In conclusion, [8] builds on these two points by affirming the importance of establishing a generally more efficient strategy for healthcare management at all levels, and incentivises the development of technologies suitable for upgrading, among the several aspects of the National Health Service, the logistics of the Italian healthcare supply chain.

1.2.2. Prior research and positioning of the present thesis

Several industry-research partnerships aiming to modernise transportation means within the Italian SSN have ventured into the globally booming market of unmanned aerial vehicles, where remotely-piloted or fully autonomous drones are being used ever more often to carry medical payloads in a plethora of scenarios: intra- vs. inter-hospital delivery [1], ordinary logistics vs. emergency missions and humanitarian applications [9], experimental/direct flight between strategic poles [10] vs. a fully-operational, inter-hospital, hub-based UAV network covering national territory (the *Zipline* case study [11]).

Following the global trend to answer a local need, in the specific case of the Piedmont region in north-west Italy, where at the time of writing there exists no city- or region-wide UAV network deployed for the delivery of medical goods, many pilot programmes (further analysed in Section 2.1) are emerging, among which a partnership between the Pharmacy department of the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex (head-quartered at Molinette Hospital in Turin), a research group within the Department of Mechanical and Aerospace Engineering (DIMEAS, *Dipartimento di Ingegneria Meccanica e Aerospaziale*) at *Politecnico di Torino*, and the Numerical Analysis Team of *FEV Italia S.r.l.*, the Italian branch of the German-founded multinational company FEV, a global leader in research and development for the automotive industry.

The six-year-long partnership between these three institutions is the backbone of the present MSc thesis project, the most recent of a chain of research projects aiming to pursue the long-term vision of flanking the current road-based transportation means with aerial alternatives within hospitals in Piedmont and, specifically, in the city of Turin (the region's capital), in favour of more rapid, more efficient, and safer transportation of medical supplies. Specifically, this research effort has broadly tackled the key aspects required to establish a complete and regulation-compliant UAV medical delivery ecosystem in the Piedmontese territory, resulting in five MSc theses and one conference paper presenting the following results:

[1] *"The type of vehicles used varies from region to region. In Piedmont it mainly consists of ambulances used only for transport plus more advanced vehicles equipped for resuscitation, replaced in some sites by mSa on a car with a doctor, nurse and rescuer driver, organised on the basis of specific needs of the territory, different in the four macro-areas of the emergency network (rEmE) in which the region was divided."*[5]

- **Cavallo (2020)** [12]: preliminary assessment of UAV platforms suitable under Italian and European regulations. Development of risk maps and definition of optimal trajectories between selected hospitals.
- **Bertelli (2021)** [13]: detailed selection of UAV based on mission specifications derived from the medical goods to be transported and regulatory constraints set by the Italian Civil Aviation Authority (ENAC). Application of the SORA (Specific Operations Risk Assessment) methodology to three feasible mission routes, yielding the safety requirements that operators must satisfy.
- **Sivieri (2021)** [14]: identification and characterisation of thermal and vibrational loads encountered during typical UAV medical missions. Experimental testing on milk-derived sera samples to determine their resistance to UAV-induced vibrational stresses. First hypothesis of medical payload container geometry and materials, implemented in the experimental campaign.
- **Fiorentino (2021)** [15]: development of a path-planning strategy for life-saving drug distribution in populated environments. Generation of risk maps and implementation of a rapidly-exploring random tree (RRT)-based algorithm to minimise global risk and flight time.
- **Angelini (2023)** [16] and **Angelini et al. (2024)** [17]: comparative analysis of the safety and economic efficiency of emergency medical deliveries performed by car, drone, and vertical take-off and landing (VTOL) aircraft. Identification of optimal pharmaceutical delivery hub locations for each transport mode to maximise logistical performance.

The design workflow presented in this thesis (informally called *inner design loop*) is therefore intended as a first attempt to bridge the gap between the research work listed above, consisting in a series of logistic, legislative, and generally high-level evaluations of drone-based missions for the delivery of medical supplies in emergency scenarios, and the long-term, broadly-defined objective of constructing a large-scale drone network over the region of Piedmont (also called *outer (design) loop*). This required preliminary work aims at formalising the project requirements at both a broad, high level and at a specific, first-step level. Having taken a legislative, logistic, market- and mission-evaluation approach until now, it was possible to delineate a scenario in which the approval of ENAC could potentially be obtained to enable this type of operation. However, as certifications are unavoidably specific to mission type, trajectory and vehicle, it appeared necessary to implement a more concrete approach rooted in *design*, whilst also adopting a *safety-first* approach that diverted the project towards a device for the *safety of the medical payload* rather, than that of the aerial vehicle, as justified in Sections 2.1 and 2.2.

1.2.3. Stakeholders

Overall, the stakeholders participating in the present project had the following roles:

- **TU Delft** (Dr. Bianca Giovanardi): primary academic supervisor overseeing the scientific quality and methodological development of the research.
- **Politecnico di Torino** (Prof. Enrico Cestino, Davide Angelini): institutional leads and co-supervisors for the broader UAV medical-delivery research project, providing access to the experimental platforms used.
- **A.O.U. Città della Salute e della Scienza di Torino / Molinette Hospital**: provider of medical and operational data related to pharmaceutical transport requirements and mission scenarios.
- **FEV Italia**: industrial partner offering technical support for advanced modelling and simulation activities, including SolidWorks-based design and LS-DYNA impact analyses.

The interaction between stakeholders, as well as the management and communication aspect of the project, is analysed in Chapter 6 - "Project Management".

1.3. Document outline

This thesis document is structured as follows: Chapter 2 gathers the necessary information from both a market analysis and a literature review, highlighting the gap between current research and development efforts and the long-term goal of establishing a UAV-based medical delivery operation over regional territories, and leads to the formulation of the research questions. Chapter 3 proposes a *safety-first* design workflow to answer the research questions, and Chapter 4 applies it to an exemplifying mission

and payload, presenting the concrete results of this thesis. Chapter 5 answers the research questions and gathers the findings into recommendations for future work, and Chapter 6 discusses the project management aspects of the thesis.

2

Background and Research-Development Gap

In technology development, it is important to gain a clear idea of two aspects:

1. **Research aspect**: if the technology that is being developed requires the exploration of an uncharted research niche, as opposed to utilising pre-existing and consolidated knowledge for the creation of a new product;
2. **Development aspect**: whether the proposed technology targets a well-defined market with unmet customer demand and no established supply.

This chapter explores both these aspects in Sections 2.2 to 2.3 and 2.1, respectively. The aim is to gather sufficient background information, from market allocation of the proposed technology to concrete understanding of airbag design criteria, to depict the choices proposed in Chapter 3 regarding operative *methodology* as an answer to the Research Questions proposed at the end of this chapter.

2.1. Regional market analysis: medical UAV logistics in Piedmont

The Italian National Health Service is regionally decentralised: The State defines the general principles and the Essential Levels of Care (LEA, *Livelli Essenziali di Assistenza*) that must be guaranteed nationwide, but each region organises and delivers healthcare through its own Regional Health Service (SSR, *Servizio Sanitario Regionale*), with wide autonomy over how it does so. Vice versa, aviation rules follow European-level enforcement mechanisms: Following the 2019 harmonisation of European drone operations across the "open" and "specific" categories [3], operators seeking authorisation for high-risk flights are required to conduct a thorough risk analysis using the *Specific Operations Risk Assessment (SORA)* methodology developed by JARUS[1], making a detailed evaluation of potential failure scenarios while proving compliance with the demanding safety standards imposed by aviation authorities.

Such discrepancy between modi operandi of the aviation and healthcare authorities has shaped the way drone-network operations have mostly been developing in the last decades in Italy: several emerging, albeit scattered, research groups, either cooperating or competing with industrial counterparts, have achieved geographically circumscribed trial deployments while being unable to scale the operation beyond testing.

A complete summary of such operations, limited to the means of investigation available for the present thesis project, can be found in Table A.2 in Appendix A.2. The table shows a clear trend: although potentially booming as several healthcare facilities are exploring the possibility of adopting drone transport and operators are proposing new drone models for them, most activities stop after the initial demonstrations, with the only exception being the region of Abruzzo where the operational phase of the drone drug-delivery service between a strategic hospital and village pharmacies was announced

[1] *Joint Authorities for Rulemaking on Unmanned Systems*

in July 2025 [18], [19]. In general, a pattern of fragmentation, short-lived pilots, and few transitions to a potentially scalable service is observed in the construction of the overall *value chain*, with many independent institutions covering equivalent roles, as shown in Table A.1 in Appendix A.1.

Table 2.1: Actors involved in drone delivery of medical goods in Piedmont, grouped by project.

Project	Role	Institution	Scope
INDOOR (2021-2024)	Promoter	Fondazione DOT	Project promoter [20]
	Funder	Centro Nazionale Trapianti	Research grant at PIC4SeR [21]
	Funder	City of Turin	Project partner [20]
	Research	PIC4SeR, Politecnico di Torino	Autonomous flight technologies for transporting biological material [22]
	Research	Università di Torino	Project partner [20]
	Technology partner	MAVTech	Technical partner; aerial surveillance and operational-support products [20]
	Operator, payload maker	ABzero (Pisa)	Supplied the drone and the Smart Capsule (sensorised container for blood, organs and drugs); handled the ENAC authorisation [21], [23], [24], [25]
	End user	A.O.U. Città della Salute e della Scienza	Test flight between hospital sites [24]
	End user	Regional Transplant Centre Piemonte–Valle d'Aosta	Organ and biological transport [20]
	Regulator	ENAC	National civil aviation authority; project partner and flight authorisation [20]
DIMEAS feasibility studies (2020-2024)	Research	DIMEAS, Politecnico di Torino (Prof. E. Cestino)	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost–safety comparison of car, drone and electric VTOL (eVTOL), hub siting [12], [13], [14], [15], [16], [17]
	Technology partner	FEV Italia	Industrial partner; numerical impact analysis
	End user	A.O.U. Città della Salute e della Scienza	Pharmacy department partner
Hydrogen logistics plan	End user	A.O.U. Città della Salute e della Scienza	Proposed hydrogen drone logistics plan [26]
	Infrastructure	Torino Caselle and Cuneo Levaldigi airports	Proposed hub and support base [26]
LastMed	Funder	City of Turin	Funded the LastMed call [27]
	Alternative (ground)	Tecnocad / Tecnomill / Bylogix	Autonomous ground rover for last-mile drug delivery [28]
Independent	Drone manufacturer	SkyEU	Autonomous high-speed eVTOL drones for first aid and critical logistics [29]
	Regulator	Regione Piemonte	Regional health authority; drone trials subject to benefit–risk analysis [30]

Adopting a region-based, rather than a national, approach for market analysis is, therefore, necessary. Table 2.1 highlights stakeholders involved in operations within Piedmont and their roles in the value chain, illustrating a fragmented scenario: Project INDOOR, launched by *Fondazione DOT* in 2021, culminated in a 500 m-long test flight by an autonomous UAV between two *Città della Salute* hospital sites, carrying medical goods in a dedicated capsule, on 1st October 2024. In this project, several other stakeholders were involved (in fact, most of the ones present in Table 2.1), among which ABzero (a spin-off from *Scuola Normale Superiore di Pisa* responsible for capsule design and handling the ENAC authorisation process for the test flight), parallel research group PIC4SeR from the Department of Electronics and Telecommunications at *Politecnico di Torino*, and MAVTech S.r.l. (UAV systems designer company based in Turin). Independently of the INDOOR project of 2021-2024, the startup SkyEU was

founded in 2025 promising to develop eVTOL drone systems for critical logistics within, but not limited to, the healthcare sector.

In parallel, and independently of INDOOR, the research group led by Prof. Enrico Cestino at DIMEAS (*Politecnico di Torino*), together with *A.O.U. Città della Salute e della Scienza* and FEV Italia, has pursued since 2020 a series of feasibility studies on the Molinette-San Giovanni Bosco route (Section 1.2.2). These covered regulatory and mission evaluation, risk mapping and path planning, payload environmental loads, and cost-safety comparisons between road, drone and eVTOL transport. No flight campaign has been carried out so far. This makes the group the only Piedmontese actor whose work so far has been entirely at the planning and assessment level.

On the other hand, the regional administration's general stance with respect to this type of operation highlights the fundamental requirement of thorough benefit-risk analyses to guarantee both the safety of the citizens and the preservation of the medical cargo, possibly needing to limit transportation to exceptional emergencies in mountain areas [30]. From an engineering perspective, these criticalities can be translated in the following barriers for the development of a full-scale, UAV-based delivery network:

1. **Authorisation of flights**: operations beyond the open category are approved by compiling the SORA. Among the different classes of mitigations, *ground risk* mitigations (one of the first steps required in the SORA) reduce the consequences of the UAV's impact on the ground after a loss of control, for example by introducing a parachute. It is key for technical stakeholders to prove the reliability of their implemented ground risk mitigations to protect civilians in case of drone failure;
2. **Payload integrity and containment**: in accordance with the regional administration's concerns, packages of Biological Substance, Category B (UN3373) must comply with Packing Instruction 650 (summarised in Appendix F), which is harmonised across transport modes (ADR for road transport in Europe [31], International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air [32] for air transport) and requires the package to pass a 1.2 m drop test with no leakage from the primary receptacle. This test represents accidental falls during manual handling and transportation, which loses its significance in the event of a free-fall from ordinary mission altitudes as the potential for biological material dispersion into the environment becomes highly relevant. The European framework for unmanned aircraft addresses this case directly: European regulation 2019/947 defines *dangerous goods* as articles or substances capable of posing a hazard to health, safety, property or the environment in the case of an incident or accident [3], and Category B infectious substances fall within this definition. The European Union Aviation Safety Agency's (EASA) Acceptable Means of Compliance allow such goods to be carried in the *specific* category only if the operator can demonstrate that they will not cause harm to third parties or to the environment in case of accident, and they accept a crash-protected container that prevents leakage or dispersion as one way to do so [4], by keeping the primary receptacles intact, so that the payload remains contained even after a loss of the UAV;
3. **Unsustainability of the current business models:** while costs of road and helicopter transport of biomedical goods such as organs, drugs and other critical supplies are expected to fall significantly with drones, most trials are funded by EU projects and one-time grants. Lacking a plan to turn the pilots into long-term, sustainable solutions, these projects generally remain confined to the trial stage.

On the basis of this analysis, several research and development gaps can be identified: SkyEU is endeavouring to enact an economically sustainable business model for large-scale UAV networks, but has a small public record; ABzero has patented capsules equipped with electronics that reliably monitor in-flight vibrations and guarantee thermal control of the medical payload, but these are only certified to the UN3373 standard, whereas in case of in-flight failure a much more conservative approach is needed; regional authorities need reassurance on the safety aspect for civilians, as a 500 m test flight on a controlled route shows that the obstacle is not drone-making, but guaranteeing safety in everyday performance rather than in single monitored missions.

In this scenario, the institutions (*Politecnico di Torino* (DIMEAS), FEV Italia, *A.O.U. Città della Salute*) that commissioned the present thesis project aim to tackle, among the three barriers, barrier 2, concerning payload integrity and containment. The proposed solution is an airbag-type device that protects the payload and prevents its dispersion in the event of an impact, especially in view of transporting

extremely valuable medicinals and/or biological materials, e.g. human organs for urgent transplant operations. This device is conceived as an add-on to existing capsules such as the one by ABzero or those mounted on drones such as SkyEU's (for which documented information remains unavailable at the time of writing), and is intended to help close the certification gap highlighted above, while acknowledging that it does not address the issues linked to SORA ground risk or the sustainability of the business model.

2.2. *Mission-first* vs. *Safety-first*: a design-philosophy gap

This section builds on the research-development gap presented earlier by connecting its root causes to a discrepancy between design philosophies applied in the aeronautical and the biomedical sectors. The proposed design solution (the airbag-type device) is, in fact, a suggestion to solve an "aeronautical problem" implementing a "medical approach".

The issue of proving compliance to match demanding safety standards imposed by the certification authorities does not appear to be a Piedmont-only (or even an Italy-only) problem, but rather a widespread trend that affects all non-standard aeronautical projects across the field. As highlighted, for example, in [2], most UAV producers focus on optimising their aerial vehicle's design, minimising mission-related vibrational loads, thermal loads, overall mass, but also proposing ever more efficient cargo-drop mechanisms and ways to overcome geographical obstacles, all partaking in a feedback loop against aerodynamic and flight-mechanics performance metrics. This machine-centric approach appears to be the default, especially considering that UAV-based missions are, naturally, radically different from standard airline and/or general aviation operations, where passenger safety and vehicle safety are inherently aligned and where, therefore, stringent certification requirements strongly limit design freedom. A common consequence of focusing exclusively on technological potential when designing UAV-based operations is, hence, neglecting the safety assurance aspect of the project, which is usually delegated to including a parachute-type device in the final design phases. As pointed out by Alulema et al. [33] in their design feasibility evaluation study, the introduction of a parachute usually increases the risk of falling into a mass-increase feedback loop: the parachute increases the drone's maximum take-off mass (MTOM), which then requires a larger, thus heavier, parachute for efficient deceleration during free-fall, which again results in an excessive mass increase, and so on. This negative feedback loop results either in never reaching design convergence, or otherwise in having to accept a smaller parachute with reduced deceleration capacity, still implying a relatively high ground-impact velocity and almost certainly destructive damage to the airframe, its cargo, and the impacted areas on the ground.

The general impression given by the brief market analysis conducted earlier and the observations gathered in the present paragraph is that, perhaps, the "mission-first" approach adopted in the majority of nascent UAV-based operations is not sustainable for the projects' longevity beyond pilot flights, and that it might be necessary to embrace a paradigm shift in design approaches. Specifically, there might be potential in drawing inspiration from the biomedical field, where medical technology designers are encouraged to shift from a techno-centred to a human-centric approach to avoid the *translation gap* or *implementation gap*, i.e. the potential to introduce or enhance health inequity when trying to implement extremely advanced technology that does not take into account all necessities and limitations of human patients [34].

Just like success is measured on the patient and not on the device in the biomedical field, in the case of UAV-based medical delivery operations success should be measured on payload integrity and protection of civilians, not (only) on the capabilities of the aerial vehicle. A drone that survives an in-flight failure while sacrificing its payload and threatening irreversible human damage has undoubtedly failed its mission, as for a biomedical device that functions correctly but does not serve the patient. This reasoning leads to the first of the contributions proposed in the present thesis, which is the *safety-first* design principle presented in Section 3.2.1. This thesis aims to serve as a test-bed for this proposed approach, where exploring the airbag design process helps bridge the gap between current development efforts and certification.

2.3. Airbag impact attenuation: background knowledge

Airbag-type devices have a well-established, enduring research-development record with contributions found across multiple industries. The information presented in Sections 2.3.1 and 2.3.2 aims to represent the State-of-the-Art practices adopted in airbag design. The scope of this investigation is developing a solid, although perhaps not comprehensive, understanding of design recommendations on which to base the proposal for the design workflow of later Section 3.2.2.

2.3.1. Established applications and common working principle

Airbag technology is well established across two primary sectors: *automotive* and *aerospace*.

In the automotive industry, these systems are implemented during vehicle crashes, distinguishing between general automobile crashes and motorcycle impacts. In the space industry, ever since the early space missions of the 1960s, airbags have been deployed during the final landing phases of space capsules. Within the aeronautical industry, specifically in the "specific" operations category, other niche applications have emerged, such as the experimental study of adaptive airbag systems for the protection of UAVs and general aviation vehicles [35] (following a long track record of exploration in the field of adaptive technologies for impact absorption [36], [37]).

In either case, the primary objective of the airbag-type device remains the same: to be stored compactly in ordinary mission conditions, inflate within the allowable time before impact (whether during an emergency impact or landing), and decelerate the target body (whether a crash test dummy or a spacecraft) far more gently than in case of direct impact. When analysing these systems, purely analytical, hard-codable solutions are typically limited to single-degree-of-freedom (1-DoF) formulations, immediately resolving to the use of far more computationally-heavy finite element analyses to capture the complex two- and three-dimensional effects of the impact. The only exception to this trend is the technical report NATICK/TR-87/042 published by T. F. Patterson in 1985[2], which presents a 6-DoF analytical simulation built on closed-form component models, but remains inaccessible at the time of writing of the present chapter.

Despite the diversity of fields, all applications share identical fundamental working principles. An example of an analytical study of airbag-based impact events, and one of the most significant papers in shaping this thesis' work, is Zhou et al. [38], in turn heavily rooted in the SAND93-2133 technical report [39] on a code for predicting the performance of the Pathfinder lander's airbag impact attenuation system, which describes the core working principle of airbag physics in a one-degree-of-freedom impact event.

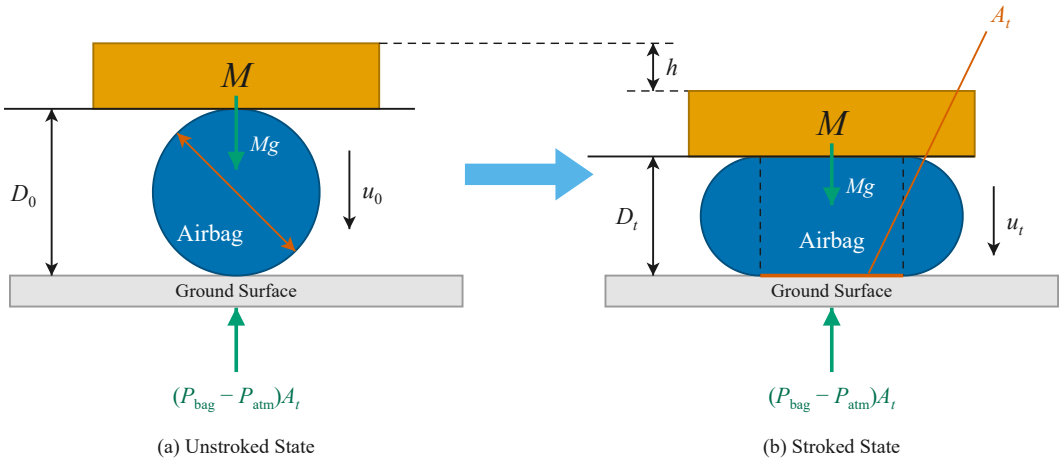


Figure 2.1: One-degree-of-freedom airbag impact model in the unstroked (a) and stroked (b) states, adapted from Zhou et al. [38].

The quantities under study for this type of phenomenon are shown in Figure 2.1, where a point load (yet,

[2]T. F. Patterson, *Design, Fabrication and Testing of an Airdrop Platform Utilizing Airbags as Shock Absorbers*, NATICK/TR-87/042, U.S. Army Natick Research, Development and Engineering Center, Aug. 1985.

with a finite extension, as to avoid penetration in the airbag and justify the crude, though analytically necessary, *uniform pressure/control volume* assumption, where pressure is uniformly distributed within the airbag volume) is protected by a single inflated airbag in incipient contact with the ground surface after a free fall. The quantities are:

- M, the mass of the payload, which remains constant throughout the analysis and results in the vertically applied gravitational force Mg.
- u_0, the initial impact velocity at time t_0, which evolves to u_t after each subsequent time increment. It is updated at every time increment through kinematic analysis.
- D_0, the initial diameter of the airbag, which evolves to D_t following the same loop update concept. Broadly speaking, its meaning can be extended to the characteristic dimension of the airbag, according to the airbag shape selected for the test case among the available options (vertically- or horizontally-oriented cylinder, sphere, toroid, etc.).
- h, the *stroke*, i.e. the (vertical) distance covered by the payload when compressing the airbag during the impact event.
- $(P_{bag} - P_{amb}) \cdot A_t$, the normal reaction force from the ground, based on the difference between the internal pressure of the airbag P_{bag} and atmospheric pressure P_{amb}, multiplied by the footprint area A_t, i.e. the contact area between the airbag and the ground surface. A_t depends on the shape of the airbag and is obtained from two quantities not marked in Figure 2.1: the length of the cross-section perimeter in contact with the ground, and the extension of the airbag in the out-of-page direction. Both are related to the stroke through geometric assumptions, so that $A_t = f(h, t)$.

The reference system is downward-positive, in accordance with the direction of gravity and the initial velocity, making the reaction force negative.

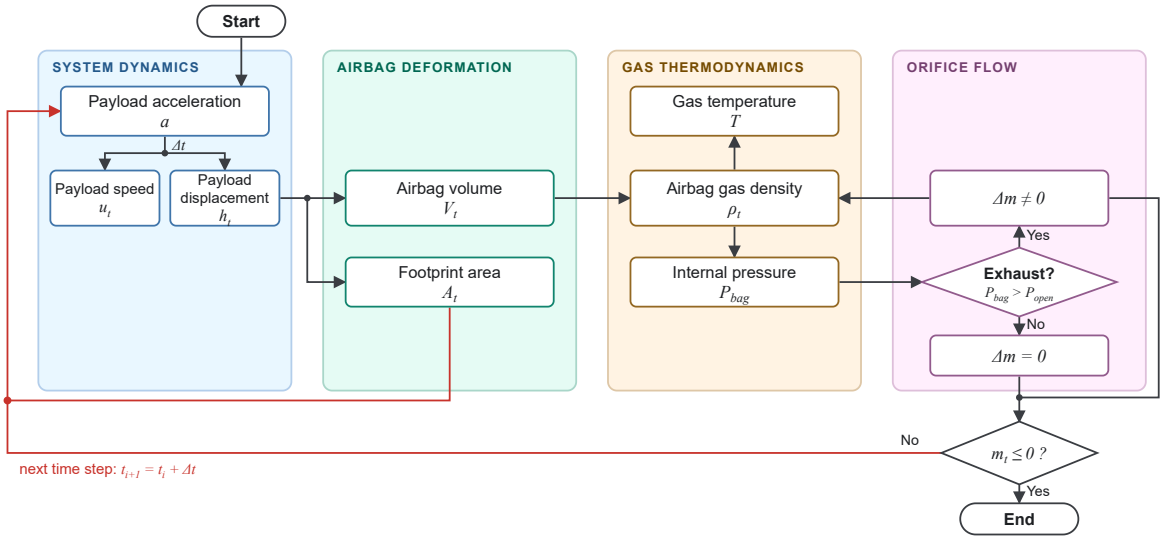


Figure 2.2: Calculation flowchart of the one-degree-of-freedom airbag impact model, adapted from Zhou et al. [38].

Except for the initial conditions and the payload's mass, these quantities are updated iteratively at each timestep to produce characteristic outputs for airbag-governed impact events such as *airbag volume*, *internal pressure*, *internal gas mass*, *payload acceleration*, *payload speed*, and *payload displacement*. The logic of this calculation loop is schematised in Figure 2.2, which is an adaptation of the original diagram presented in the paper. The calculation loop is divided into four blocks, run sequentially at each timestep:

1. **System dynamics step**: the resultant acceleration a acting on the payload is calculated based on the reaction force from the ground calculated in the previous timestep, and then integrated using the *central difference scheme* to obtain payload velocity u_t and displacement (or stroke) h.

2. **Airbag deformation step**: the airbag volume V_t and footprint area A_t on the ground surface are updated according to the kinematic information from the previous timestep. In this step, the hypothesis linking the stroke to the footprint area is enforced based on geometric assumptions, as mentioned earlier.
3. **Gas thermodynamics step**: the state variables of the gas are updated assuming adiabatic compression, as it is safe to assume that little to no heat is exchanged during the rapid compression process.
4. **Orifice flow step**: this critical step accounts for energy dissipation during the impact event. By modelling the outward airflow as flow through a one-way valve, treated as an orifice embedded in an impermeable fabric, typical nozzle fluid dynamics equations are applied to calculate the Δm vented at each timestep. In the case of this specific paper, mass is vented only if the internal pressure P_{bag} exceeds threshold pressure P_{open}, greater than atmospheric pressure P_{amb}. Within a single timestep, venting takes place as long as the exhaust condition withstands, updating gas density ρ_t and recalculating the gas' temperature T and pressure in an internal feedback loop spanning over a manually controlled number of venting sub-steps.

In the framing of Zhou et al. [38], the code stops either when the manually-set duration of the phenomenon is reached, or when all mass is vented off. Testing of the code indicates that the no-rebound condition shown in the paper is numerically enforced by introducing an if-check managing the application of the reaction force.

The core work of the present thesis resides in turning the working principle of airbag impact events as presented by Zhou et al. [38] into an informed, multi-variable problem that requires numerical optimisation. The code presented in later Section 4.5 is an expanded version of this implementation, incorporating additional criteria and alternative termination mechanisms.

2.3.2. Design guidelines, requirements, and variables

Framing

In essence, an airbag can be considered as an energy absorber and dissipator that must bring the payload from its impact velocity u_0 to rest within an available stroke h, without exceeding an allowable deceleration a. This relation is expressed in its most basic form by the kinematic relation [40], [41]

$$h = \frac{u_0^2}{2a} \tag{2.1}$$

which shows the central trade-off in airbag design: the lower the allowable deceleration, the longer the stroke needed to decelerate the payload and, therefore, the larger the airbag. During compression, the airbag behaves as a non-linear spring whose stiffness depends on its geometry and venting characteristics [38], [39], [42]. Venting is what dissipates the absorbed energy [42]; whatever energy remains stored in the gas at the end of the stroke is returned to the payload as rebound [43]. Increasing the volume of the airbag and inflation pressure raises its energy dissipation capacity, but at the cost of higher membrane loads on the fabric [40], [42].

Configuration concepts

Airbags are generally classified into three types depending on whether they contemplate the presence of venting in their design: *non-vented*, *vented*, and *combined* [38]. A combined airbag usually consists of an outer, main-venting airbag and an inner, non-venting, anti-bottoming airbag. This "bag within a bag" concept prevents the payload from bottoming out once the outer bag has vented [42], [44], [45], and commonly relies on complex sequential inflation of the outer and inner bags [45]. An alternative to this design is using multiple airbags side by side rather than one within the other, following a "fail-safe" design philosophy by sharing impact loads and providing redundancy in case of airbag failure [40]. Rather than completely enveloping the payload, the airbag can act as a buffer cushion beneath it [45], or inflate between plates attached to the sides of the cargo [46]. Moreover, venting can be passive, activated pneumatically, or active, activated electronically [37], [43]. Active venting is the basis of adaptive impact absorption, in which the airbag responds to the actual, case-specific impact load [35]: for example, in multi-chamber designs, controllable valves allow gas to flow between adjacent chambers and to the environment, redistributing a non-uniform load across the structure [36], [37].

Design requirements

Notable airbag design requirements and recommendations identified in literature are summarised in Table 2.2 and relate to impact conditions, such as the vertical and horizontal velocity, impact attitude, and landing surface [40], [42], [47]; to the allowable deceleration of the payload; and to general constraints on mass, packaging volume and deployment time [35], [45], [48].

Table 2.2: Design requirements for airbag landing and impact-attenuation systems identified in the literature.

Category	Requirement	Sources
Dynamics	Limit the peak acceleration transmitted to the payload	[35], [42], [45], [49], [50]
	Reduce the touchdown velocity to a target value	[35]
	Cope with horizontal velocity (non-nominal impact attitudes coupling)	[40], [42], [47]
Landing conditions	Cope with different landing surfaces	[44], [47], [51]
	Avoid fabric rupture and tearing	[42], [48]
	Prevent bottoming-out of the payload	[38], [42], [44], [45]
	Limit rebound and prevent rollover	[43], [45]
Inflation pre-landing	Synchronise venting and payload contact	[52]
	Deploy fully within the available time	[35], [45]
	Retain pressure for the required duration	[45], [48]
Packaging	Survive long-term stowage in the packed state	[48]
	Integrate with the host vehicle without interference	[35]
	Withstand the environmental temperature range	[47], [48]
General	Minimise system mass	[35], [45], [48]
	Allow reuse	[45], [47]
	Provide redundancy against single-bag failure	[40]

Design variables

All airbag design variables found in the literature analysed for the present review can be found in Table 2.3. The variables in the table are divided into two umbrella categories: *Energy absorption at impact* and *Deployment and packaging*. These categories can be mapped almost directly onto two distinct impact phases (at-impact and pre-impact behaviour, respectively) and, consequently, into two separate, although not entirely uncoupled, design stages: *phase 1*, focusing on the inflated configuration during impact to *limit peak acceleration* below the allowable limit set by the payload's integrity requirement, and *Phase 2*, focusing on the pre-impact inflation process by mostly *optimising out-of-stowage deployment time* and leading to prototyping.

Table 2.3: Airbag design variables identified in the literature, grouped by design focus and subcategory, with the associated trade-offs.

Group	Variable	Trade-off	Sources
Energy absorption at impact			
Geometry	Shape	Pressure-vessel shapes carry membrane loads well; faceted shapes are easier to build	[38], [40], [42], [44], [51], [53]
	Characteristic size	More volume absorbs more energy but raises membrane loads, chance of rebound, and overall structural weight	[40], [42], [43], [54]
	Payload contact area	Larger contact spreads the load; wrapping the entire payload wastes material and increases weight	[44], [45], [53]
	Number of chambers	Inner anti-bottoming and multiple bags add margin and redundancy (fail-safe philosophy), but increase weight and complexity	[37], [38], [40], [42], [44], [45]
	Tethers and webbing	Divert inertial loads from the fabric and control the shape, at the cost of added parts	[44], [51], [55]

continued on next page

Group	Variable	Trade-off	Sources
Fabric	Fabric type	Strength, tear resistance, low permeability, temperature flexibility vs. mass, packability	[44], [45], [48], [50], [51], [52], [55], [56], [57], [58], [59]
	Fabric thickness	Thicker fabric raises the rupture margin but slows deployment, enlarges the package, and increases weight	[42], [49], [60]
Venting	Vent area	Larger orifices dissipate more energy and limit rebound, but risk bottoming-out	[39], [40], [42]
	Vent location	Sets the outflow and whether the ground or payload blocks the vent during the stroke	[44]
	Vent burst pressure	Early opening lowers the peak pressure and resultant acceleration but spends stroke; late opening raises the peak pressure and acceleration	[40], [52]
	Fabric porosity	Distributed gas leakage dissipates energy without vents but reduces pressure retention	[48], [49], [60], [61], [62]
	Vent actuation	Active valves adapt to the actual impact; passive valves need no sensing or power	[35], [36], [37], [43], [44], [45]
Gas	Initial bag pressure	Higher pressure stops the payload in a shorter stroke but raises membrane loads	[40], [42], [52]
	Gas composition	Specific-heat ratio sets compression stiffness; mixtures cut temperature sensitivity	[44], [50], [52], [56], [63]
Deployment and packaging			
Inflation	Inflator type	Pyrotechnic vs. stored compressed gas: drives gas output, package size and triggering mechanism	[44], [52], [56], [63], [64]
	Inflator supply pressure	Faster filling vs. higher deployment loads; the flow chokes at sonic conditions	[52], [61]
	Inflator mass flow rate	Shorter injection time vs. higher deployment loads	[49], [60], [61]
	Inflator gas temperature	More pressure per unit gas mass vs. thermal load on the fabric just after inflation	[44], [48], [49], [60]
Housing	Housing size and geometry	Compact housings save mass and volume but pack the bag tighter	[49], [65]
	Housing material	Plastic saves mass over metal; metal bases give a stiff mounting	[41], [49], [56], [65]
Folding	Folding pattern	Governs unfolding kinematics, deployment time and pre-stress in the packed fabric	[48], [58], [60], [61], [65], [66], [67], [68], [69]
	Number of folds	More folds give a more compact package, but lengthen the unfolding sequence and make it more sensitive to small variations in the folding and the gas flow.	[65], [68], [69]
Seams	Seam locations	Seams concentrate stress and are the main failure site; patterning minimises their number	[48], [55], [70]

The table conveys the complexity of the design problem at hand, highlighting the large number of design variables. Multi-variable problems typically require an optimisation process where an optimal combination of variable values is achieved through guided iterations based on penalty concepts. This consideration forms the basis for the following chapters, in which the airbag design process is distilled into an ad-hoc workflow tailored for a specific payload application and impact scenario, focusing on a single design phase among the two presented. It is now possible to define the research-development objectives of the present thesis, leading to the proposed solution in Chapter 3 and the specific implementation choices presented in Chapter 4.

2.4. Research-development objective

Having presented the context in which the research project is allocated and having gathered the necessary design information, it is possible to address the requirements set by the project commissioners:

- Identify and characterise the mechanical allowables of typical medical payloads based on general packaging requirements for UAV-transported medical goods;
- Develop an informed conceptual design of an airbag system capable of deploying during emergency

release conditions to absorb impact energy and preserve payload integrity.

These requirements can be summarised in the following Research-Development Objective and Research Questions:

Research-Development Objective

Develop and demonstrate a design workflow to model and simulate an airbag-based protective system for UAV medical payloads using LS-DYNA, with the primary objective to ensure the structural integrity of pharmaceutical materials and their containment during drone failure and emergency payload release, thereby preventing environmental dispersion and supporting certification requirements.

Research Question(s)

How can a systematic multi-phase simulation workflow be developed to design an airbag-based protective system for medical payloads during UAV emergency-release events?

Sub-questions:

1. What are the allowable mechanical limits (acceleration, stress, etc.) for medical products during emergency release, and how should these define the design envelope?
2. How can the inflated airbag configuration be optimised to satisfy impact-absorption requirements while balancing competing design requirements?
3. What recommendations facilitate transforming the inflated geometry into a compact, functional deployment system?

3

Methodology: The Airbag Design Workflow

3.1. Problem statement

To properly outline and, more importantly, test the workflow proposed as a main objective of this thesis, it is necessary to identify the specific context in which this project is embedded, as it originates from prior research work conducted within the specific framework of Turin. The design inputs presented below were gathered from analysing the results presented by the group ([12], [13], [14], [15], and [16]) and categorising them into mission aspects on which to root the proposal of Section 3.2.

3.1.1. Reference mission and design inputs

Route and flight profile

The reference mission is based on a hospital-to-hospital route as initially suggested by Cavallo [12], connecting two hospitals across the city of Turin as shown in Figure 3.1, representing a city map with route schematics. In particular, the starting point is the Molinette Hospital in the southern region of the city, and the destination is the San Giovanni Bosco hospital in the northern sector. As they are located far apart, these two hospitals require medical deliveries to traverse a broad part of the city's territory. The hospitals are connected through two possible routes:

- **Route A**: initially hypothesised by Cavallo and confirmed by Bertelli [13], it takes advantage of the above-water air corridor over the river Po and its tributary Stura, mostly skirting the city and flying over city ground only in the relatively short segments connecting the hospitals to the rivers. Among the two routes, it is the safest (and, thus, most certifiable according to Cavallo) as it reduces the probability of ground impact in case of drone failure due to the UAV's permanence in the above-water air corridor. This route was selected as viable within the concept of operations (ConOps) that informed the preliminary drafting of the SORA. Approximate length: 11 km. Approximate duration time: 15 min, comparable to ambulance-based delivery times.
- **Route B**: a crude alternative to Cavallo and Bertelli's Route A, connecting the two hospitals in a straight line over the city. Angelini [16] deemed this route infeasible based on their comparative safety analysis of delivery means, primarily due to the lack of proper risk mitigation measures concerning in-flight failures. However, as the approach proposed in this thesis aims at partially mitigating these risks, it was chosen to consider this route (at least for preliminary evaluation).
- **Alternative routes *(omitted from analysis)***: the work of Fiorentino [15] and Angelini [17] is rooted in minimum risk-based route selection considering territories that extend beyond city borders, specifically including the entire province[1] in case of Angelini's study. Either using the RRT*

[1] In Italy, local government is organised across three administrative levels: regions, provinces, and municipalities. Here, assessing the Province of Turin corresponds to evaluating the territorial sub-region of Piedmont administered by the provincial capital of Turin.

algorithm in Fiorentino's case or the Dijkstra and A* algorithms in Angelini's case, these two studies share a common approach of considering multiple layers (such as population density, shelters, and critical infrastructure) to calculate the probabilistic minimum-risk paths for different means of transportation (eVTOL, UAV, etc.), either with predefined start and end points within the city of Turin and path calculation with constraints on mission time, or without predefined start and end points thereby determining the optimal drone hub location across the entire province. The results of these works were omitted from the definition of the reference mission for three main reasons:

- Angelini's work does not specifically target hospital-to-hospital operations, focusing on optimising general logistics rather than facilitating inter-hospital transport, contrarily to one of the primary objectives of the present thesis;
- The inter-hospital trajectories proposed by Fiorentino offer little advantage in terms of time with respect to the over-the-river trajectory proposed by Bertelli. Depending on the UAV platform considered in the specific analysis, these trajectories either move inside the city territory, similarly to Route B, or extend far outside it, passing over the fields around the city, which minimises the risk but without any time constraint. Both remain valid alternatives, and the in-city variant is conceptually comparable to Route B, which is already analysed in this work. For a preliminary analysis, however, the over-the-river trajectory represents the safer option within reasonable flight times, and it is therefore adopted here, while the route over the fields may be worth considering for less time-critical missions;
- Angelini's findings indicate that drone operations, when evaluated against the merit indices adopted in their analyses, are deemed unviable due to safety constraints. In contrast, the objective of this research is precisely to identify a mission scenario that is unfeasible under current conditions and implement risk-mitigation strategies to ultimately enable drone transit.

In either Route A or B, as specified in Bertelli's ConOps, the cruising height is about 75 m, at 17 m/s cruising velocity (approximately 70% of the maximum cruising speed allowed by the UAV platform of 25 m/s, unsustainable for the entire duration of the mission at the time of evaluation), with possible speed variations according to mission urgency if specified in the ConOps.

Accounting for the presence of wind in the operating environment, wind speed in Turin averages about 1.8 m/s (6.5 km/h) [71], with a peak of about 28.6 m/s (103 km/h) in 2013 [72].

Ambient conditions are taken from the International Standard Atmosphere (ISA) at sea level, i.e. a reference temperature $T_{\text{amb}} = 288.15\,\text{K}$ (15 °C) and pressure $P_{\text{amb}} = 101\,325\,\text{Pa}$, consistent with the city's mean annual temperature of about 15 °C over the 2017-2019 period [73]. Given Turin's elevation (about 240 m) and the 75 m cruising height, the deviation from these values is negligible for the purposes of the preliminary design.

UAV platform

The UAV platform choice is inherently linked to the reference mission specifications above, meaning that the platform is required to sustain the cruising speed requirements specified in the ConOps. An MTOM of 9 kg and a 0.81 m characteristic dimension were also specified in Bertelli's ConOps, leading to the suggestion of a potential drone model suitable for the situation: DJI's *Matrice 300 RTK*. Despite having characteristics suitable for the speed requirements, the drone has a payload limit of 2.7 kg and a wind resistance of 15 m/s in cruise. While this drone was initially selected for the preliminary SORA, its specific parameters are not binding for the present study. In particular, the payload-mass and wind-resistance limitations may be disregarded, given that the ConOps set a much broader requirement for MTOM. In addition to these requirements, Cavallo, in their UAV platform market evaluation, considered several other factors: maximum flight time (usually corresponding to half the autonomy), geometry, glide speed, and frontal area, the latter covering an important role in free-fall drag calculations. By ConOps, the platform is equipped with a parachute, whose contribution is ignored in the calculations regarding the free-fall event.

[2] Imagery: Esri, Maxar, Earthstar Geographics, and the GIS User Community. Basemap: Web Mercator; grid: WGS84 / UTM 32N.

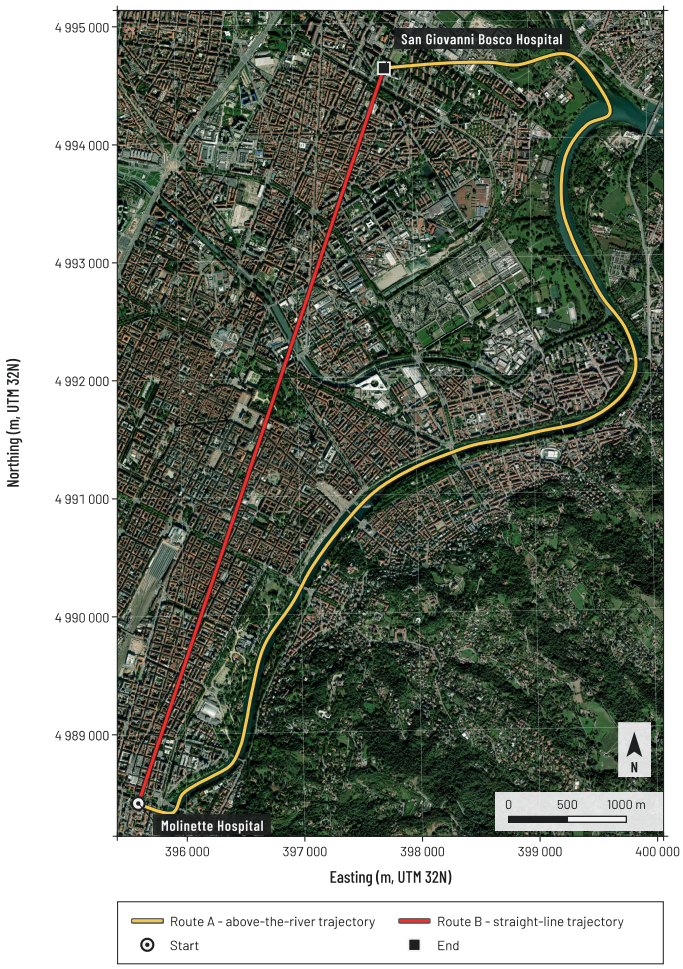


Figure 3.1: Reference mission routes across Turin, from the Molinette Hospital to the San Giovanni Bosco Hospital: Route A (above-the-river trajectory) and Route B (straight-line trajectory).[2]

Failure scenario and impact conditions

The chosen failure scenario implies an in-flight malfunction or fatal failure of the UAV in combination with a failure of the parachute mounted on the drone by ConOps. Excluding the nominal, non-conservative parachute descent of 3 m/s estimated by Bertelli, in this extreme failure scenario the payload is dropped in free air, separated from the drone at an altitude of 75 m above ground. After the failure, a *ballistic* descent mode is chosen for calculations of impact time and point depending on the release position on the map, including the effects of gravity and aerodynamic drag (as opposed to *uncontrolled glide* descent mode, which is applicable to the UAV platform but not to its detached payload), used in Angelini's impact probability study for ground risk evaluation and inspired by [74]. Wind enters the calculations (as mentioned in the reference route definition) as a horizontal velocity component.

Payload

Considering the wide variety of medical goods potentially suitable for UAV-based air transport, it was necessary to identify a payload representative of standard shipments, excluding human organs and specialised equipment (e.g. defibrillators) to focus specifically on pharmaceuticals. Therefore, glass phials of the medicinal product *Akineton Fiale*, containing *biperidene lactate* at a concentration of 5 mg/ml in injectable solution and shown in Figure 3.2[3], were chosen. The product is commercially supplied in boxes of five phials, each individually packaged in a plastic blister tray.

Akineton is a pharmaceutical product commonly used in the treatment of Parkinson's disease and

[3]Source of photo: `https://www.guttafarm.net/akineton-cena`.

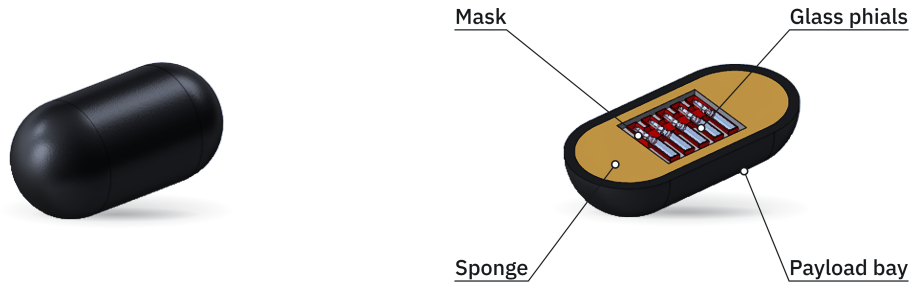
(a) Original packaging. **(b)** Single Akineton phial.

Figure 3.2: Akineton Fiale, 5 mg/ml injectable solution: original packaging and single phial.

drug-induced extrapyramidal symptoms, and is usually administered by intravenous or intramuscular injection. For the purposes of the present thesis, only the mechanical properties of the glass container are of interest, as the pharmacological action of the medical fluid is not relevant to the type of structural analyses implemented in this study.

Payload container

The design of the payload container to be carried by the UAV platform in delivery missions was proposed, but never fully prototyped, by Sivieri [14]. Its geometry is shown in Figure 3.3 as modelled for the scope of the present thesis: the central body is a cylinder capped with two hemispherical domes. In Sivieri's original design, this container had to be constructed via stratifying four layers, from inside out: first layer of 3d-printed PETG with thickness 2 mm, second layer of manually applied thermal insulation of unknown effective thickness, third layer of 3D-printed PETG and 2 mm thickness, and a final layer of manually-applied thermo-reflective paint. This sandwich-type structure was designed specifically for evaluating the thermal and vibration loads of a reference mission similar to the one presented earlier, and thus avoiding making considerations on transportability, attachment mechanisms, and other application-specific parameters. From a structural perspective, the first and third layer of Sivieri's design were connected by four radial ribs, but this detail was deliberately omitted in the exemplification of the payload container for modelling simplicity and redundancy, as explained later in Section 4.4.



(a) Closed container. **(b)** Sectioned view.

Figure 3.3: Payload container geometry: cylindrical central body capped with two hemispherical domes. The sectioned view shows the outer shell, the foam filling, and the housed payload.

Due to the artisanal nature of this design, linked to the difficulty in modelling specific 3D-printed PETG and the unquantifiable specifications of the insulation material, the payload container was modelled by retaining its shape with a 5 mm thick shell of injection-moulded acrylonitrile butadiene styrene (ABS), commonly used in packaging. The cavity is filled with polyurethane foam, chosen to simulate a protective wrapping (such as bubble wrap) that medical operators would use as established praxis to mitigate vibrations. The thermal paint is omitted from the payload container modelling process.

3.1.2. Information gap

The only missing inputs required for a full design workflow evaluation are:

- The derived *impact velocity*, *angle*, and *mass* for the selected scenario;
- The quantification of the phials' fragility through the characterisation of their material's mechanical *allowables*;

These failure scenario data must be fed into the design loop(s) proposed below to enable an effective design of the airbag-based impact attenuation system.

3.2. Proposed approach

3.2.1. Outer design loop: formalising the *safety-first* principle

In response to the gap identified in Section 2.2, the first contribution of this thesis is to propose an alternative design approach adopting (and later testing) the following guidelines:

Safety-First Design Principle

1. The UAV platform and its payload must be considered as two distinct entities;
2. In the type of operations considered, saving the payload takes higher priority than saving the UAV (carrying organs, expensive vaccines, emergency medicines, or high-value medical equipment is worth more than the drone itself);
3. Overall, the objectives are to minimise or mitigate:
   - ground impact consequences,
   - payload damage (preserving its integrity),
   - environmental dispersion;
4. These considerations should be applied at the beginning of the design process, possibly by integrating multiple safety devices (e.g., an airbag and a parachute working in tandem) and adjusting flight-mechanics considerations accordingly.

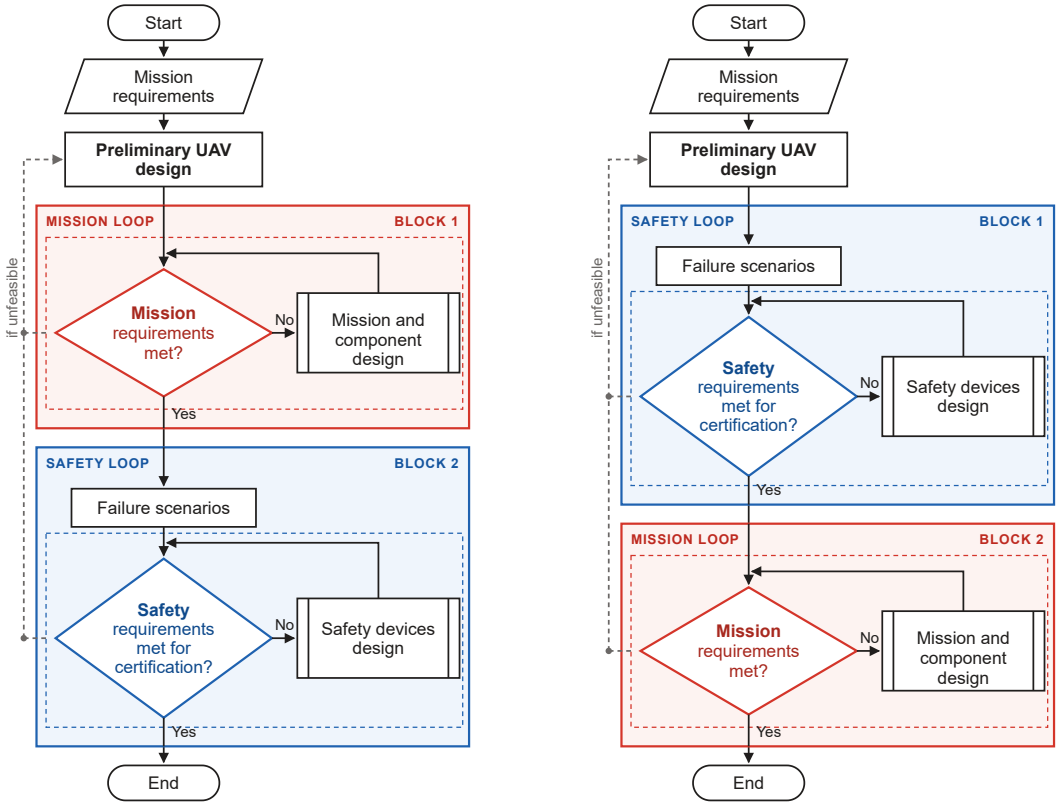
A direct consequence of applying the *safety-first* design principle in UAV-based operations' design is changing the sequence of design events, as exemplified in the simple flowcharts of Figure 3.4, where Figures 3.4a and 3.4b compare scenarios in which the mission loop can be interchanged with the safety loop to achieve design convergence.

3.2.2. Inner design loop: a payload-protection subsystem design workflow

This section aims to expand the *Safety loop* concept presented in Section 3.2.1 by informing it with the design guidelines distilled in Section 2.3.2. The proposed workflow can be seen in Figure 3.5, and is divided into seven steps that bridge the gap between payload allowables' evaluation and safety device (airbag) prototyping. The loop is also split into two main branches, highlighting the boundary between the present MSc thesis' deliverable and steps left to future developers. The first branch (Steps 0-4) is partially payload-specific, meaning that some of the considerations made throughout these steps depend on the specific medical payload considered, while also generalisable to a broader spectrum of medical payload receptacles; the second branch (Steps 5-7) are exclusively geometry-specific, implying that they require information on receptacle design, UAV-container hooking mechanism, and general information on system bulk, making them unattainable for the DIMEAS feasibility studies group, as they currently lack access to a functional medical payload container beyond their initial artisanal prototype by [14]. Completing the workflow should lead directly to the *Mission loop* as shown earlier in Section 3.2.1.

Step 0 - Preliminary mission analysis

This step covers the preliminary analysis of the payload's impact conditions. It evaluates the drop scenario, incorporates statistical analysis of wind direction and cruising conditions, and computes the trajectory and landing location of the payload. Serving as a general free-fall analysis, this phase yields



(a) *Mission-First* approach **(b)** *Safety-first* approach

Figure 3.4: Sequence of design events in the two approaches: *Mission-first* and *Safety-first* loops.

key design inputs: the free-fall time (t_{fall}) and the horizontal and vertical impact velocities (u_{x0}, u_{y0} respectively).

Step 1 - Axial compression test (experimental)

This step covers the experimental characterisation of the glass phials' (the selected medical payload) breaking point through laboratory testing. The goal is to conduct a test campaign to characterise the phial's mechanical behaviour, specifically providing key data, such as the failure force (F_{fail}, better defined in Section 4.2), which feeds into the next phase to accurately calibrate the finite element (FE) model.

Step 2 - Axial compression test (FEM)

This step focuses on matching the experimental behaviour observed in Step 1 with an FE model built to accurately reproduce the material's mechanical response. Specifically, both experimentally and in the FE model, the phials are subjected to axial compression loads, resembling the stress type to which they would be subjected to during an impact event. To replicate the laboratory results and build a calibrated material card that captures the phial's failure behaviour, the simulation mirrors the physical testing setup as closely as possible. This yields a generally realistic phial model, complete with its brittle failure threshold (FT_{fail}), that can be integrated into the full sled-test impact simulation in Step 3.

Step 3 - Sled test (FEM)

This step involves simulating a sled test, heavily inspired by automotive industry practices, to observe the mechanical response of a test article (such as a dummy in vehicle crash testing, or in this case, a glass phial) measured through perceived accelerations and damage under an input acceleration profile. The simulated test subjects the payload container (based on the artisanal design proposed by [14]) to realistic impact scenarios by reproducing an acceleration profile equivalent to a free-fall impact under the conditions defined in Step 0. This approach relies on the principle that the accelerations experienced

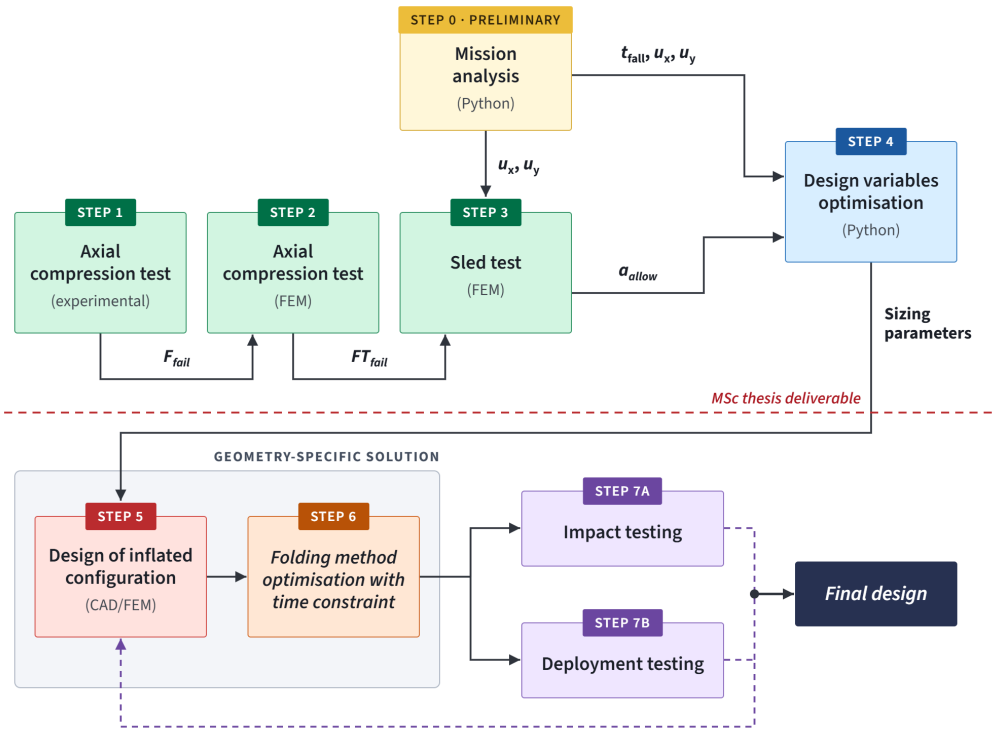


Figure 3.5: Inner design loop, specific for the airbag-based payload-protection subsystem.

by the payload directly correlate with the velocity that must be dissipated during impact, linking the input acceleration profile to the resulting output. The primary output of this phase is the payload's allowable acceleration (a_{allow}).

Step 4 - Design variables optimisation

This fourth and final step of the thesis deliverable translates the established boundary conditions evaluated in previous steps (failure loads, allowable peak accelerations, impact velocities) into a numerical Python script for airbag sizing. Building upon the framework established in Section 2.3.2, the simulation optimises the airbag design against critical criteria, including total mass, bottoming-out speed, and rebound behaviour. Ultimately, this tool yields fundamental sizing parameters that will inform the future geometry-specific design of the system.

Steps 5-7: Design of inflated configuration, folding method optimisation, impact and deployment testing

Steps 5 and 6 shift the focus to geometry-specific development. Step 5 details the fully inflated configuration tailored directly to the geometry of the payload and UAV, matching the gas generator mechanism to the drone layout and impact timelines. Step 6 then builds upon this complete configuration by incorporating real-world manufacturing constraints, such as folding patterns, seaming details, and stowage volume, to optimise the deployment mechanism. While inherently interconnected, Step 6 specifically targets deployment time optimisation, as defined in Table 2.3.

Once a preliminary design and folding mechanism are established, the prototype undergoes a two-part experimental validation campaign: Step 7A (deployment testing) and Step 7B (impact testing). This phase validates the FE models of Steps 5 and 6 by subjecting physical prototypes to real-world impact conditions, operating in an iterative feedback loop between simulation and testing until design convergence is achieved.

4

Workflow Implementation: Application and Deliverables

This chapter presents an example of how the airbag design workflow can be applied following the steps presented in Section 3.2.2. Each step requires its own ad-hoc procedure and analysis of the results. Hence, the procedure of each step is described first, its results are presented, and these are then fed into the next step. This structure is meant to show how the output of each step is used in the next, starting from a preliminary mission analysis, continuing with the characterisation of the medical payload and resulting allowables, and ending with recommendations on the design of an airbag intended to protect the payload in case of a drone failure.

The inputs used in this example are partly experimental and partly hypothesised. The material model of the glass phials is calibrated against experimental compression tests, while the geometry of the payload bay and the behaviour of the sponge are representative assumptions. Therefore, the numerical results should be read as a demonstration of the workflow, not as the qualification of a specific design.

4.1. Step 0 - Preliminary mission analysis

4.1.1. Implementation

Map extraction and preprocessing

The first step of the mission analysis consists in classifying and representing real-life geographical information at the scope of using it for probabilistic impact calculations. This implies processing OpenStreetMap (OSM) data extracted for the relevant geographical area of Turin, as identified earlier in Figure 3.1. The OSM file is usually a plain XML[1] document containing three types of elements, namely *nodes* (individual points with a latitude and a longitude), *ways* (lines or polygons connecting the nodes), and *relations* (groups of ways that together form larger features, e.g. a river). The system of reference used for OSM data is the *WGS 84* (World Geodetic System 1984) based on the traditional definitions of latitude and longitude.

A Python code was implemented to extract the city map's schematised information and classify it, so that all the geographical features present in the territory of Turin could be divided into five different macro-categories established for the purpose of this analysis:

1. `WATER`: these are "way" features that are schematised as one-dimensional paths with a thickness assigned in degrees (°) to match the WGS 84 coordinate system used, and they include waterways such as the Po river (60 m width = 0.0007° buffer), the Dora Riparia tributary (18 m width = 0.0002° buffer), and smaller streams (9 m width = 0.0001° buffer).
2. `BUILDINGS`: anything tagged with the `building` key in the OSM file, e.g. schools, churches, houses, hospitals, etc.

[1] An XML file is a plain-text document that uses custom tags to store, organise, and transport structured data.

3. `ROADS`: the ways tagged as `highway` across the city. Any transportation corridor or railway line is also classified as such.
4. `GREEN/NATURAL`: all natural terrain present in the region of interest, such as gardens, pitches, and parks, but also grass, farmland, meadows, recreation grounds, and woods.
5. `GROUND`: a collection of everything else that is not in the first four categories. It is assigned to any cell that is still unclassified at the end of this block.

Following the classification of the individual elements in the OSM file, a change in reference system was necessary to project the three-dimensional geographical features expressed in degrees latitude and longitude onto a two-dimensional map in metres. This was achieved through a Universal Transverse Mercator (UTM) projection, expressed in:

- Easting (x): distance in metres from the zone's central meridian (9 deg E for Zone 32), with a false offset of 500 000 m to avoid negative values;
- Northing (y) : distance in metres from the equator.

The conversion from geoidal to planar geometry was done according to the specific *zone* of the globe in which Turin is to account for specific deformation and correction. Since Turin is in Zone 32N (EPSG:32632), the coordinates expressed in metres become about 4990 km north of the equator and about 400 km west of the central meridian of the zone (9°E). The bounding box shown in Figure 3.1 measures approximately 4757 m (easting) × 6924 m (northing).

The steps of OSM information classification and reference system conversion were preliminary to building the city map's grid. The resolution chosen was 500 × 500 (250 000 cells), each cell measuring approximately 9.5 m × 13.8 m. Individual cells were assigned one of the five feature categories described above based on what feature lied at their centre. For example, if the centre of a cell falls on a `BUILDING` feature, then the entire cell was assigned to `BUILDING`, despite the variety of features found within the cell's bounds. Naturally, this crude approximation was adopted to simplify (thus, shorten calculation times for) the process of grid-building. The resulting map can be seen in Figure 4.1.

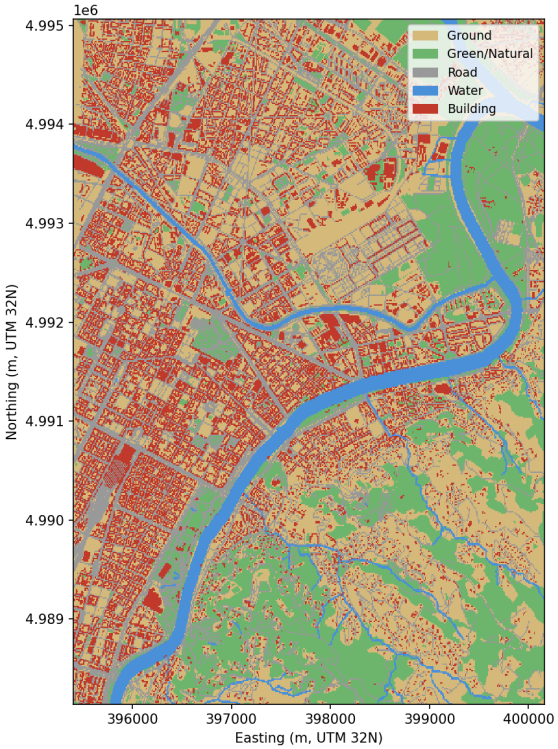
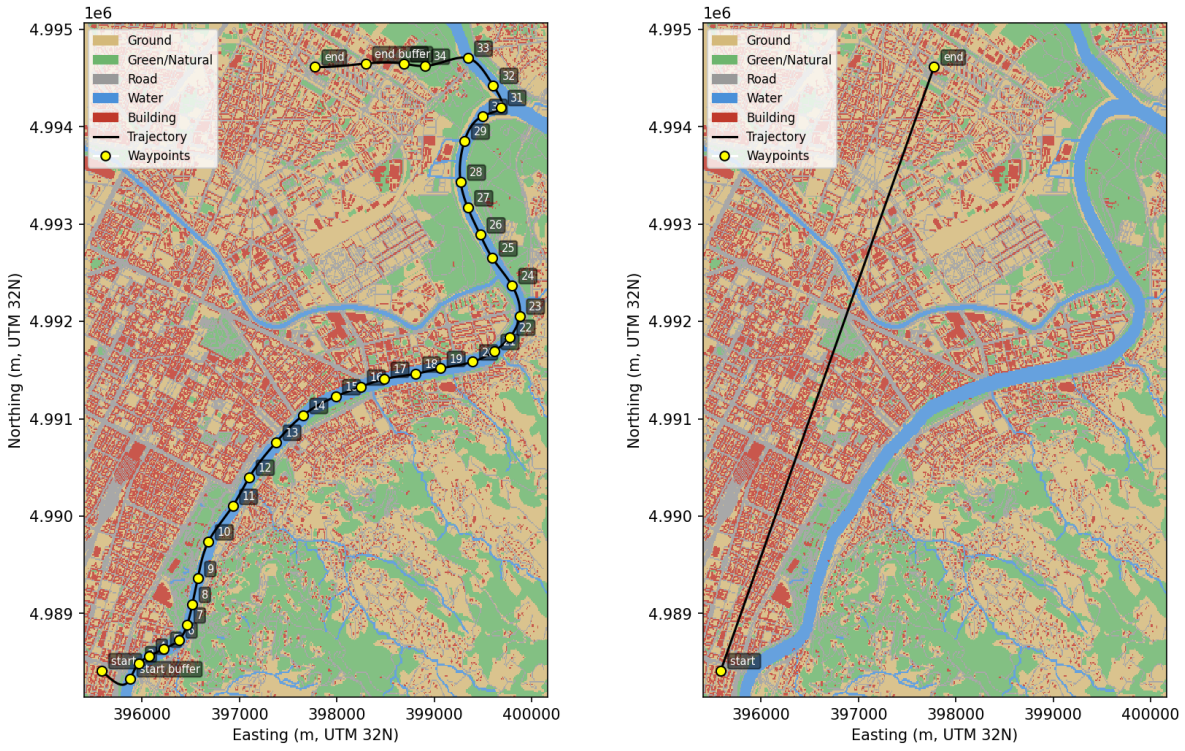


Figure 4.1: Classified terrain grid of the area of interest in UTM coordinates (Zone 32N), with 500 × 500 cells of approximately 9.5 m × 13.8 m, each assigned to one of the five categories `WATER`, `BUILDINGS`, `ROADS`, `GREEN/NATURAL`, and `GROUND`.

Mission trajectory definition

This step aims to define the trajectory above the map produced earlier. The waypoints of the trajectory (a collection of manually-selected points retracing the routes proposed in Section 3.1.1, defined in degrees of latitude and longitude and stored in an external Excel file) are loaded and checked against the bounding box of the map, ensuring that any waypoint falling outside it is discarded. The waypoints are then converted to UTM metres (zone 32N, EPSG:32632), with the same transformation used for the map, and interpolated with a cubic B-spline using the `splprep` function of SciPy. No smoothing is applied ($s = 0$), so the curve passes exactly through every waypoint. The spline is parametric, i.e. x and y are both expressed as functions of a parameter $t \in [0, 1]$, and the curve is made of cubic polynomials joined at the waypoints with continuous first and second derivatives. If fewer than four valid waypoints are available, the code falls back to straight line segments ($k = 1$).

Then, points every 10 m along the curve are extracted through an arclength parametrisation. The spline is evaluated at 100 000 equally spaced values of t, and the cumulative arclength is computed by summing the lengths of the straight segments between consecutive points. The values of t that correspond to every 10 m of distance are found by linear interpolation on this cumulative arclength and saved with the new label t^*, and the spline is evaluated at these values to give the points where the evaluation is conducted later. The same goes for the heading angle, computed at every sample point from the local direction of the trajectory and measured counterclockwise from East. The sample points guarantee a high level of accuracy because, for a total length of about 11 km, consecutive spline points t are about 0.11 m apart, so there are about 90 of them for every sample point t^*. The trajectories following Route A and Route B are shown in Figure 4.2.



(a) Route A (above-the-river trajectory along the Po and the Stura).

(b) Route B (straight-line trajectory over the city).

Figure 4.2: Mission trajectories over the classified terrain grid in UTM coordinates (Zone 32N), from the Molinette Hospital (start) to the San Giovanni Bosco Hospital (end): (a) Route A, interpolated with a cubic B-spline through the numbered waypoints; (b) Route B, a straight segment between the two endpoints.

Failure and fall model

As described in Section 3.1.1, the reference mission scenario assumes that, in case of UAV failure with uncontrolled descent, the payload is detached from the drone and falls with no propulsion or control mechanism. Therefore, the equations of motion that determine its trajectory must include only the

gravitational contribution and the aerodynamic contribution, the latter through a drag term. Given the air density ρ, the frontal area of the payload S and the drag coefficient C_D, the drag force is

$$F_d = c\,u^2, \qquad c = \frac{1}{2}\,\rho\,S\,C_D, \tag{4.1}$$

where u is the instantaneous speed of the payload.

For every combination of parameters, the drag constant c is calculated first, and then the terminal velocity

$$\Gamma = \sqrt{\frac{M\,g}{c}}, \tag{4.2}$$

i.e. the maximum vertical speed the payload can reach, at which drag balances gravity. For the payload considered here ($M = 1.0\,\text{kg}$, $\rho = 1.225\,\text{kg/m}^3$, $S = 0.04\,\text{m}^2$, $C_D \approx 1.0$), $\Gamma \approx 20.0\,\text{m/s}$.

The descent is then divided into three phases. Firstly, there is an "upward motion phase", present only if the payload has an upward vertical velocity at separation. Secondly, there is a primary descent phase in which the vertical velocity is lower than the horizontal velocity, which ends when the two become equal. Finally, there is the final descent phase from this crossing to the ground, in which the vertical velocity catches up and approaches Γ while the horizontal velocity keeps decreasing. Each phase has its own approximate closed-form expressions for position and velocity based on the model presented by Angelini [16]. The horizontal range is the sum of the distances covered in the three phases, and the effects of the wind and drag are taken into account throughout the calculation for both the vertical and the horizontal velocities.

The conditions at the moment of drone-payload separation cannot be predicted exactly for each single flight. For this reason, some parameters are modelled through a Gaussian distribution, each described by a mean μ and a standard deviation σ, and sampled randomly at each simulation, as listed in Table 4.1.

Table 4.1: Uncertain parameters of the drop model, each sampled from a Gaussian distribution.

Parameter	Unit	μ	σ	Remarks
Altitude at separation h	m	75	5	Small variations in flight altitude
Drag coefficient C_D	-	1.0	0.2	Payload tumbling or rotation
Horizontal velocity at separation $u_{x,i}$	m/s	17	6	Inherited from the drone cruise speed
Vertical velocity at separation $u_{y,i}$	m/s	0	0.5	Zero mean assumes level flight at failure
Wind speed V_w	m/s	15	0	Assuming a constant, high wind speed for conservative calculations
Wind direction ψ_w	deg	0	360	Assuming completely random wind direction, with no preferential direction

Monte Carlo impact loop

For each point t^* of the trajectory calculated earlier, i.e. for every point corresponding to a distance of 10 m along the curve, a Monte Carlo simulation is run with 5000 combinations of the stochastic parameters, selected randomly from the Gaussian distributions of Table 4.1. Having established these parameter combinations coupled with information on UAV heading direction at single t^* points, the function `impact_point_ballistic_descent` is called to calculate where the trajectory and the landing location of the payload. The function computes the horizontal range and the impact time of all 5000 scenarios at once and rotates the resulting cloud of impact points by the heading, so that it is oriented along the flight direction. Each impact point is then shifted by the wind drift $V_w\,t_{\text{fall}}$ in the wind direction, assuming that the wind carries the payload at its own speed for the whole duration of the fall, and the cloud is finally translated to the UTM coordinates of the trajectory point.

The impact points are then counted on a grid of cells [2000 × 2000], finer than the one used for the terrain map to guarantee higher accuracy in computing the impact distribution. As each trajectory point this gives a local distribution, the local distributions of all the t^* points are summed up with equal weight

and normalised over the total number of impact events, i.e. 5000 times the number of t^* points on the trajectory. At the end of the Monte Carlo simulation, the heatmap in Figure 4.3 is obtained, which gives, for each cell, the probability that the payload falls in that cell in case of a failure along the trajectory.

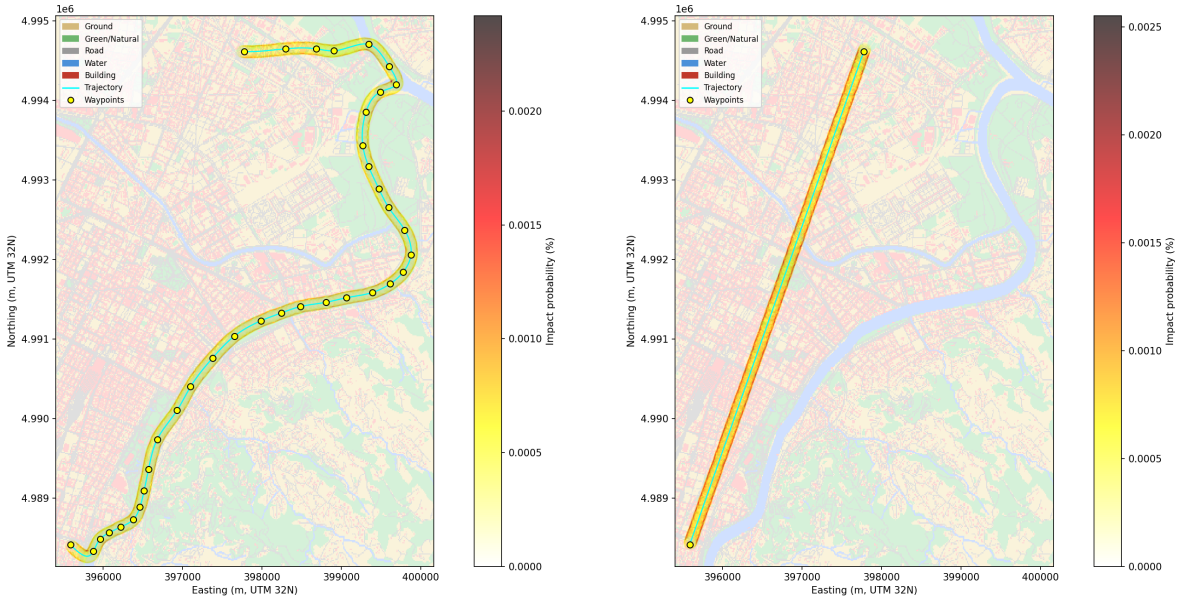
4.1.2. Results

Note on statistical to deterministic results conversion

The results of Step 0 presented in this section mostly come in the form of probability distributions, where the diversity of impact scenarios is entirely influenced by the statistical variability set at the beginning of the analysis. However, as the inputs required for the subsequent phases of the workflow must be deterministic for how the workflow was designed, it is necessary to adopt a meaningful datum selection criterion. The compromise lies in choosing traditionally *conservative* values, usually found at the extremes of the probability distributions, while also having them represent a statistically significant number of events. This consideration is valid for all results regarding the analyses conducted in Step 0.

Impact location distribution and surface statistics

The results of the Monte Carlo simulations described earlier are shown in Figure 4.3 and Figure 4.4. Respectively, they show impact probability maps over the selected geographical bounds of the city of Turin and the impact surface distributions along the mission trajectories.



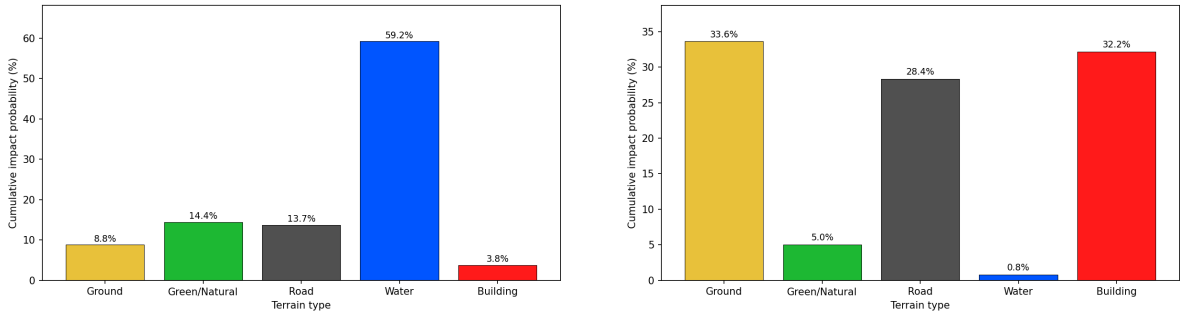
(a) Route A (above-the-river trajectory along the Po and the Stura).

(b) Route B (straight-line trajectory over the city).

Figure 4.3: Impact probability maps over the classified terrain grid in UTM coordinates (Zone 32N), obtained with 5000 Monte Carlo simulations per trajectory point and expressed as the probability (%) that the payload falls in each cell in case of a failure along the route: (a) Route A, 1099 trajectory points; (b) Route B, 658 trajectory points.

Figure 4.4 presents the principal reason for having previously excluded Route B from the certification process of the UAV-based medical delivery network: in case of a drone failure event, payload impact would often happen over populated territories, i.e. 32.2% over buildings, 28.5% over roads, and 33.6% over the broad *ground* category. On the contrary, in the case of Route A, 59.2% of the impact events would take place over water (corresponding to the water corridor used for this mission type), and 14.4% over green field, which is a generally preferable scenario when considering the concerns presented and explored in Section 2.1.

These results do not necessarily contribute to the airbag design workflow, but rather illustrate the type of scenarios that designers need to consider when designing the product and subjecting it to the required certification process.



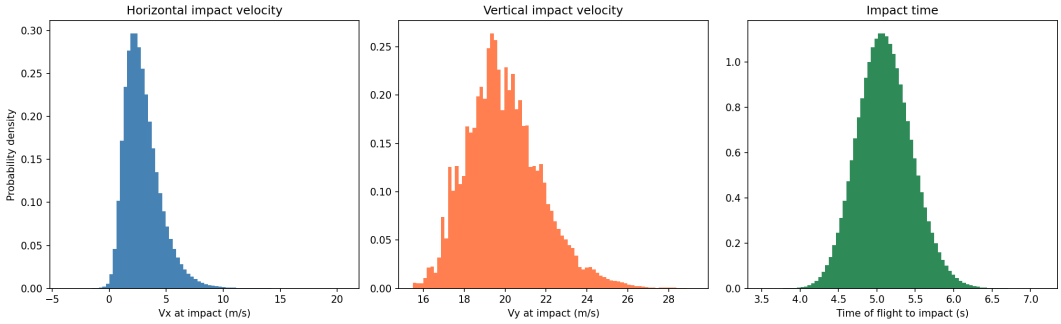
(a) Route A (above-the-river trajectory along the Po and the Stura).

(b) Route B (straight-line trajectory over the city).

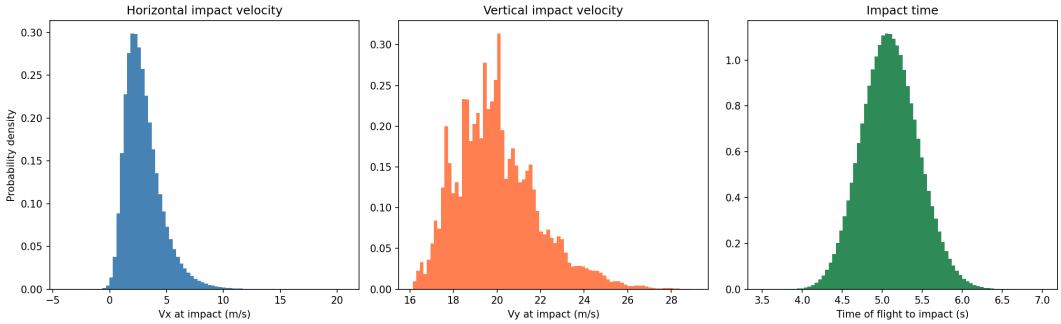
Figure 4.4: Impact distribution by terrain category (`GROUND`, `GREEN/NATURAL`, `ROADS`, `WATER`, `BUILDINGS`): (a) Route A; (b) Route B.

Impact conditions

Figure 4.5 shows distributions of horizontal impact velocity, vertical impact velocity, and time of flight to impact. As the descent's physics is computed with the same model on both routes, starting from the same distributions of drone-payload separation conditions, the impact conditions appear comparable for Route A and Route B. The horizontal velocity at impact ranges between 0 and 10 m/s, while the vertical velocity between 16 and 27 m/s, with little difference between the two scenarios. As for the time of flight to impact, the distributions are practically identical for both cases, with flights lasting between 4 and 6.5 s. This datum is the hard requirement to impose on the inflation mechanism of the airbag, but is not necessary for the design of the airbag's inflated configuration. Therefore, it reported here for future uses but will not be considered for the scopes of this thesis.



(a) Route A (above-the-river trajectory along the Po and the Stura).



(b) Route B (straight-line trajectory over the city).

Figure 4.5: Distributions of horizontal and vertical impact velocity, and time of flight to impact: (a) Route A; (b) Route B.

Based on the consideration presented earlier about statistical to deterministic results conversion, the *98.5th percentile* of the vertical and horizontal impact velocity distribution was adopted as the design value, corresponding to values of $u_x = 7.5\,\mathrm{m/s}$ and $u_y = 25\,\mathrm{m/s}$. These data include a relevant number of impact events (between about 49k - 82k), but are still characteristic of a rather conservative scenario, with relatively high levels of kinetic energy to dissipate in the impact event. In fact, it is likely that the

two velocities do not necessarily correspond to the same impact event, making this conservative choice lose its physical significance. However, based on the observation that these impact velocities are of the same order of magnitude of the terminal velocity calculated in Equation (4.2) with the contribution of wind drift as described in Table 4.1, they are kept as the reference impact event exemplification to be adopted throughout the workflow.

Step 0 deliverable: impact velocities and free-fall time

A summary dataset presented in Table 4.2 is selected as the starting input for the following phases of the airbag design workflow, namely Step 3 and Step 4 as shown in Figure 3.5.

Table 4.2: Impact conditions selected from the Monte Carlo mission analysis.

Quantity	Value	Source / note
Horizontal impact velocity (u_{x0})	7.5 m/s	98.5th percentile of the impact velocity distribution (Figure 4.5)
Vertical impact velocity (u_{y0})	25 m/s	98.5th percentile of the impact velocity distribution (Figure 4.5)
Free-fall time (t_{fall})	4-6.5 s	Percentile to be defined for future analyses; not required by the following steps

4.2. Step 1 - Axial compression test (experimental)

4.2.1. Implementation

The pharmaceutical product selected for the characterisation of medical payloads for UAV-based transportation is one of the most fragile phial types currently available on the pharmaceutical market. This choice was deliberate: by characterising a worst-case, low-strength test article, the resulting airbag design is expected to provide adequate protection also for more robust payload cases. The test consisted of a simple compression test, through which the maximum compressive load sustainable by the phial was measured.

Therefore, the parameter of interest for this test was the compressive force at which the phial broke, hereafter referred to as the ***failure load*** F_{fail}, expressed in Newtons. This was the only quantity of interest measured during the test. The failure load constitutes the experimental datum that is used to calibrate the FE model of the compression test in the design workflow's Step 2.

Test article

The test articles were glass phials containing 5 ml of saline solution, 70 mm in height, 15 mm in diameter, with an assumed wall thickness of 0.4 to 0.5 mm (see Figure 4.6). The saline solution phials represented safe alternatives to phials containing medicinal products of the type used in the typical failure scenario described in Section 3.1.1, Figure 3.2. The choice deliberately addressed safety concerns related to the toxicity of *Akineton Fiale*, to which the testing operators would have been exposed when the tested phials reached their breaking point (see paragraphs on *Test arrangement* and *Test procedure* for clarification, and Appendix B.1 for information on safety precautions). The assumption at the basis of this test procedure was that *the material of which the test articles were made corresponded to the material of the medicinal glass phials* relevant for the reference mission described in Section 3.1.1. This assumption was supported by previous investigations of standard pharmaceutical manufacturers in the Italian and European pharmaceutical landscape, where ampoules of this kind are made of Type I borosilicate glass in accordance with the ISO 9187-1 standard [75]. For the purposes of this test, only the mechanical properties of the glass container were of interest.

For the purpose of efficient data analysis, it is important to note that the phials originated from two distinct medicinal batches, designated as Batch A (36 phials) and Batch B (40 phials), provided by different pharmaceutical producers. The initial assumption that all phials share identical geometry and material properties stems from their classification as standardised containers designed to hold 5 ml of solution. Within the pharmaceutical supply chain, glass container manufacturers typically produce and dispatch these standardised phials to pharmaceutical producers, who subsequently fill them with

Figure 4.6: Glass phial containing saline solution, annotated with dimensions.

medicinal product and seal them through specific heat treatments. This implies that parameters such as exact wall thickness variation and specific thermal treatment histories cannot be directly measured prior to testing, and the assumption of the standardised geometric and material properties is ultimately proven incorrect by the experimental results presented in Section 4.2.2.

Test arrangement

The compression test was performed using an *MTS QTest/10* testing machine, equipped with a 10 kN load cell for force measurement and control software *TestWork*. The test setup is shown in Figure 4.7. The phial was held vertically by means of a custom 3D-printed PETG adaptor, which was clamped at its base by the machine's lower clamp. The compression head of the machine, mounted on the moving crosshead, was lowered from above until contact was established with the tip of the phial, after which the compressive load was applied along the phial's longitudinal axis.

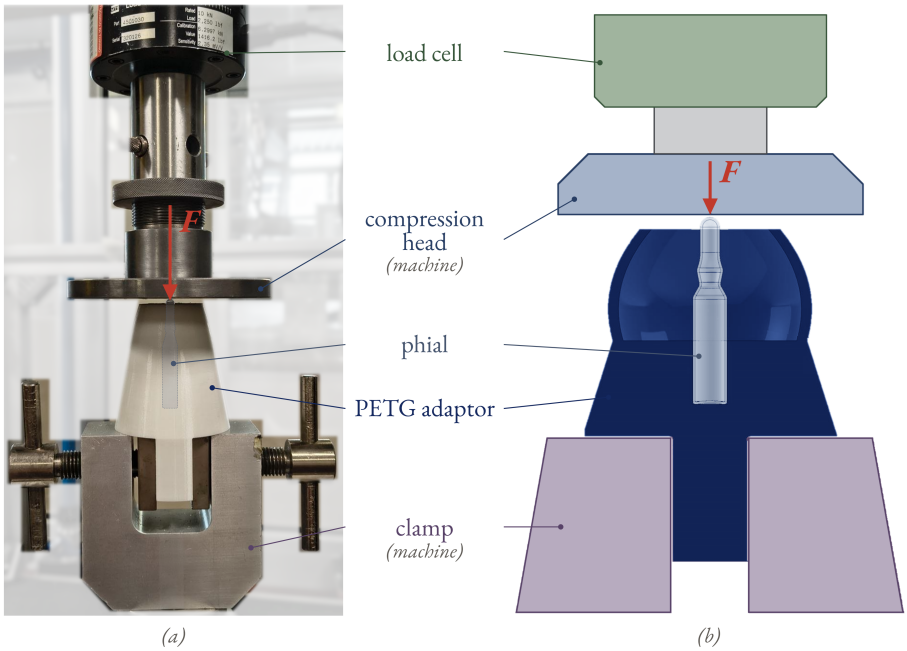


Figure 4.7: Test arrangement: phial held by a PETG adaptor and lower clamp, with the compression head applying the load from above.

The machine was operated under velocity control, measured in mm/min. During the test, the applied

force was continuously monitored as a function of crosshead displacement, until phial breakage occurred. Breakage was detected by the testing machine as a sudden drop in the measured load, which triggered an immediate, automatic stop of the crosshead motion.

Test procedure

The failure load was measured at three displacement rates: 1, 2, and 3 mm/min, which guaranteed the quasi-static nature of the test as the measured failure load was observed to be independent of the applied displacement rate (the variation in failure load across the tested rates fell within the expected scatter of the material).

The compression test was repeated on a batch of 76 nominally identical phials (although belonging to two different medicinal batches) to obtain a statistically meaningful population of failure load values. Each phial was visually inspected for pre-existing damage (e.g., cracks or chips) that could affect the test results and was discarded if any defect was found. The phial was then inserted into the PETG adaptor (cleaned from previous phial tests), ensuring correct vertical alignment and full seating within the adaptor up to the height of the phial's main body. Next, the PETG adaptor, with the phial inserted, was positioned within the lower clamp of the testing machine and secured. The compression head was lowered until contact was established with the tip of the phial, taking care not to apply any preload. Finally, the displacement rate was selected, and the test was monitored until phial breakage occurred (triggering an automatic stop of the crosshead). Displacement and applied load data were continuously recorded for each phial.

4.2.2. Results

The medicinal phials under study are not designed as standard coupons to test material properties. In addition, it is important to consider a second complication related to the material of *glass* itself, notoriously characterised by great scatter in mechanical properties. For these two reasons, it is key to introduce some statistical analysis in analysing the experimental results, specifically adopting a two-parameter Weibull distribution parameters to study the phials behaviour. The Weibull distribution is widely recognised in the literature as the most appropriate statistical model for describing the strength of brittle materials like glass due to its ability to capture the weakest-link nature of brittle fracture, which is governed by the size and distribution of surface flaws [76]. Several of the observations were made empirically, i.e. just by looking at the failure modes or even listening to sound cues in the laboratory, as the almost artisanal setup comes with strong limitations.

Phial failure modes and statistical observations

Prior to conducting the experimental campaign on the glass phials, it was hypothesised that the phials would break in five areas of potential stress concentration along their geometry. Figure 4.8 illustrates these critical areas: the phial's dome *A*, the first bulge *B*, the necking *C* (the failure point designed to open the phial), the base *D*, and the main body wall *E*. During the tests, it was noticed that not all failure modes would manifest and that a significant portion of the broken samples would show a combination of two or three failure modes, described as *hybrid* failure modes from this point onwards. Figure 4.8 is a qualitative diagram of various failure modes and exemplifies their nature through pictures taken throughout the testing campaign to accompany empirical observations.

A quantitative synthesis of the data gathered experimentally about phial failure trends can be seen in Figure 4.9, which categorises the 76 tested phials depending on the observed failure mode. In addition to stiffness and failure load ranges observed for the individual modes, the table in Figure 4.9 also gives some insights about the correlation between the parent medicinal batch (A or B) and the failure mechanisms. For example, the first row of the table shows that among all the 76 tested phials, 27 of them broke in mode A (dome). Within these 27 phials, 70% belonged to Batch A, and the remaining 30% to Batch B ("Batch mix" column). Furthermore, 53% of all phials of Batch A broke in failure mode A, while only 20% of Batch B showed the same failure mode ("% of batch with this mode" column). Looking instead at the second row of the table, it is possible to see that 18 phials broke in mode E (main body), 89% of them belonging to medicinal Batch B and representing 40% of the entire batch. These two rows give an important insight regarding the *design* of the two medicinal batches. Although nominally identical, Batches A and B show different typical failure modes, where Batch A favours mode A, and Batch B favours mode E. Generalising this line of reasoning, geometrically identical (as supposedly

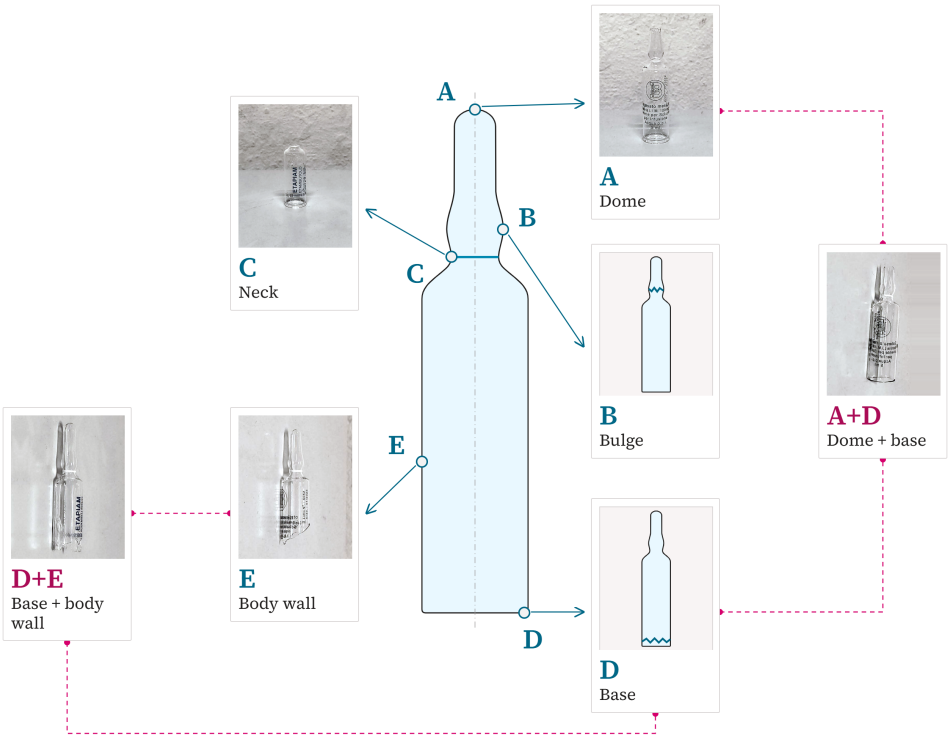


Figure 4.8: Observed phial failure modes: *Dome* (A), *Bulge* (B), *Neck* (C), *Base* (D) and *Body Wall* (E), together with examples of hybrid modes (A+D and D+E).

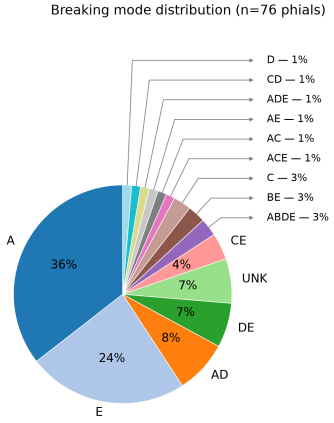
standard) phials coming from different pharmaceutical producers show radically different mechanical behaviour.

Diverging phial behaviour can be retraced to their manufacturing processes. The phials are originally manufactured open, featuring an aperture in place of the dome to allow them to be filled with the medicinal product. At a later stage, the upper part of the glass is melted and reshaped into the dome. Depending on the medical content, varying levels of control are implemented into the thermal treatment of the phial, as the primary requirements are functionality (breaking in the designed necking when opening the phial, also known as one-point cut (OPC) breaking systems, regulated by the ISO 9187-2 standard [75]) and transportability (withstanding handling and transport loads, such as the 1.2 m drop test prescribed by UN3373 for packages of biological substances [32]), over process control for all load cases. Therefore, it is entirely acceptable for different phials to exhibit distinct mechanical behaviours, even under identical loading conditions and nominally identical geometries, at the cost of needing to differentiate between pharmaceutical batches to draw any significant conclusions on material mechanics.

Stiffness data - glass phials and PETG adaptor

Out of the total 76 phials tested, only 65 phials were considered for the present stiffness analysis, with the remainder omitted due to unprocessability of the gathered data. Load-displacement curves were extracted from the raw dataset to construct the stiffness plot shown in Figure 4.10. To isolate the elastic part of the structural response, data recorded after the first breaking point was discarded for each phial, as the load-bearing capacity of the phials became compromised beyond this threshold. Moreover, knowing that the primary objective of the investigation is to characterise the phials failure load to avoid medical content dispersion during a phial drop event, it is acceptable to deem post-failure behaviour irrelevant.

Categorising the results by batch reveals that Batch B exhibits a visually distinct, generally higher stiffness compared to Batch A, reinforcing the observation that different batches display significantly different mechanical behaviour. Given that thermal treatments typically do not influence stiffness, this discrepancy suggests potential diversity in geometry, disproving the assumption made in the *Test*



Mode	n	Stiffness range (N/mm)	Failure load range (N)	Batch mix (of this mode)	% of batch with this mode
A	27	318-1506 (n=26)	61-497	A: 70%, B: 30%	A: 53%, B: 20%
E	18	675-2007 (n=17)	37-933	A: 11%, B: 89%	A: 6%, B: 40%
AD	6	259-1321 (n=6)	140-516	A: 83%, B: 17%	A: 14%, B: 2%
DE	5	940-1698 (n=4)	306-915	A: 40%, B: 60%	A: 6%, B: 8%
UNK	5	506-2107 (n=5)	208-1088	A: 40%, B: 60%	A: 6%, B: 8%
CE	3	1304-2239 (n=3)	518-972	B: 100%	A: 0%, B: 8%
ABDE	2	565-722 (n=2)	389-458	A: 100%	A: 6%, B: 0%
BE	2	271-1254 (n=2)	127-598	A: 100%	A: 6%, B: 0%
C	2	890-1726 (n=2)	442-712	B: 100%	A: 0%, B: 5%
ACE	1	587-587 (n=1)	232-232	B: 100%	A: 0%, B: 2%
AC	1	1200-1200 (n=1)	484-484	B: 100%	A: 0%, B: 2%
AE	1	n/a	133-133	B: 100%	A: 0%, B: 2%
ADE	1	1079-1079 (n=1)	602-602	A: 100%	A: 3%, B: 0%
CD	1	1685-1685 (n=1)	607-607	B: 100%	A: 0%, B: 2%
D	1	245-245 (n=1)	239-239	A: 100%	A: 3%, B: 0%

Mode codes: A = Dome B = Bulging C = Indentation D = Base E = Cylinder | combined letters (e.g. "AD") = multiple modes on the same phial | UNK = mode not identified

Figure 4.9: Failure modes observed over the 76 tested phials. Left: share of each single and hybrid mode. Right: number of phials, stiffness range and failure load range for each mode, with its split between the two batches (*Batch mix*) and the fraction of each batch that failed in that mode (*% of batch with this mode*).

article section presented earlier. Specifically, a thicker wall in Batch B's phials would account for both higher stiffness and an increased failure load, assuming a comparable failure stress under the current compressive loading state.

Across all curves, an initial compliant branch is observed. This could be attributable to experimental setup constraints as the phials settle and slide into proper contact before effective load transfer occurs. The response remains predominantly elastic across the observed stiffness range, with minor softening observed around 800 N, which is associated with the stiffness compliance of the PETG adaptor (a similar "step" is observed in Figure 4.11 at comparable loads, indicating some sort of elastic instability in the response of the adaptor). Moreover, a degree of random error must be considered regarding perfect alignment within the PETG adaptor. Deviations from perfect verticality (caused either by clearance between the adaptor and the phial or by subtle manufacturing phial shape imperfections) have a significant influence on the measured stiffness. This aspect is further investigated in the FE model presented in Section 4.3, which proposes considerations on initial geometric defects to evaluate their impact on overall phial stiffness.

In retrospective of completing the present experimental campaign and its FE modelling in Section 4.3, further attention had to be given to the strong limitations of the experimental setup. To validate the stiffness data gathered, it was necessary to conduct a separate investigation on the mechanical compliance of the setup, to determine experimentally the stiffness of the PETG adaptor. For this scope, the adaptor was subjected to the same loading configuration used for the glass phials (refer to Figure 4.7 for clarification), using a 15cm steel bar of comparable diameter to the glass phials (considered infinitely rigid compared to the PETG structure) to compress the component directly without the risk of inducing instabilities or buckling. The PETG adaptor's load-displacement data is shown in Figure 4.11, which evaluates the adaptor stiffness at a nominal value of 2037 N/mm obtained by a linear fit within the load band of mode A failures. Furthermore, it is possible to notice that slight softening behaviour occurred at approximately 800 N, introducing a step-like feature that matches the response observed in the high-load regime of the glass phial tests.

Overall, PETG adaptor stiffness serves as a critical datum for reconciling experimental stiffness with numerical predictions of the FE model presented later. Notably, calculating the adaptor stiffness was not originally planned during the test campaign, but rather it was performed post-hoc to resolve a significant mismatch observed between the empirical stiffness measurements and the initial numerical predictions.

Failure load extraction

Figure 4.12 displays the brittle failure points extracted from the experimental load-displacement curves for all 76 tested phials, categorised by batch and fitted to a two-parameter Weibull distribution. The

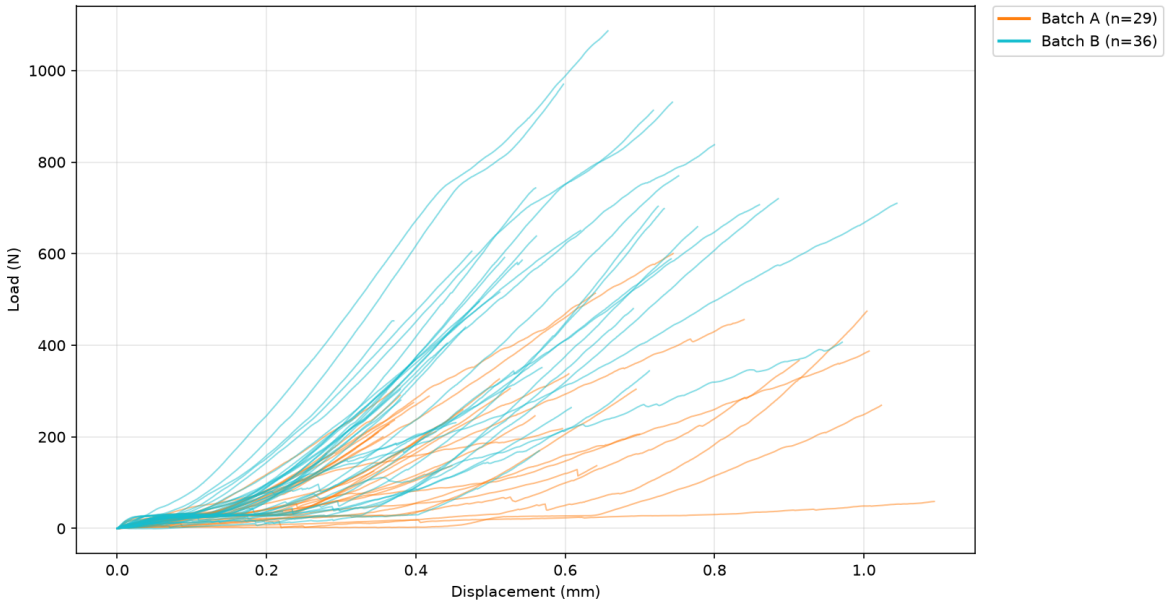


Figure 4.10: Experimental load-displacement curves of all tested phials up to the failure point, grouped by batch (Batch A, $n = 29$; Batch B, $n = 36$).

top plot illustrates the fitted two-parameter Weibull probability density functions, highlighting the 1st and 5th percentiles of each curve, while the bottom plot details the individual failure loads grouped by batch and displacement rate. An analysis of these Weibull distribution plots indicates that while clear stiffness data could not be extracted for every sample, a definitive failure load was successfully obtained across the entire dataset of 76 phials. These failure loads were plotted collectively, regardless of the observed failure mode, to provide an objective picture of the failure loads independent of the manual, visual classification of individual failure modes.

The resulting Weibull distribution highlights that Batch A exhibits a lower characteristic failure load but lower scatter, whereas Batch B shows a higher mean failure load alongside higher dispersion. Consequently, Batch A offers greater reliability for statistical and mechanical analysis and is therefore the only batch considered for the analysis from this point forward. Furthermore, these plots confirm that the testing rate had a negligible effect on the results, as evidenced by the lack of correlation between experimental outcomes and load application speed.

Step 1 deliverable: design failure load of the phial

Based on the considerations made previously, the summary dataset presented in Table 4.3 is selected as the baseline datum to feed into the subsequent stage of the airbag design workflow.

Table 4.3: Experimental data selected for the calibration of the FEM glass material.

Quantity	Value	Source / note
Target failure load (F_{fail})	98.7 N	Batch A, Weibull 5th percentile ($\beta = 2.44$, $\alpha = 332.7$ N, $n = 36$)
Mode A stiffness	318 N/mm - 1506 N/mm	Dome failure; 53 % of Batch A
Mode AD stiffness	259 N/mm - 1321 N/mm	Dome + base failure; 14 % of Batch A
PETG adaptor stiffness	2037 N/mm	Linear fit, 20 N-316 N band (Figure 4.11)

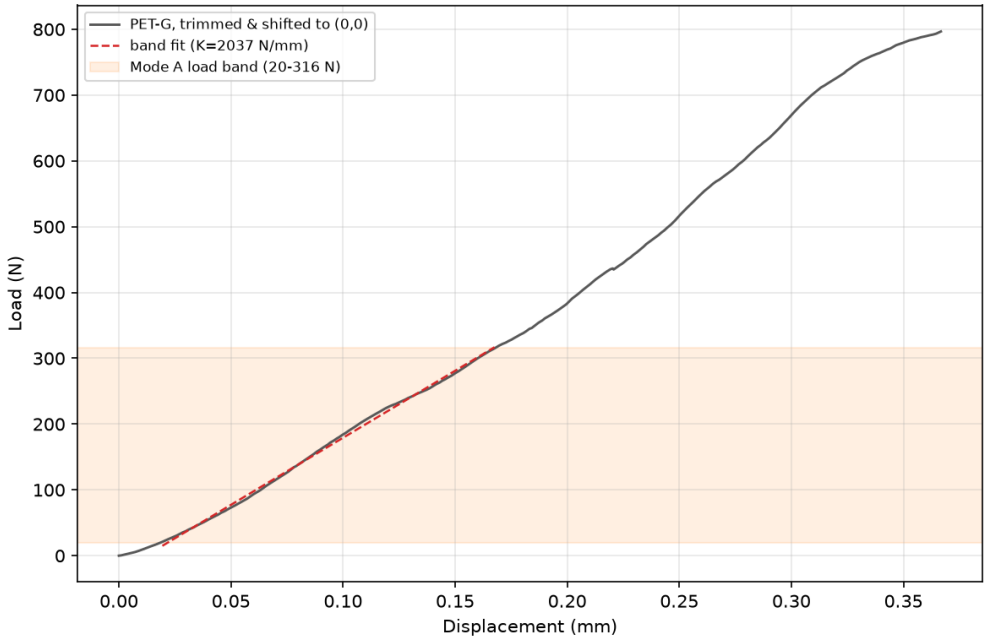


Figure 4.11: Load-displacement curve of the PETG adaptor, trimmed and shifted to the origin. The adaptor stiffness, $K_{PETG} = 2037\,\mathrm{N/mm}$, is obtained by a linear fit within the load band of mode A failures (20-316 N).

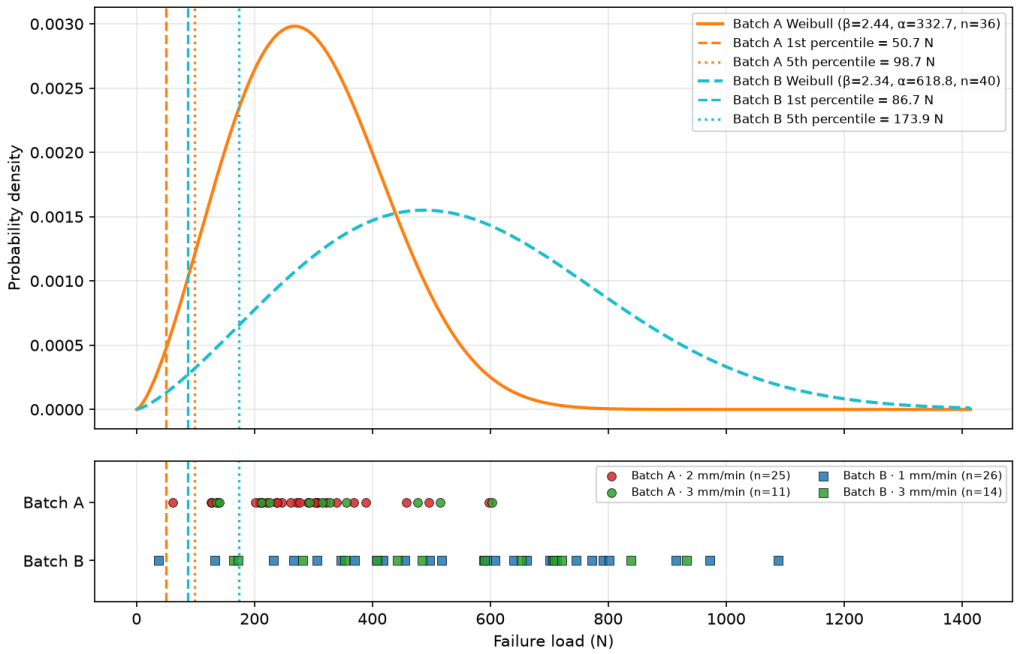


Figure 4.12: Failure load distributions of Batch A and Batch B. Top: fitted two-parameter Weibull probability density functions, with the 1st and 5th percentiles marked. Bottom: individual failure loads, grouped by batch and displacement rate.

4.3. Step 2 - Axial compression test (FEM)

The quantity of interest for the calibration of the glass phial's mechanical behaviour is the *tensile strength* `FT` of the glass material card in LS-DYNA (*`MAT_GLASS`, corresponding to the material model *`MAT_280`), which governs the failure load observed in the present FE model. As the model was designed to simulate as closely as possible the loading condition to which the phials were subjected in Step 1 of the workflow, the parameter `FT` is tuned iteratively until the failure force predicted by the FEM matches the value of 98.7 N measured experimentally. The calibrated model is then used to characterise the behaviour of the same glass phials in a separate FE model in Step 3, aimed at predicting the maximum impact acceleration that the phials can withstand. This value is one of the key design parameters for sizing the airbag protection device.

The model is built using *explicit* time integration. The unit system used in the present model is as follows:

Table 4.4: Engineering Units

Time	Distance	Mass	Stress / Pressure	Force	Energy / Work	Velocity	Acceleration	Density
ms	mm	kg	GPa (kN/mm^2)	kN	J ($kN \cdot mm$)	mm/ms	mm/ms^2	kg/mm^3

The current modelling choices are the result of several model iterations, carried out by trial and error, from a first implicit analysis (using Abaqus as a solver) to the present LS-DYNA model. These iterations are summarised in Appendix C.

4.3.1. Implementation

Model setup and glass material modelling

The model is composed of three parts, as shown in Figure 4.13: the *base*, the *crosshead* and the *glass phial*. The numerical setup aims to imitate the experimental test, where the glass phial is compressed in the axial direction, i.e. the vertical (Y) direction. Two considerations need to be made regarding the construction of the model:

1. The glass phials virtual twin was reproduced on the computer-aided design (CAD) software SolidWorks by taking manual (destructive) measurements on the specimens offered by the Molinette Hospital. As the phials' geometry contains curvatures, fillets, domes, and bulges, using a simple Vernier calliper certainly lowers the fidelity of the virtual representation and induces challenges in the evaluation of the numerical results;
2. The absence of the PETG holder in the present numerical model is a decision rooted in the risk of inaccurate or excessively complex modelling, as it was 3D printed with a reticular filling that would require a destructive intervention to be observed. The consequences of omitting the PETG adaptor in the FE model are another aspect that is explored and accounted for in the results presented later.

Within the FE model, the crosshead and the base belong to the same super-part, named *machine*, and they are meshed identically. Both are made of solid hexa elements with an average side length of 0.2 mm, extruded to a thickness of 1 mm in the vertical direction. In plan, they measure 18 x 18 mm (base) and 14 x 14 mm (crosshead). Both have enough surface area to cover the base and the dome of the glass phial, so that the breaking pressure and the reaction force from the base are applied correctly.

The two parts are modelled with a material that treats them as infinitely rigid bodies (*`MAT_RIGID`, material model *`MAT_020`), with only one free degree of freedom, in this case the displacement in the Y-direction. The base is then constrained in Y by a boundary condition that imposes a displacement of 0 mm, while the crosshead moves downwards and applies pressure on the phial in a quasi-static regime. The numerical workaround of leaving the Y direction free and then imposing a zero-displacement boundary condition on the base makes it possible to measure the reaction force exerted on it in the direction of the degree of freedom that was chosen as liable. With a fully constrained rigid body, this

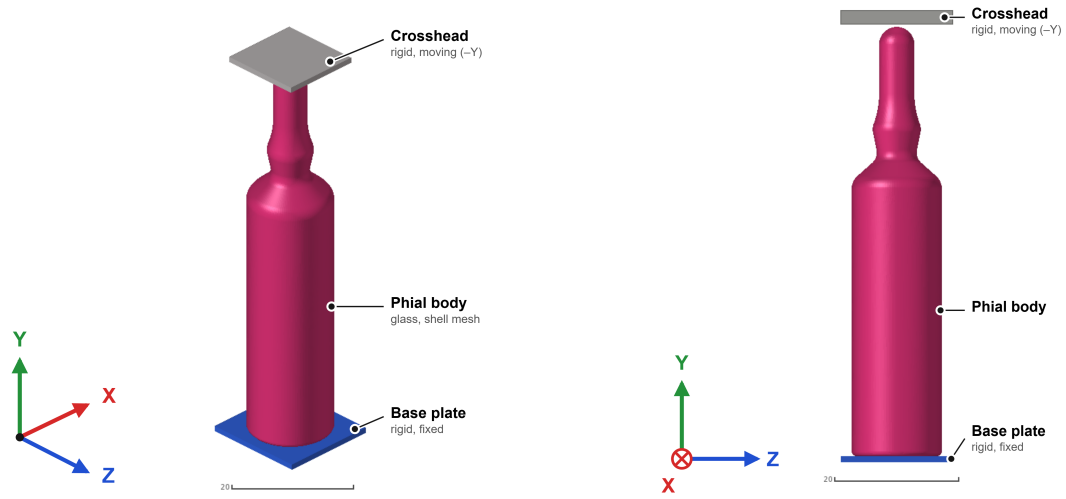


Figure 4.13: FE model of the axial compression test in isometric (left) and front (right) view: the *glass phial* is compressed between the fixed rigid *base* and the rigid *crosshead* moving in the −Y direction.

measurement would not be so straightforward, needing a slightly different procedure. The material card applied to the machine's parts can be found in LS-DYNA input 4.1.

LS-DYNA input 4.1: Rigid material card for the base and crosshead

```
*MAT_RIGID
$HMNAME MATERIAL         3acciaio_matrigid_y-free
$---+----1----+----2----+----3----+----4----+----5----+----6----+----7----+----8
$       MID       RHO         E        PR         N    COUPLE         M     ALIAS
          3   7.8E-06     210.0       0.3
$       CMO      CON1      CON2
        1.0         6         7
```

The glass phial is meshed with shell elements with `ELFORM = 2`[2] of thickness 0.4 mm, with 5 integration points along the thickness. The structured mesh has a regular size of 0.1 mm and is mostly composed of quads, with trias where needed to follow the shape, so that it keeps the curvatures of the dome and the fillets of the rounded-corner geometry. The material used for the glass phial is shown in LS-DYNA input 4.2, which highlights the *tensile strength* `FT` as the parameter to fine-tune the numerical prediction of the phial's failure load. The card has several parameters of interest that govern the post-fracture behaviour, which is not of interest for the present analysis as the objective is to identify the material's point of brittle failure, aligned with the broader aim to avoid medicinal dispersion during an impact-failure event. However, an informed attempt to model glass behaviour with an acceptable degree of fidelity in both the pre- and post-fracture regimes was made by setting material parameters based on a separate literature search across worked examples of `*MAT_GLASS` implementations [77] and the LS-DYNA user manual [78, p. 1989-2001]:

- `RHO`: density, 2.6E-06 kg/mm^3 (2600 kg/m^3);
- `E`: Young's modulus, 70 GPa, a relatively well-established value for commercial soda-lime glass, adopted here as an approximation of the Type I borosilicate glass of the phials (see Section 4.2), whose elastic properties differ only slightly. As the manufacturer of the glass phials is unknown, this value carries some uncertainty;
- `PR`: Poisson's ratio, 0.23, chosen within a range between 0.20 to 0.25 for soda-lime glass, with the same approximation and uncertainty as for `E`;

[2] Reduced-integration elements with one in-plane integration point and through-thickness points chosen manually, with the default of two. No hourglass control implemented.

- ILAW: flag to choose the degradation procedure (among two options, 0 and 1) when critical stress is reached. The default value of 0 signifies that the family of parameters relative to NCYCR (found in lines 8-9 of the material card) have to be defined;
- FMOD: flag to choose between failure criteria (among six options). The default value of 0 is chosen, selecting a *Rankine maximum stress* criterion[3];
- FT: tensile strength, knob to fine-tune the glass phial's failure load, expressed in GPa. Common values found in literature span from 40 MPa to 70 MPa.
- FC: compressive strength, usually one order of magnitude greater than the tensile strength in glass, here arbitrarily chosen to 1 GPa to guarantee tensile-strength driven failure when fine-tuning the parameter FT;
- FTSCL: scale factor for the tensile strength (where the default = 1.0). $\text{FT}_{\text{mod}} = \text{FTSCL} \times \text{FT}$, thus, in this case, the glass' actual tensile strength needs to be scaled by the chosen factor of FTSCL = 1.2. An effective value of tensile strength $\text{FT}_{\text{mod}} > \text{FT}$ implies that the tensile strength drops to its original value of FT as soon as the first crack happens in the phial. Although it adds numerical complexity, having FTSCL > 1.0 can be helpful in modelling high-force peaks in impact events according to the LS-DYNA user manual, which suits the intended application of all the FE models produced in this thesis;
- SFSTI and SFSTR: scaling factors for elastic stiffness and stress after failure, respectively, to be applied at the end of the post-crack softening. It dictates how much residual stress the element can hold after it has been damaged, scaling the post-crack strength and stiffness down to SF-XXX × FT or E. In this case, SFSTI = SFSTR = 0.01 means that both stiffness and stress are reduced to 1% of their values at failure;
- ICRIN: flag for crack strain initialisation. The default value of 0 indicates that initial crack strain is the strain reached at failure;
- NCYCR: number of cycles in which the stress is reduced to SFSTR × failure stress, in this case chosen to 50.
- NIPF: number of failed through-thickness integration points needed to label the element as failed, in this case chosen as 5 to consider all through-thickness integration points set in the phial's shell elements;
- EPSCR: the effective critical strain to trigger element deletion, which can be useful to eliminate highly distorted elements. However, this value was set to the extremely high value of 0.1 to purposefully avoid premature element deletion and having to implement an internal surface generation criterion for contact modelling.
- CTRACK: flag for crack tracking algorithm, set to the non-default value of 1 to track cracks.

LS-DYNA input 4.2: Glass material card

```
*MAT_GLASS
$HMNAME MATERIAL         5glass_shell
$---+----1----+----2----+----3----+----4----+----5----+----6----+----7----+----8
$      MID       RHO         E        PR                           IMOD      ILAW
         52.6000E-06      70.0      0.23                                        0.0
$     FMOD        FT         FC        AT        BT        AC        BC     FTSCL
        0.0     0.060        1.0                                                1.2
$    SFSTI     SFSTR     ICRIN     ECRCL     NCYCR      NIPF
       0.01      0.01       0.0                50.0       5.0
$    EPSCR    ENGCRT    RADCRT    RATENL    RFILTF    FRACEN    CTRACK
     0.100                                                          1
```

Boundary conditions

The phial simply rests on the base's surface. Its interaction with the base and the crosshead is governed by the contact card *CONTACT_AUTOMATIC_SINGLE_SURFACE_ID, the most general contact model in LS-DYNA, as shown in LS-DYNA input 4.3. It allows all surfaces to come in contact with each other without

[3]The *Rankine Maximum Stress* criterion consists in bounding principal stresses, σ_1 and σ_2, between tensile strength, f_t, and compressive strength, f_c, as follows: $-f_c < \{\sigma_1, \sigma_2\} < f_t$. It is the least complex maximum stress criterion available in LS-DYNA.

defining single master-slave pairs. The tangential behaviour is defined by a friction coefficient of 0.3 for both static (FS) and dynamic (FD) friction. The phial is liable and only feels the reaction force from below and the force applied by the moving crosshead.

LS-DYNA input 4.3: Automatic single surface contact card

```
*CONTACT_AUTOMATIC_SINGLE_SURFACE_ID
$HWCOLOR GROUPS       1       57
$       CIDHEADING
          1contact_group
$      SSID               SSTYP             SBOXID    MBOXID       SPR       MPR
          6                   2                                        1         1
$        FS        FD        DC        VC        VDC    PENCHK        BT        DT
         0.3       0.3
$       SFS       SFM       SST       MST       SFST      SFMT       FSF       VSF

$      SOFT    SOFSCL    LCIDAB    MAXPAR      SBOPT     DEPTH     BSORT    FRCFRQ
```

A velocity-control boundary condition is applied to the crosshead in the vertical direction Y to induce a total displacement of approximately 0.2 mm. Velocity is applied with an initial ramp that scales up to a constant negative velocity of 0.0266 mm/ms. The choice of imposing the motion of the crosshead through a velocity boundary condition, rather than the more usual displacement boundary condition, is rooted in the idea of emulating the experiment, in which the test machine moves the crosshead at a constant, relatively slow speed of 1 to 3 mm/min. Furthermore, velocity cannot be applied as a step as an instantaneous jump from zero to the final velocity would correspond to an infinite acceleration at the beginning of the analysis. In an explicit simulation, this impulse would be transmitted to the phial through the contact as a numerically unwelcome spike in the reaction force, followed by spurious oscillations that could potentially even trigger premature glass failure. With the initial ramp, the velocity increases linearly, so the acceleration remains finite and non-zero only during the ramp. The application velocity is low enough to avoid a fully dynamic regime, as explained in the modal analysis section, but still higher than in the actual experiment, as applying a similar loading speed would increase the runtime excessively due to the stability condition imposed on the timestep.

LS-DYNA input 4.4: Crosshead velocity curve

```
*DEFINE_CURVE
$HMNAME CURVES         2curve_velocity
$HMCURVE     2    2 curve_velocity
$     LCID      SIDR       SFA       SFO      OFFA      OFFO
         3                 1.0       1.0
$                  X                   Y
                 0.0                 0.0
                 5.0              -0.0266
                10.0              -0.0266
```

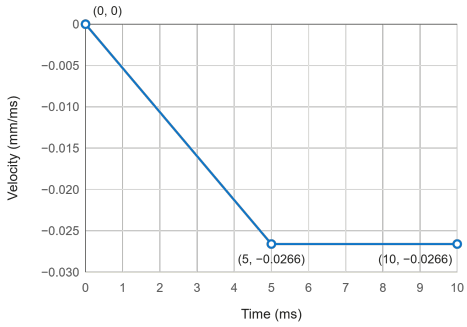


Figure 4.14: Velocity-time curve applied to the crosshead.

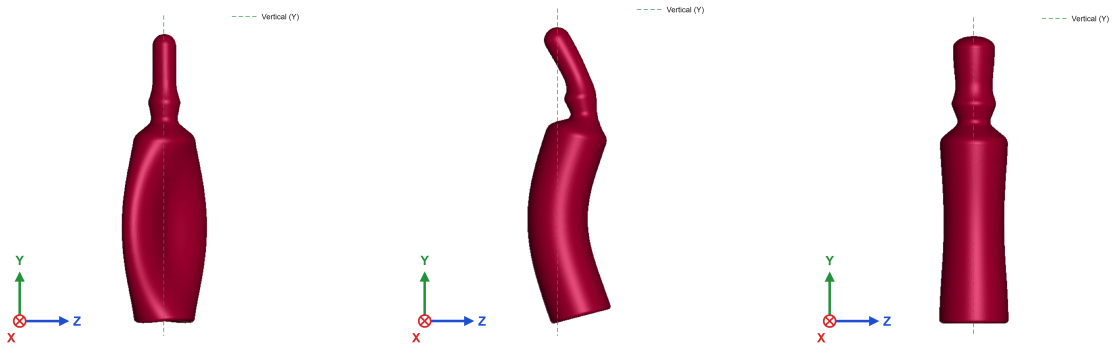
Modal analysis

An implicit modal analysis (*CONTROL_IMPLICIT_EIGENVALUE) was run on the free glass phial to obtain its natural frequencies and mode shapes. The first 30 modes were requested in the analysis, including the first six rigid-body modes.

The first elastic mode (eigenmode 7) is of particular relevance for the present analysis. It is a bending mode, with a natural frequency of about 8.179 kHz, and its period T_1 is the characteristic time of the structure's dynamic response:

$$T_1 = 1/f_1 \approx 122\ \mu\text{s} \tag{4.3}$$

If the load were applied over a time of the same order as T_1, the structure's inertia would contribute to the



(a) Eigenmode 7 (8.179 kHz) **(b)** Eigenmode 18 (26.202 kHz) **(c)** Eigenmode 27 (32.890 kHz)

Figure 4.15: Mode shapes of the free glass phial from the implicit modal analysis.

response and the test would fall in the dynamic regime. Therefore, to keep the simulation quasi-static, the load application time must be at least ten times T_1, i.e. above about 1.22 ms. For this reason, the crosshead was driven by a velocity curve that ramps up over 5 ms to a constant value and then holds it for the remaining 5 ms of the simulation. The ramp time is therefore about 40 times the period of the first elastic mode, well inside the quasi-static range.

Geometric imperfections

The phials tested experimentally showed a large dispersion in their strength properties, as well as in their failure modes. This phenomenon is usually ascribed to the loosely controlled manufacturing process (mentioned in Section 4.2) which naturally introduces geometric imperfections. These imperfections are not influential in the context of the medical applications for which the phials were designed, but are certainly exposed in destructive testing such as what was adopted in Step 1. For these reasons, it was necessary to introduce initial defects into the FE model by perturbing the phial's geometry. Two options were considered:

1. Perturbing the nodes along a primarily bending (see eigenmode 18 in Figure 4.15) mode shape from the modal analysis, as to introduce a specific defect of controlled amplitude (measured in mm) over the nominal geometry;
2. Leaving the geometry unchanged and rotating the phial by 1° about its centroid in the YZ-plane, which introduces a misalignment between the phial axis and the loading direction (as shown in Figure 4.16)

After a series of iterations verifying the numerically observed failure mode, the second option was chosen as it was found that this misalignment triggers an additional, hybrid failure mode, which had also been observed experimentally. The results presented later therefore confirm a correlation between initial deviations from the nominal geometry and the different failure modes observed in the tests.

Requested outputs

The monitored outputs are the displacement of the crosshead and the displacement of the glass node on top of the dome (monitored through the `NODOUT` card), the reaction forces on both the crosshead and the base (`RCFORC` card), added mass/energy data (`GLSTAT`, `MATSUM` cards), and the `MAT_GLASS` *Extra History Variable #1*, i.e. the crack flag `HISV#1`, stored at each integration point in the `d3plot` file to visualise crack initiation and the crack pattern across the glass phial. To obtain results comparable with the experimental campaign, a force-displacement curve was extracted from each run of the simulation, using the reaction force on the crosshead and the displacement of the glass node at the top of the phial dome (considering its displacement only until its eventual brittle failure). The objective is to reproduce the experimentally-observed brittle failure, in which the glass behaves linearly elastically up to fracture, with no plastic deformation before it breaks.

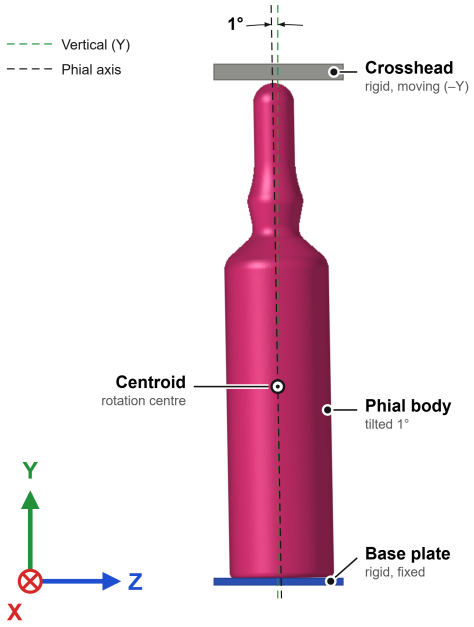


Figure 4.16: Phial rotated by 1° about its centroid in the YZ-plane, introducing a misalignment between the phial axis and the loading direction.

Calibration procedure

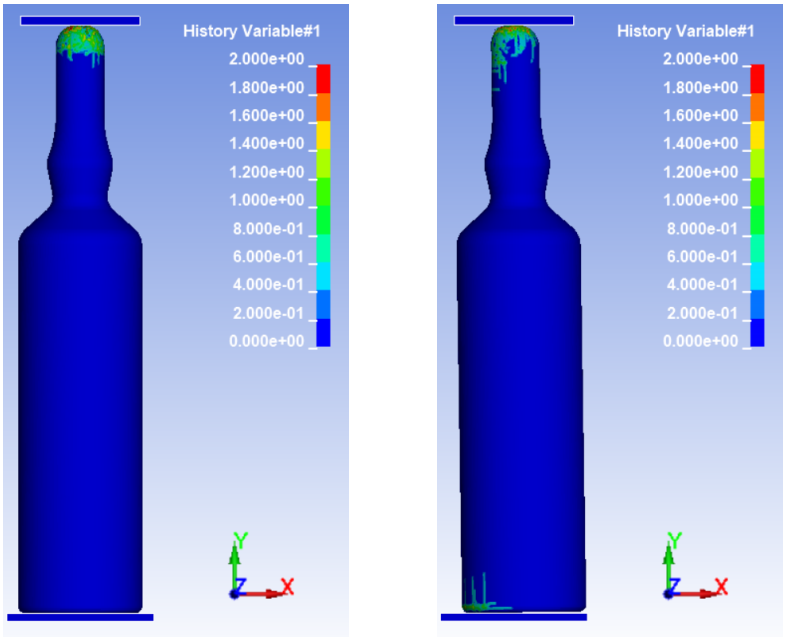
As shown in the LS-DYNA input 4.2, the tensile strength `FT` is the parameter used as a calibration knob to adjust the breaking point of the phial until it matches the failure load value measured experimentally at 98.7 N (5th percentile of the Batch-A Weibull distribution of failure loads). Therefore, several simulations were run iteratively testing different values of `FT` until the simulated phial broke within a 2% tolerance from the experimental failure load. The resulting value of `FT` should therefore be read as a calibrated model parameter rather than as the actual tensile strength of the glass, since it also absorbs the effects of mesh size, imperfections and misalignment. This calibrated material model characterises the phial and is used as input to the next step of the workflow.

4.3.2. Results

Observed failure mode and energy balance

As shown in Figure 4.17, some preliminary results of the FEM analysis display the presence of failure mode A in the model with perfect glass phial geometry, and of failure mode AD with the introduction of an initial imperfection as explained earlier. Based on this qualitative observation, it is already possible to state that the FE model generally matches the mechanical behaviour of the majoritarian portion of glass phials, as the failure modes reproduced numerically are the most common ones recorded in medicinal Batch A (see Figure 4.9).

Further confirmation regarding the simulation's partial reliability can be found in the energy balance of Figure 4.19, for both mode A and mode AD. Figure 4.19a shows a steady increase of all the displayed energies, except the kinetic energy, throughout the elastic loading until 3.80 ms. On one hand, the negligible variation in kinetic energy confirms that the analysis remains strictly quasi-static and, thus, comparable to the experimental loading conditions. On the other hand, the *external work* matches the *total energy*, i.e. the sum of the *internal energy* and *sliding interface energy*, within a 2% tolerance only up to 3.80 ms, before the brittle failure point at 4.30 ms. At this point, the total energy has substantially deviated from the external work by about 14%, indicating that the work done by the moving crosshead is not accounted for in the energy balance and some energy dissipation is appearing, which limits the reliability of the analysis from this point on.

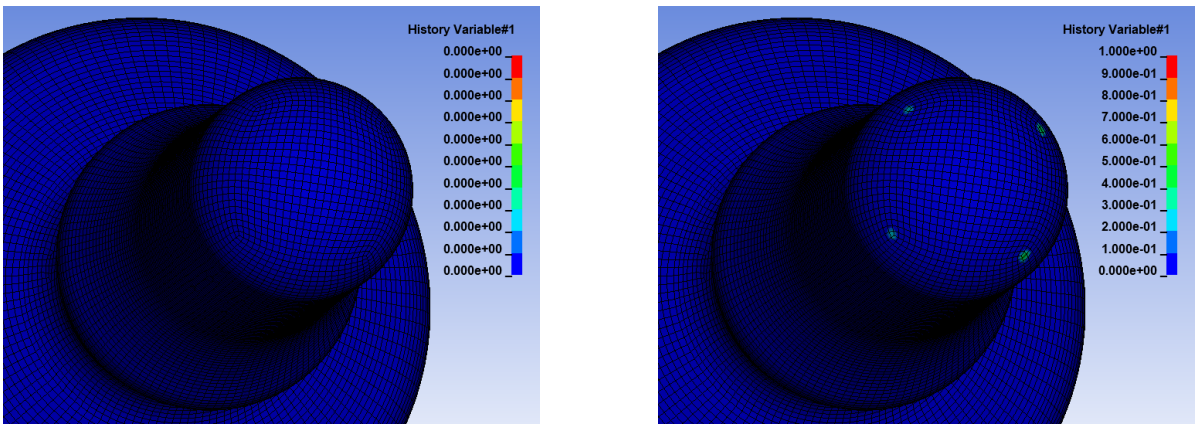


(a) Mode A (dome). **(b)** Mode AD (dome and base).

Figure 4.17: Broken phial at failure in the FE model: (a) mode A; (b) mode AD.

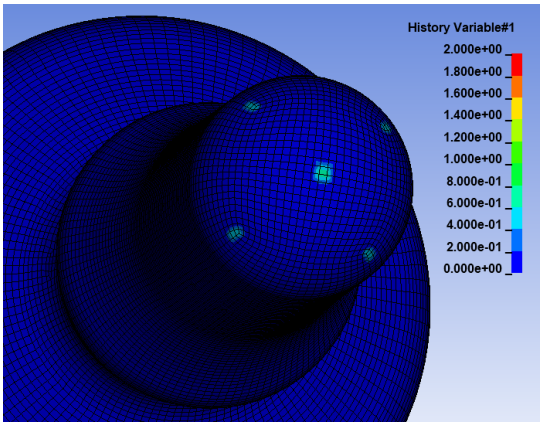
The separation at 3.80 ms can be compared with the cracking sequence of the dome in Figure 4.18, shown through the crack flag `HISV#1` of `*MAT_GLASS` (equal to 0 for an intact integration point, 1 for one tensile crack, 2 for a second, orthogonal crack, and −1 for compressive failure; plotted with the max-IP fringe):

1. 3.80 ms: `HISV#1` becomes non-zero in four small spots of the dome, located at junction points where the mesh is more distorted. The spots do not grow and the force keeps increasing, so this is treated as local cracking induced by the mesh and not as the failure of the phial;
2. 4.30 ms: a crack opens at the centre of the dome and the force reaches its peak. This is the instant taken as brittle failure and used for the calibration. This is also the moment where sliding interface energy becomes negative;
3. 4.50 ms: the dome is almost completely cracked;
4. 6.25 ms: the first elements start being deleted;
5. 8.25 ms: a large opening appears in the dome with substantial element deletion, which matches the drop in internal energy in Figure 4.19a.

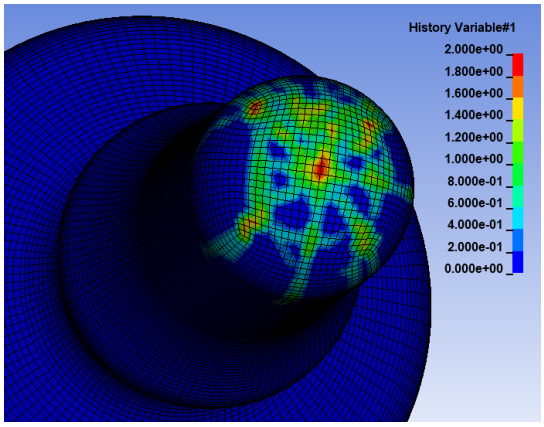


(a) $t < 3.80$ ms. **(b)** $t = 3.80$ ms.

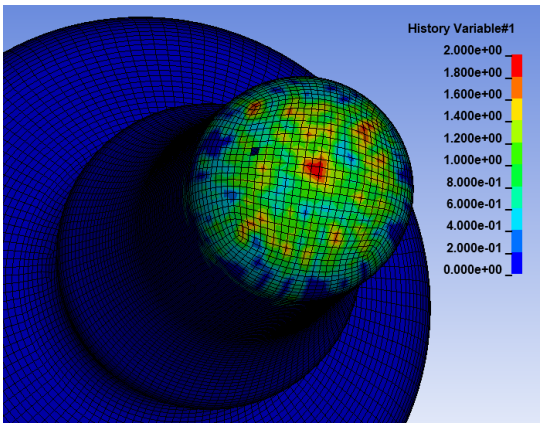
Figure 4.18: Cracking sequence of the dome in the mode A model, shown through `HISV#1` of `*MAT_GLASS`.



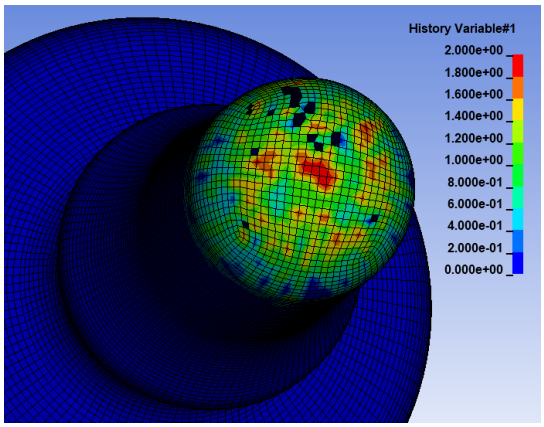
(c) $t = 4.30\,\text{ms}$.



(d) $t = 4.50\,\text{ms}$.



(e) $t = 6.25\,\text{ms}$.



(f) $t = 8.25\,\text{ms}$.

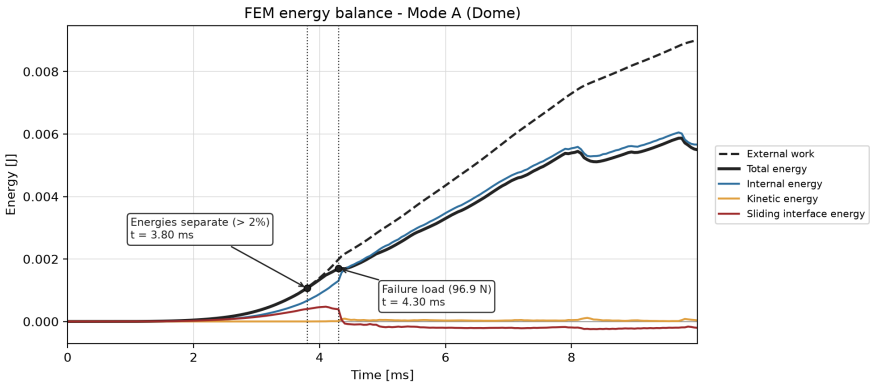
Figure 4.18: Cracking sequence of the dome in the mode A model (continued).

The separation between external work and total energy starts at 3.80 ms, at the same time as the first non-zero values of `HISV#1` emerge. These appear in four small spots of the dome, at nodes where only three elements meet, i.e. at mesh irregularities locations. This cracking is therefore attributed to the mesh and not to the failure of the phial. Nevertheless, these first cracks affect the subsequent response: since `FTSCL` = 1.2 and `RATENL` = 0, the tensile strength of the whole phial is 72 MPa (1.2× `FT`) until the first crack forms anywhere in the part, after which it drops to `FT` = 60 MPa. The dome failure at 4.30 ms therefore occurs with `FT` = 60 MPa. Therefore, the actual phial cracking point is taken at 4.30 ms, where the drop in sliding interface energy to negative values (see Figure 4.19a again). This drop could potentially be justified by the fact that the sudden collapse of the dome unloads the penalty contact faster than it was loaded, so the incrementally-computed contact energy changes sign. This is not caused by element deletion, which only starts at 6.25 ms as shown in Figure 4.18.

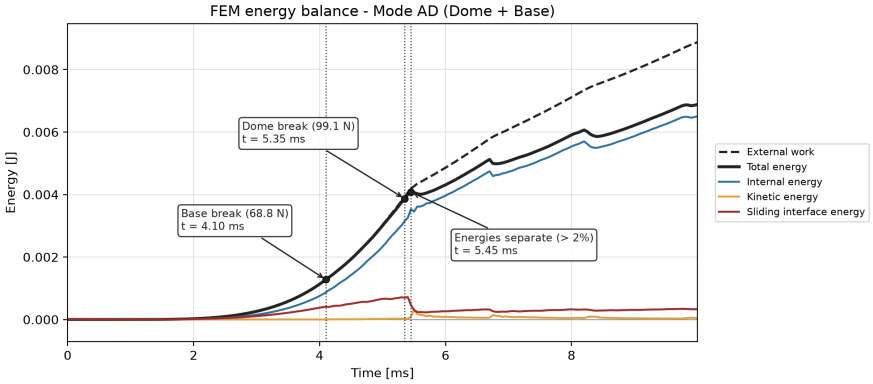
Different observations can be made for mode AD (Figure 4.19b), where the *external work* matches the *total energy* within the 2% tolerance until about 5.45 ms, which is 0.1 ms *after* the dome failure point at 5.35 ms. Sliding interface energy remains positive and kinetic energy remains negligible throughout the simulation, with spikes appearing periodically both in the internal and the kinetic energy likely indicating small instability events linked to element deletion. In general, this energy balance confirms the reliability of the mode AD analysis up to and including the dome failure (i.e. the second failure point of this mode), unlike the mode A model with perfect geometry, whose energy balance already deviates before failure. As in mode A, the first crack, i.e. the initial fracture at the base, lowers the tensile strength of the whole phial from 72 MPa to `FT` = 60 MPa, so that the dome failure at 5.35 ms also occurs with `FT` = 60 MPa in fact.

Overall, for the scope of material strength finetuning, these results are deemed of sufficient quality to

make reliable observations for the characterisation of the phials' glass tensile strength, as post-failure crack evolution is not of interest for the airbag design workflow. This is supported in particular by the mode AD simulation, in which the energy balance remains closed within 2% up to the dome failure at 5.35 ms, i.e. through the whole range of loads relevant for the calibration of FT.



(a) Mode A (dome).



(b) Mode AD (dome and base).

Figure 4.19: Energy balance of the axial compression FE model: (a) mode A; (b) mode AD.

Stiffness verification

A further check needed to assess the reliability of the FE model is the comparison between the experimental stiffness of the phials and the numerical one, where stiffness is intended as the slope of the load-displacement curve of the glass phial relative to this specific load case. As highlighted in Figure 4.20, the model alone has a higher stiffness than the one measured experimentally, with a ratio of ≈ 3.7 for mode A and ≈ 2.8 for mode AD, with respect to the median measured stiffness of each mode. The additional test campaign on the PETG adaptor documented in Section 4.2.2 showed that the holder cannot be treated as infinitely rigid, and that its compliance must be taken into account when measuring the displacement imposed on the phial. The two are therefore combined as stiffnesses in series:

$$\frac{1}{K_{\text{eq}}} = \frac{1}{K_{\text{FEM}}} + \frac{1}{K_{\text{PETG}}} \tag{4.4}$$

where K_{FEM} is the stiffness of the phial model and K_{PETG} the stiffness of the PETG adaptor. Approximating the stiffnesses to the constant values of 3655 N/mm (mode A) or 2767 N/mm (mode AD) and 2037 N/mm, the equivalent stiffness is 1308 N/mm and 1173 N/mm respectively. Since the measured response of the PETG adaptor is not perfectly linear, these values are only a first estimate. For this reason, in the post-processing of the FEM data, the same combination is applied at every load point: for a given load F, the displacement computed by the FE model and the displacement of the PETG adaptor measured experimentally are added together as in the following equation:

$$\delta_{\text{eq}}(F) = \delta_{\text{FEM}}(F) + \delta_{\text{PETG}}(F) \tag{4.5}$$

which gives the corrected curves shown as thick lines in Figure 4.20. With this normalisation, the equivalent stiffness of the model breaking in mode A and in mode AD falls into the experimental range highlighted in Figure 4.9. This comparison confirms the consistency of the model with the experiment and validates the stiffness of the FE model, at least in terms of *range* of values. For mode AD, however, this agreement should be read with caution, as the experimental range is based on only six phials and its upper limit of 1321 N/mm comes from the single phial of Batch B in the group. Considering Batch A alone (five phials, 259 N/mm - 1049 N/mm), the equivalent stiffness of mode AD, 1154 N/mm, lies about 10% above the experimental range, whereas that of mode A, 1251 N/mm, remains within it (318 N/mm - 1506 N/mm).

However, these results are still affected by the limitation mentioned earlier, i.e. the geometry of the phial, and especially its thickness, is not exactly known. A campaign on different thicknesses was carried out and is reported in Appendix C. It did not yield any significant results, except for confirming slight changes in the overall stiffness, not large enough to radically change the results presented here. A summary of these observations is given in Table 4.5.

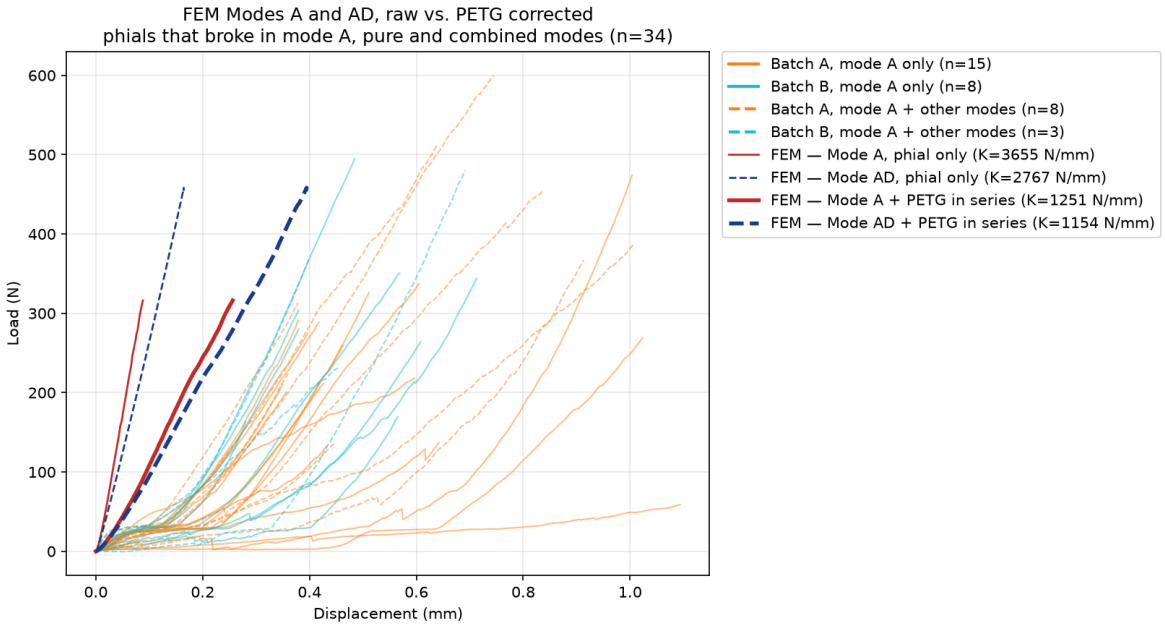


Figure 4.20: Experimental load-displacement curves of the phials that broke in mode A (pure and combined modes), compared with the FE phial model breaking in mode A (red) and mode AD (blue). Thin lines show the phial model alone; thick lines show the model in series with the PETG adaptor, as in Equation (4.5). Curves with a preload at the start of the test are omitted, whereas one phial without a reliable linear stiffness fit is included; for this reason, the number of phials differs from that in Table 4.5.

Tensile strength calibration

Previous observations confer the FE model enough reliability to conduct the tensile strength finetuning process. Analogous to the load case of two springs in series as illustrated above, the load needed for the phial's glass to locally reach the critical stress remains unchanged compared to the experimentally measured value. Owing to the higher stiffness of the FE model, the critical load is merely reached over a smaller displacement, leading to the results presented in Figure 4.21.

The two graphs in Figure 4.21 show the load-displacement curves for mode A and mode AD as parameter FT varies. Each curve shows an initial soft branch indicating a transient phase in which machine-phial contact fully engages, followed by a linear elastic behaviour up to the brittle failure point. Since the phial's stiffness is naturally independent of FT, the failure points of the different curves lie along the same slope, at increasing loads for increasing values of FT. For mode A, a value of FT = 60 MPa gives a

Table 4.5: Summary of the stiffness verification for failure modes A and AD.

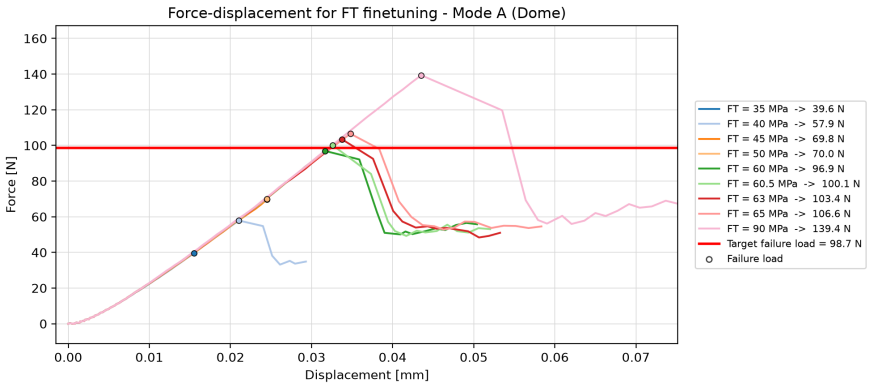
Quantity	Mode A	Mode AD	Source / note
K_{FEM}	3655 N/mm	2767 N/mm	Linear fit of the FEM load-displacement curve
K_{PETG}	2037 N/mm	2037 N/mm	Linear fit, 20 N-316 N band (Figure 4.11)
K_{eq}, constant stiffnesses	1308 N/mm	1173 N/mm	Equation (4.4)
K_{eq}, point by point	1251 N/mm	1154 N/mm	Equation (4.5)
Median measured stiffness	975 N/mm	972 N/mm	Phials that broke exclusively in that mode. Mode A: $n = 26$ (18 Batch A, 8 Batch B); mode AD: $n = 6$ (5 Batch A, 1 Batch B)
Ratio K_{FEM} / median	≈ 3.7	≈ 2.8	Model alone, without the PETG adaptor
Experimental stiffness range	318 N/mm - 1506 N/mm	259 N/mm - 1321 N/mm	Figure 4.9

breaking load of 96.9 N, compared with 98.7 N established experimentally. While at first sight mode AD might point to a different FT value to match the 98.7 N failure threshold, it can be useful to isolate the load-displacement curves with FT = 60 MPa for both mode A and mode AD, as illustrated in Figure 4.22. The FE model captures the hybrid failure mode AD remarkably well: initial fracture occurs at the base due to tilting-induced stress concentrations, but the structure retains partial load-bearing capacity until ultimate failure occurs at a load comparable to the dome failure in mode A. This aligns with the first-hand laboratory experience concerning the hybrid mode AD, which was characterised by distinct acoustic emissions: a minor energy release during initial base fracture, followed by a much larger release upon final dome collapse.

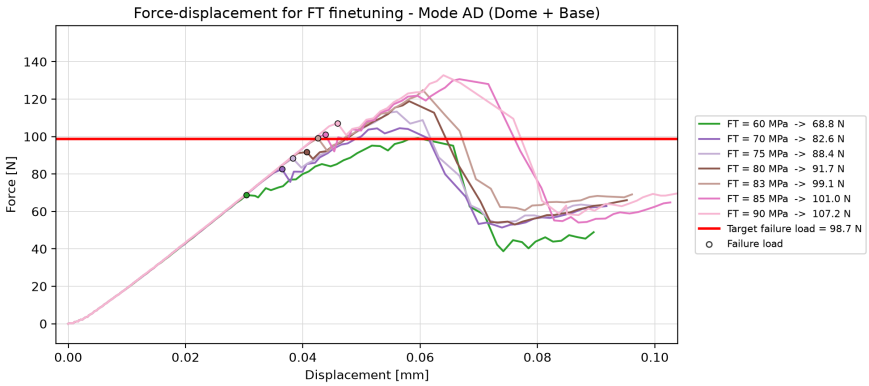
Overall, it is possible to conclude that the calibrated value of FT = 60 MPa successfully reproduces the experimental behaviour for Batch A, closely aligning with the 5^{th} percentile estimate derived in Section 4.2. While this value lies at the boundary between the bending strengths reported for annealed (21.2 - 58.6 MPa) and heat-strengthened (54.9 - 167.0 MPa) float glass [79], using a value from such a low percentile suggests that the true material strength is likely higher. The fact that a low tensile strength is required to match the experimental load suggests that the modelled phial geometry may be overly thin, as a thicker geometry would offer greater compressive resistance. The combination of thin wall and low breaking load is therefore a generally conservative reproduction of phial behaviour, which suits the purposes of this workflow.

The results also show a remarkable sensitivity to initial defects and to the variability of the phials' geometry. For this reason, the quantitative predictive capability of the model is limited, but the qualitative indication is clear: it is fundamental to improve ways of characterising medical payloads. Coming from a different industry, pharmaceutical phials are produced and certified according to standards that focus on requirements such as chemical resistance and dimensional tolerances, rather than on the structural behaviour under extreme impact loads (outside the context of ordinary transportation). As a consequence, the mechanical data needed in an aerospace design process are generally not available, and studying the phials with the means typically used for aerospace structures makes their behaviour hard to predict.

Therefore, these results should not be interpreted as a rigorous material characterisation. The testing conditions were non-standard, including a custom adaptor that required stiffness calibration and a batch-to-batch variability stemming directly from differences in pharmaceutical phial manufacturing parameters. Furthermore, this evaluation was strictly limited to axial compression, omitting other critical loading cases such as pure bending or direct tension, which would typically exhibit even higher scatter in failure strength. Ultimately, this testing served as a pragmatic, preliminary demonstration of the phials' structural behaviour rather than a comprehensive material modelling campaign. Nevertheless, with appropriate adjustments and considerations, these results provide a reliable foundation for the subsequent steps in the airbag design workflow.



(a) FT calibration for failure mode A.



(b) FT calibration for failure mode AD.

Figure 4.21: Tensile strength (FT) calibration of the FE phial model: simulated force-displacement curves for a range of FT values, compared with the experimental target failure load of 98.7 N (red line). Circles mark the breaking load of each simulation.

Step 2 deliverable: calibrated glass material card

Based on these results, the calibrated LS-DYNA material card reported in LS-DYNA input 4.5 is adopted as the input for the following steps of the workflow.

LS-DYNA input 4.5: Calibrated glass material card

```
*MAT_GLASS
$HMNAME MATERIAL         5glass_shell
$---+----1----+----2----+----3----+----4----+----5----+----6----+----7----+----8
$      MID       RHO         E        PR                          IMOD      ILAW
         52.6000E-06      70.0      0.23                                       0.0
$     FMOD        FT        FC        AT        BT        AC        BC     FTSCL
        0.0     0.060       1.0                                              1.2
$    SFSTI     SFSTR      CRIN     ECRCL     NCYCR      NIPF
       0.01      0.01       0.0                50.0       5.0
$    EPSCR    ENGCRT    RADCRT    RATENL    RFILTF    FRACEN    CTRACK
     0.100                                                          1
```

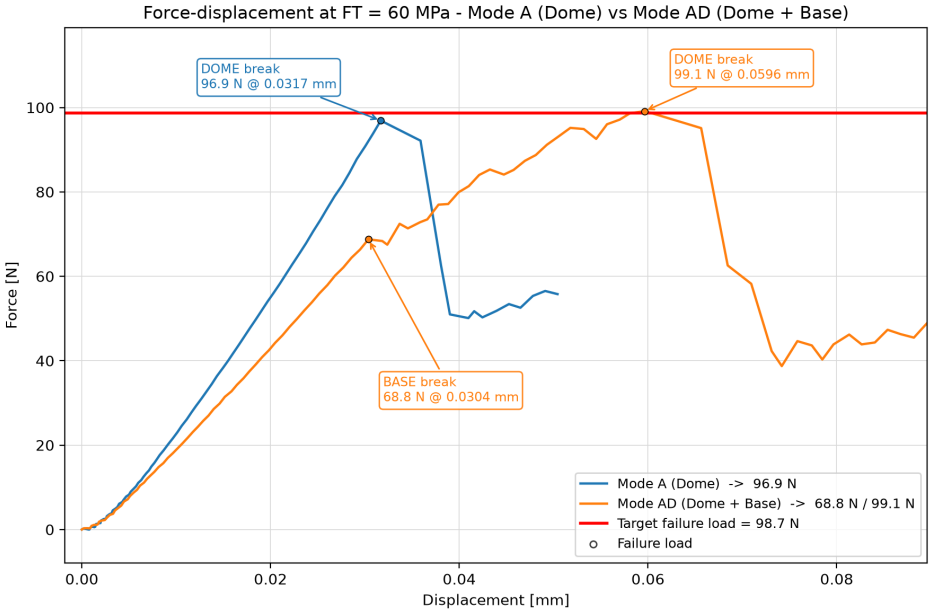


Figure 4.22: Overlay of the FT calibration results for failure modes A and AD.

4.4. Step 3 - Sled test (FEM)

The scope of this third step within the workflow is to subject the payload bay, i.e. the container of the medical payload analysed until now, to the type of acceleration it would possibly undergo in an impact event. Specifically, the idea is to imitate a *sled test*, which is common practice in the automotive industry: an ATD[4] crash dummy is strapped onto a sled (a trolley sliding on almost frictionless rails) that is subjected to a prescribed acceleration pulse, and the response of the dummies, governed predominantly by their own inertia and the forces exerted by the restraint system, is checked against damage thresholds [80].

The model built in this Step 3 aims to simulate a similar phenomenon: here, an acceleration boundary condition is imposed on the external shell of the medical payload container, leaving its contents to be governed by their inertial response and by the contact forces exchanged between each other. The model aims to verify whether the imposed acceleration impulse damages the medical payload inside, i.e. the glass phials. The outcome is, therefore, binary: either the phials are damaged, or they are not. The information related to crack initiation and material failure is monitored through the same LS-DYNA variable `HISV#1` activated in Step 2 to evaluate phial failure load under axial compression. The apparition of damage implies that the acceleration pulse to which the container was subjected was too high for the phials to resist and the *allowable* (i.e. maximum withstandable) peak acceleration must correspond to a lower value. The test is in itself can be reduced to a threshold problem, where exceeding a certain peak acceleration immediately induces damage (with due considerations on tolerance). Finding the acceleration allowable a_{allow} is an iterative process and results in obtaining the last input necessary for the airbag design optimisation code of the workflow's Step 4.

Just like the FE model for the axial compression test, the present model is built using *explicit* time integration with an engineering unit system consistent with the one adopted in Step 2, reported here again for clarity:

4.4.1. Implementation

Model setup

As illustrated in Section 3.1.1, the payload container considered in this model is inspired by the designed proposed by Sivieri [14]. The payload assembly presented here is composed of four parts, as shown in Figure 4.23: the *payload bay* (the outer shell of the receptacle), the *sponge* (an internal buffer between the

[4] *Anthropomorphic Test Devices*.

Table 4.6: Engineering Units

Time	Distance	Mass	Stress / Pressure	Force	Energy / Work	Velocity	Acceleration	Density
ms	mm	kg	GPa (kN/mm^2)	kN	J ($kN \cdot mm$)	mm/ms	mm/ms^2	kg/mm^3

medical payload and the receptacle, conceptually representing a protective layer such as bubble wrap, of the kind an operator would use to attenuate in-flight vibrations), the medicinal *mask* (a plastic tray commonly found in pharmaceutical packaging to hold the phials in place during transport), and the glass phials (as modelled in previous Step 2, and only including one in the model setup for exemplification), overall representing a realistic carry-on configuration for typical mission scenarios.

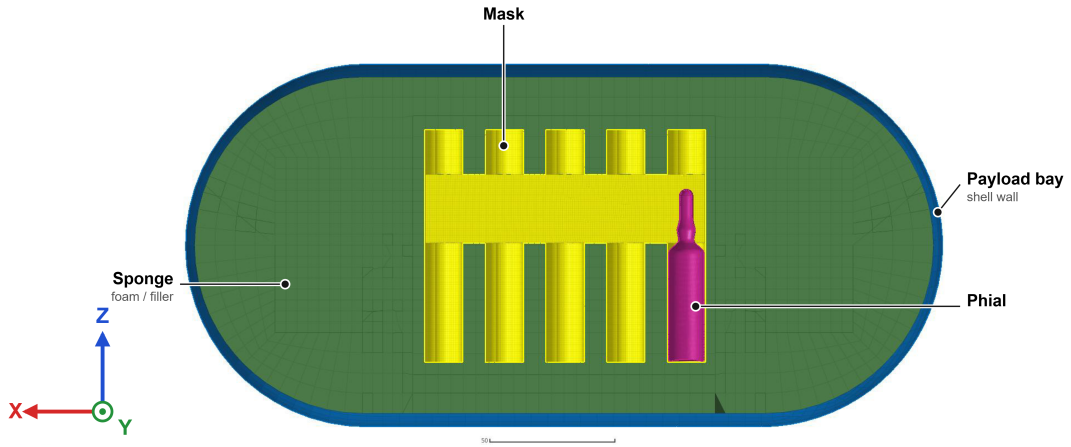


Figure 4.23: FE model of the payload assembly, viewed along the Y axis: the *payload bay* shell contains the *sponge*, the *mask* and a single *glass phial*, shown for exemplification.

Two considerations should be made regarding the modelling choices:

1. The presence of the sponge layer strongly affects the value of a_{allow} as it heavily contributes to the attenuation mechanism protecting the phials. Therefore, to capture an accurate response, it is key to study the phials' behaviour in relation to their interaction with the assembly they belong to;
2. Apart from the phials' glass, only the payload bay's material has a failure criterion implemented, wrongly assuming that the sponge and the mask remain intact during an impact event. This non-realistic choice was made to purposefully focus on the accurate modelling of the phials' behaviour, supported by concrete experimental data. The other materials' data on mechanical properties is extracted from an ad-hoc literature review.

The outer shell of the payload bay is meshed with shell elements of average length of 6 mm size and 5 mm thickness, default ELFORM = 2 with two integration points through the thickness. Its material is injection-moulded ABS, modelled with `*MAT_PIECEWISE_LINEAR_PLASTICITY` (material model `*MAT_024`), as shown in LS-DYNA input 4.6.

LS-DYNA input 4.6: ABS material card for the payload bay

```
*MAT_PIECEWISE_LINEAR_PLASTICITY
$HMNAME MATERIAL         2ABS_inject-mould
$---+----1----+----2----+----3----+----4----+----5----+----6----+----7----+----8
$       MID       RHO         E        PR      SIGY      ETAN      FAIL      TDEL
          2 1.050E-06      2.10      0.38     0.040     0.150     0.025
$         C         P      LCSS      LCSR        VP       LCF

$      EPS1      EPS2      EPS3      EPS4      EPS5      EPS6      EPS7      EPS8

$       ES1       ES2       ES3       ES4       ES5       ES6       ES7       ES8
```

where the parameters set in the material card, with `E` and `PR` consistent with [81], are:

- `RHO`: density, 1.05E-06 kg/mm^3 (1050 kg/m^3, close to the 1040 kg/m^3 reported in [81]);
- `E`: Young's modulus, 2.10 GPa;
- `PR`: Poisson's ratio, 0.38;
- `SIGY`: yield stress, 0.040 GPa (40 MPa), assumed slightly below the tensile strength of 45 MPa reported in [81];
- `ETAN`: tangent modulus, 0.150 GPa (150 MPa), assumed as a typical value for ABS, defining the slope of the hardening branch after yield. As no stress-strain curve is given (`LCSS` left blank), the material follows a bilinear elastic-plastic law;
- `FAIL`: effective plastic strain at failure, 0.025, assumed as a typical value for ABS, beyond which the element is deleted.

The mask holding the glass phials is meshed with shell elements of 0.5 mm thickness, with the material card in LS-DYNA input 4.7. The mask material is polypropylene (PP), modelled as linear elastic with *`MAT_ELASTIC` (material model *`MAT_001`), as shown in LS-DYNA input 4.7. No failure criterion is defined, so the mask cannot fail.

LS-DYNA input 4.7: PP material card for the mask

```
*MAT_ELASTIC
$HMNAME MATERIAL         3polypropylene_PP
$HWCOLOR MATERIAL         3      25
$---+----1----+----2----+----3----+----4----+----5----+----6----+----7----+----8
$       MID       RHO         E        PR        DA        DB         K
          3 9.1000E-7     1.350       0.4
```

where the parameters set in the material card, chosen consistently with the tensile tests on injection-moulded PP reported in [82], are:

- `RHO`: density, 9.10E-07 kg/mm^3 (910 kg/m^3), typical value for PP;
- `E`: Young's modulus, 1.35 GPa (1.375–1.491 GPa in [82]);
- `PR`: Poisson's ratio, 0.40 (0.411–0.420 in [82]).

The sponge is meshed with constant-stress, 8-node hexa elements with single-point integration (`ELFORM = 1`) and average side length of 6 mm. Its loading curve was built to describe the typical behaviour of a porous foam: an initial soft branch, followed by hardening as the porosity closes and the material densifies. The curve is based on [83] and is shown in Figure 4.24, with the material card in LS-DYNA input 4.8.

The sponge material is a polyurethane foam, modelled with *`MAT_LOW_DENSITY_FOAM` (material model *`MAT_057`), as shown in LS-DYNA input 4.8.

LS-DYNA input 4.8: Polyurethane foam material card for the sponge

```
*MAT_LOW_DENSITY_FOAM
$HMNAME MATERIAL         1polyurethane_foam
$---+----1----+----2----+----3----+----4----+----5----+----6----+----7----+----8
$       MID       RHO         E      LCID        TC        HU      BETA      DAMP
          1 2.500E-08       5.0         1                 1.0                 0.05
$     SHAPE      FAIL    BVFLAG        ED     BETA1      KCON       REF
```

where the parameters set in the material card are:

- RHO: density, 2.50E-08 kg/mm^3 (25 kg/m^3), close to the 27 kg/m^3 open-cell polyurethane foam tested in [83];
- E: Young's modulus used in tension, 5.0 GPa;
- LCID: ID of the load curve defining the nominal stress versus strain in compression, i.e. curve 1, given in LS-DYNA input 4.9 and plotted in Figure 4.24;
- HU: hysteretic unloading factor, 1.0. With this value there is no hysteretic unloading: the foam unloads along the loading curve and no energy is dissipated through hysteresis;
- DAMP: viscous coefficient, 0.05, which models damping effects (recommended range 0.05-0.50 according to the LS-DYNA user manual [78, p. 453]).

LS-DYNA input 4.9: Sponge loading curve

```
*DEFINE_CURVE_TITLE
$HMCURVE     1     2 polyurethane_curve
$NAME
polyurethane_curve
$      LCID      SIDR       SFA       SFO      OFFA
        OFFO
         1                 1.0       1.0
$                   X                   Y
                  0.0                 0.0
                 0.05             3.0E-06
                  0.1             5.0E-06
                  0.2             6.0E-06
                  0.4             7.0E-06
                  0.6             9.0E-06
                  0.7             1.5E-05
                  0.8             4.0E-05
                  0.9             1.5E-04
```

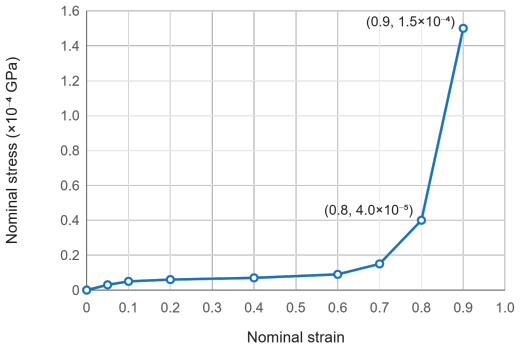


Figure 4.24: Nominal stress-strain loading curve of the polyurethane sponge.

The parts interact through *CONTACT_AUTOMATIC_GENERAL in LS-DYNA, which again treats every part as potentially in contact with every other part without master-slave couples hierarchy. In addition, an interior contact option was activated through the *CONTACT_INTERIOR keyword to prevent negative volume formation during large compression events.

Loading through an acceleration pulse

The displacement forced onto the payload bay is applied through an acceleration pulse prescribed through *BOUNDARY_PRESCRIBED_MOTION_RIGID on a selection of nodes on the bay's mesh forming two rigid rings through the keyword *CONSTRAINED_NODAL_RIGID_BODY (see Figure 4.25), virtually representing the strapping mechanism that would be implemented in the corresponding laboratory sled test. The rest of the bay remains deformable. The prescribed acceleration-time curve has a trapezoidal shape: an initial ramp of duration t_r, a plateau at constant acceleration a of duration t_p, and a descending ramp of duration t_r again[5]. The ramps avoid discontinuities in the acceleration, which would excite spurious high-frequency oscillations, so that the velocity and displacement histories remain smooth. The acceleration is therefore applied at the attachment points, with the possibility to choose the degree of freedom in which to displace the bay (namely X, Y, or Z, and sweeping to detect the most critical one), and reaches the phials only after being transmitted through the deformable bay, the sponge and the mask.

[5]The duration of t_r with respect to the entire duration of the phenomenon is governed by the parameter RISE_FRACTION $\in (0, 0.5)$ that is arbitrarily set to 0.2 to generate the present family of curves. A value of RISE_FRACTION = 0 implies a fully rectangular pulse, while RISE_FRACTION = 0.5 corresponds to a triangular-shaped pulse.

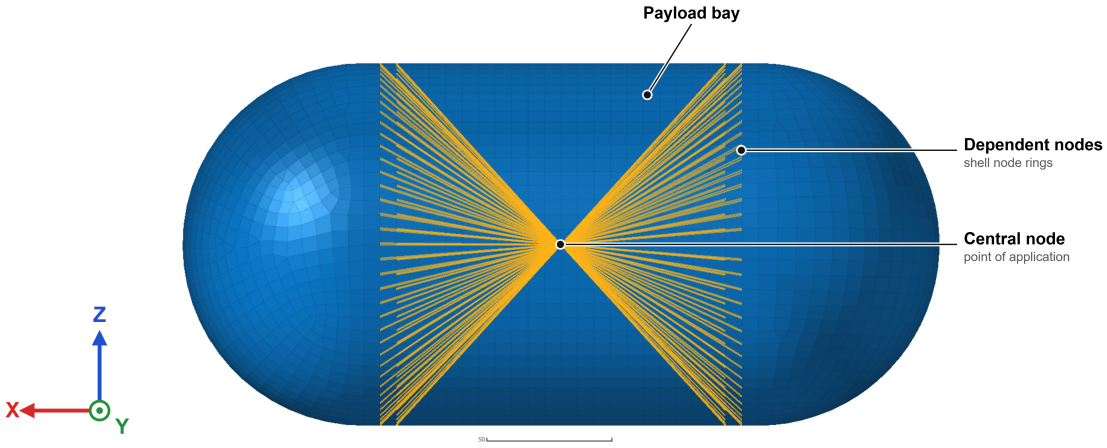


Figure 4.25: Two rings of dependent nodes on the payload bay mesh, tied to a central node through `*CONSTRAINED_NODAL_RIGID_BODY`. The acceleration pulse is prescribed on the central node.

The impact is characterised by an initial velocity $\mathbf{u}_0 = (u_{x0}, u_{y0})$, which includes both horizontal and vertical components. Over the deceleration process, this velocity must be brought completely to zero. Therefore, the integral of the acceleration curve described above must equal the velocity shed by the system:

$$\int_0^T a(t)\,\mathrm{d}t = a\,(t_r + t_p) = \|\mathbf{u}_0\|, \qquad T = 2t_r + t_p. \tag{4.6}$$

This condition allows to define a family of acceleration curves as shown in Figure 4.26: for a given amplitude a, Equation (4.6) fixes the plateau duration $t_p = \|\mathbf{u}_0\|/a - t_r$, and therefore the characteristic time scale of the event. The aim is to reproduce the pulse felt by the payload bay during an impact as closely as possible, applying an acceleration profile similar to the peak observed in such an event.

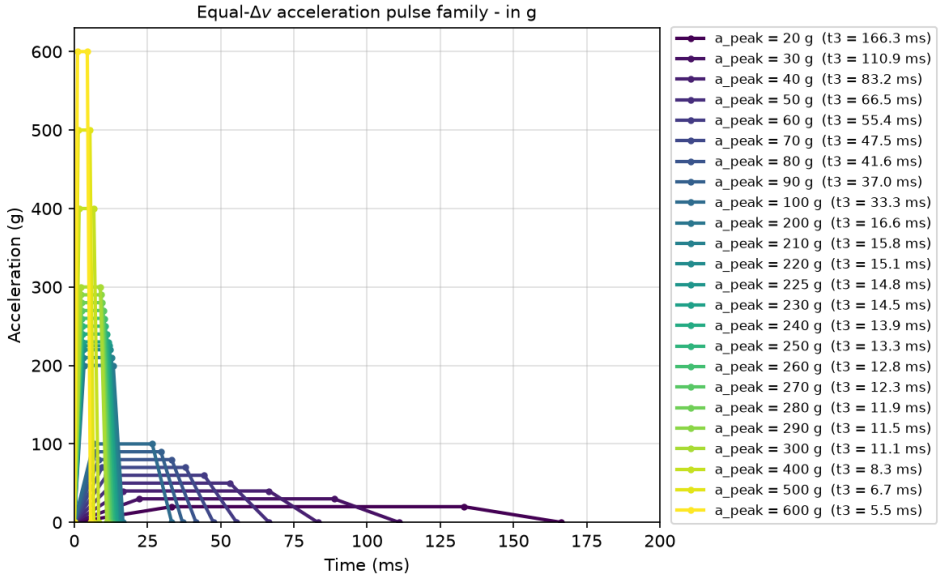


Figure 4.26: Family of trapezoidal acceleration pulses with equal area $\|\mathbf{u}_0\|$: a higher plateau amplitude a corresponds to a shorter plateau duration t_p.

The maximum acceleration (the value of the acceleration plateau) that the phials can withstand is found with a search procedure analogous to the calibration of FT for Step 2 of the workflow: the amplitude a

is varied step by step, with the duration of the simulation adjusted through Equation (4.6), until the threshold for the initiation of phial failure is found. Since a larger amplitude also implies a shorter pulse, the resulting limit refers to this pulse shape and to the given initial velocity and is expressed in terms of the acceleration prescribed at the rings.

Mass scaling

Explicit analyses must abide to the Courant-Friedrichs-Lewy (CFL) condition, which limits the possible timestep amplitude Δt adopted in the calculation due to numerical stability constraints. Specifically, the condition is based on the most critical combination of two factors, shown in the following equation:

$$\Delta t \leq \Delta t_{\text{crit}} = \frac{L_e}{c} \tag{4.7}$$

where:

- L_e is the characteristic length of the critical element in the model (e.g., shortest side or minimum height of a shell or solid element);
- c is the acoustic wave speed (speed of sound) in the material of that element.

The element-specific combination of L_e and c that minimises their ratio dictates the size of the allowable timestep and, therefore, the speed at which the calculations can continue. In the case of the sled test model, the 0.1 mm phial shells set the stable timestep ($\Delta t \approx L_e/c \approx 1.9 \times 10^{-5}$ ms), yielding 10^6 to 10^7 calculation cycles for phenomena of the duration between 10 and 100 ms such as the simulated event, demanding significant computational capacity and runtime.

As the speed of sound in elastic materials is inversely proportional to the square root of the material's density ($c = \sqrt{\frac{M}{\rho}}$, where M represents the appropriate elastic modulus, such as Young's modulus E or bulk/shear moduli K and G), a common procedure adopted in FEM practice to raise the stable timestep is to locally increase material density in the critical elements by artificially adding non-physical mass, directly reducing c and subsequently increasing Δt_{crit} as per Equation (4.7). In LS-DYNA, *mass scaling* is activated through the keyword `*CONTROL_TIMESTEP`, parameter `DT2MS`: given the target Δt, and choosing whether scaling applies only to elements below the target (negative `DT2MS`), the solver calculates the density increase and, therefore, the local non-physical mass increase to bring the stable timestep up to the hard-coded value. While mass scaling accelerates solver runtime, excessive artificial mass addition might compromise physical realism of the simulated phenomenon. While this is negligible in static or quasi-static conditions where dynamic effects do not influence numerical results, it becomes particularly relevant during dynamic impact events, where the inertial component significantly dictates stress magnitudes and transient structural responses. An ideal timestep to impose is one that minimises calculation runtimes while keeping the model's physics realistic and the numerical results reliable.

Due to the glass phials' fine mesh, artificial mass is added entirely on the phial. Through a process of trial-and-error, it was observed that increasing the timestep by one order of magnitude would result in a ≈50 to 60x local mass increase, which in turn would reduce the phial's fundamental frequency by a factor of $\sqrt{[50,\ 60]} \approx [7.1,\ 7.7]\times$ (considering the relation $\omega_1 = \sqrt{\frac{k}{m}}$, where k and m are the equivalent stiffness and mass of a mass-spring system representing the payload assembly). In transient impact dynamics, reducing the natural frequency shifts the system more out of phase with the input pulse width, artificially smoothing or delaying the peak contact forces between the phial and the sponge. To avoid incurring in this possibility, it was decided to settle for a timestep value of 2×10^{-5} ms, increasing the phial's mass only by 2.7×10^{-4} g and, thus, maintaining overall reliability.

Requested outputs and failure detection

Numerically, an event counting as a failure of the glass phial corresponds to observing the crack flag `HISV#1` ≥ 1 at any integration point, i.e. at least one tensile crack has formed in at least one through-thickness integration point. This is a crack-initiation criterion that does not depend on `NIPF`, which only governs when the whole element is considered failed (all 5 integration points failed). Verification of crack formation can be observed in the results (`d3plot`) file activating the `HISV#1` contour plot and

using the max-IP fringe. This criterion is stricter than the one adopted in Step 2 of the workflow (see Section 4.3.2), where localised mesh-induced cracks were disregarded and failure was taken at the force peak; the resulting allowable computed in this step is therefore conservative.

The failure event must be comparable to the one produced experimentally in Step 1, i.e. a similar failure mode should be observed in the sled test simulation to guarantee the reliability of the observed allowable acceleration. The allowable is therefore taken as the highest acceleration level without failure, which lies within the bracket between the last pass and the first fail, whose width equals the sweep step and is stated in the results section below.

4.4.2. Results

Failure location and mode

The payload was accelerated along the axes of a local reference system, in which Y and Z are rotated by 35° with respect to the global ones, as shown in Figure 4.27. Rotated directions were preferred to pure ones to exemplify impacts in which two main directions are coupled, e.g. the global Y and Z. Among the three possible directions in which the payload could be accelerated in a preliminary sweep, direction Y' was chosen. Although this is not coherent with the loading mode of Steps 1 and 2 of the workflow, i.e. the axial compression, it was selected as the case yielding the lowest allowable acceleration, and hence the most conservative of the three directions, which was established in a qualitative evaluation of observed damage after applying an acceleration pulse of identical amplitude (600 g) in X, Y', and Z' (Figure 4.28). In particular, the Z' case does not fail at 600 g, while the damage obtained along X is not comparable with the other two. Therefore, the dilemma was whether to accelerate the payload in the same direction as the compression, keeping the model consistent with the calibration step and its loading case, or to choose a different direction, corresponding to a more critical, and thus more conservative, case.

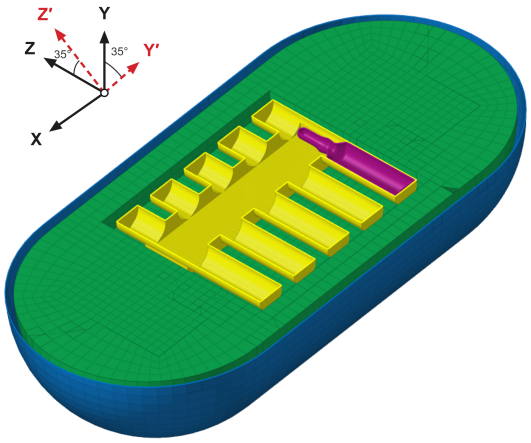
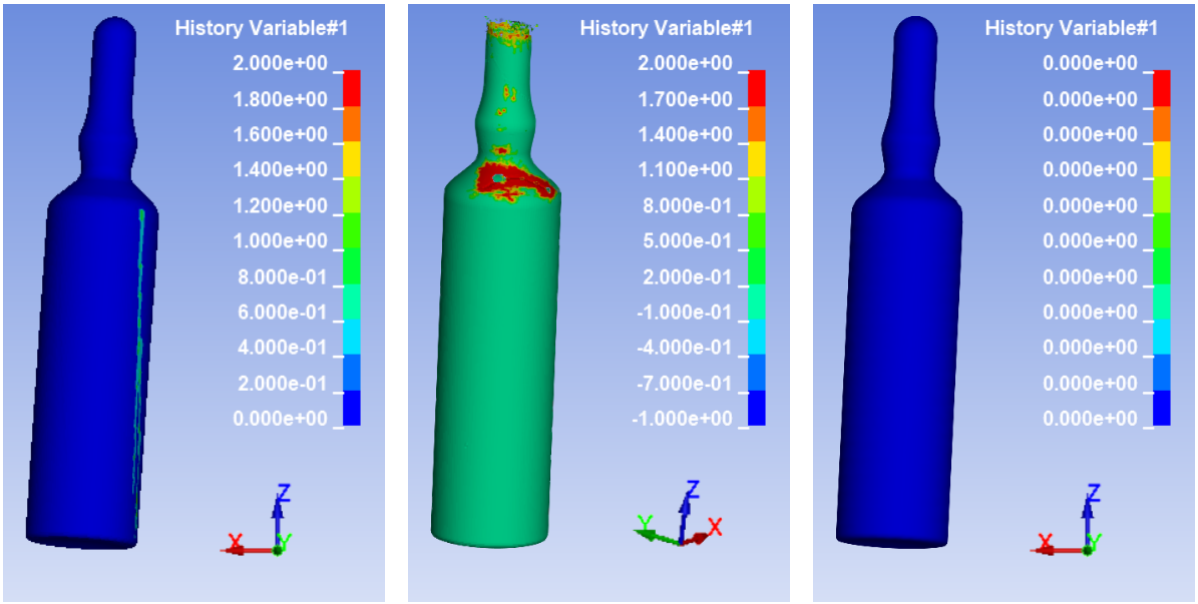


Figure 4.27: Global (X, Y, Z, black) and local (X, Y', Z', red dashed) reference systems of the payload. The local axes Y' and Z' are obtained by rotating Y and Z by 35° about the X axis, which is common to both systems.

In a complete airbag design workflow, this choice would naturally follow from a complete material characterisation and from a full sweep to identify the most critical loading direction, i.e. the one in which failure is anticipated first. However, just as a complete material characterisation is not available, the full sweep could not be carried out because of the computational effort required. For the purpose of exemplification, the most critical direction was therefore adopted, as it would be within an ideal workflow, under the assumption of a material characterisation valid for any loading direction. The resulting failure mode partially resembles the one observed experimentally in Step 1, as it involves the dome but also engages part of the phial body.

Acceleration sweep

The acceleration pulses used for the simulations are shown in Figure 4.26. The peak acceleration was swept with steps progressively refined from 100 g to 5 g. Through manual iterations, no damage was observed at a peak acceleration of 225 g, while damage occurred at 230 g. The allowable peak



(a) X direction. **(b)** Y' direction. **(c)** Z' direction.

Figure 4.28: Glass phial damage in the sled test simulation for loading in the X, Y' and Z' directions: `*MAT_GLASS` `HISV#1` contour plot (max-IP fringe), where `HISV#1` ≥ 1 marks crack initiation. All three cases were run with the trapezoidal pulse of Figure 4.26 at a peak of 600 g. The reference frame shown in each image is the global coordinate system, not the local one.

acceleration therefore lies between 225 g and 230 g. Consistently with the 5 g resolution adopted for the sweep, the value of 225 g, i.e. the highest acceleration level without damage, is used as the allowable for the purpose of testing the airbag design workflow presented in this thesis.

Step 3 deliverable: allowable payload acceleration

The results of the acceleration sweep conducted in this step of the workflow are summarised in Table 4.7 and are adopted as the input for the final step of the airbag design workflow developed in this thesis.

Table 4.7: Allowable payload acceleration obtained from the sled test simulation.

Quantity	Value	Source / note
Allowable peak acceleration (a_{allow})	225 g	Highest plateau without failure, 5 g sweep step
Critical loading direction	Y' (local frame, Figure 4.27)	Most damage among X, Y', Z' at equal 600 g pulse (Figure 4.28)
Main failure mode	Dome breaking, extending into the phial body	Partially comparable to mode A of Step 1 (Table 4.3)
Velocity shed ($\|\mathbf{u}_0\|$)	26.1 m/s	Resultant of u_{x0} and u_{y0} from Step 0 (Table 4.2)
Pulse shape	Trapezoidal	`RISE_FRACTION` = 0.2, area from Equation (4.6)

4.5. Step 4 - Design variables optimisation

The methodology applied in this section represents the last step of the workflow developed for the preliminary sizing of an airbag device. It gathers the results obtained in the previous steps and feeds them to a Python code organised in two layers: an inner core, the *oracle*, which models the physics of impact events involving airbag devices, and an outer *optimisation wrapper*, which finds suitable low-mass candidates capable of protecting the payload under the given impact conditions. The inherited inputs are:

1. **Payload mass**: 1 kg (initially used in Step 0, Preliminary mission analysis in Section 4.1, and numerically estimated in Step 3, Sled test (FEM) in Section 4.4)

2. **Impact velocity**: $u_{x0} = 7.5\,\mathrm{m/s}$, $u_{y0} = 25\,\mathrm{m/s}$ (from Step 0)
3. **Allowable peak acceleration**: $a_{allow} = 225\,g$

The outputs of this section, i.e. the candidate design parameters produced by the code, can be turned into specific design recommendations for practical implementation and prototyping of the airbag device, which constitute the final deliverable of the present MSc thesis project.

4.5.1. Implementation choices

The airbag design variables optimised by the code used in Step 4 are shown in the right-most column of Table 4.8, which also highlights the *bounds* within which the search for the optimum candidate was conducted (dictated either by manufacturer information or by geometrical constraints). The design variables can be subdivided into three categories: *continuous* (usually pointing to geometric features), *discrete, integers* (in this case, the number of airbag fabric plies), and *discrete, categorical* (such as fabric type, shape, and gas type, to which the concept of searchable bounds cannot be applied).

Table 4.8: Inputs of the preliminary airbag optimisation: fixed inputs and boundary conditions (left) and optimisable design variables (right). Fabric properties are rounded, representative values, not measurements of a specific product; their sources are compared in Appendix G.

Fixed inputs and boundary conditions		Optimisable design variables	
Impact scenario		**Continuous (searched by the optimiser)**	
M	1.0 kg, payload mass	D_0	$D_{0,\min}$ (Equation (4.16)) to 1.5 m, diameter
u_{y0}	25.0 m/s, vertical impact speed	L_0/D_0	0.5 to 4.0, aspect ratio (cylinder only)
u_{x0}	7.5 m/s, horizontal impact speed	P_0	P_{amb} to 1.4×10^5 Pa, inflation pressure
u_{n0}	≈ 26.1 m/s *along* $\mathbf{n}$, *at* $\theta \approx 16.7°$ *(derived)*	φ_{vent}	0 to 0.10, vent open-area fraction
Environment and fixed parameters		**Discrete - integer (searched by the optimiser)**	
P_{amb}	101 325 Pa, ambient pressure	n_{ply}	$\{1, \ldots, 4\}$, number of fabric plies
g	9.806 65 m/s^2	**Discrete - categorical (enumerated)**	
T_0	288 K, average gas temperature	Shape	sphere ($k = 0.5$), cylinder ($k = 1$)
d_{or}	0 m, orifice diameter (fabric venting only)	Fabric	Nylon 6,6, silicone-coated: 600 N/cm, 180 g/m^2 [84]
Requirements and safety factors			Polyester, silicone-coated: 550 N/cm, 190 g/m^2 [85]
a_{allow}	2206.50 m/s^2 (225 g), allowable payload acceleration		Aramid (Kevlar 29), elastomer-coated: $\approx$ 1000 N/cm, $\approx$ 260 g/m^2 [86][10]
SF	1.2, safety factor on a_{allow}	Gas	N_2: $R = 296.8$ J/(kg K), $\gamma = 1.400$
a_{lim}	$a_{allow}/SF \approx 1839$ m/s^2 *(derived)*		He: $R = 2076.9$ J/(kg K), $\gamma = 1.667$
$u_{\mathrm{res,max}}$	0.5 m/s, 1.0 m/s, 1.5 m/s and 2.0 m/s, residual speed cap (swept)		Ar: $R = 208.1$ J/(kg K), $\gamma = 1.667$
η_{seam}	0.70, seam efficiency on fabric strength		CO_2: $R = 188.9$ J/(kg K), $\gamma = 1.289$
SF_{burst}	1.50, design factor on membrane tension		Dry air: $R = 287.0$ J/(kg K), $\gamma = 1.400$
PEN	10^4, penalty weight on constraint violations		

In their most basic configuration, optimisation codes consist in two elements:

- **Oracle Function** f: serves as a high-fidelity evaluator of the underlying physics. It receives a candidate design vector $\mathbf{x}$ and computes the system response $\mathbf{y} = f(\mathbf{x})$, serving as a deterministic representation of the phenomenon;
- **Optimisation wrapper** O: operates as an algorithmic controller over the oracle. Through an iterative oracle interrogation process ($\mathbf{x}_{k+1} = O(\mathbf{x}_k, \mathbf{y}_k)$), it navigates the design space to drive the system towards an optimal response defined by a user-specific objective function $\mathcal{J}(\mathbf{y})$ and constraint set $\mathcal{C}$.

[10] Extrapolated from lighter two-ply Kevlar 29 laminates to a heavier coated fabric, and therefore the least certain of the three entries (Appendix G).

4.5.2. `simulate_airbag` oracle

Airbag impact model updates

In the case of the present thesis work, the oracle is based on the airbag simulation function proposed by Zhou et al. [38] as introduced in Section 2.3.1. The underlying calculation loop shown in Figure 2.2 remains virtually the same, but is enriched with some additional considerations to allow for a more complete simulation (and subsequent evaluation) of the airbag impact phenomenon:

- **Bidirectional energy dissipation**: while in Zhou et al. the impact phenomenon only involves the vertical (impact direction) degree of freedom, in this code the horizontal component of the velocity is included, increasing the number of degrees of freedom to two. In practice, this means calculating the magnitude of the velocity, $u_{n0} = \sqrt{u_{x0}^2 + u_{y0}^2}$ and the angle θ at which the impact happens, and then updating the mechanics considering the coupled vertical and horizontal components;
- **Stroke vs. geometry correlation assumptions**: Zhou et al. considers several possibilities regarding the assumptions correlating contact area A_t and airbag volume V with stroke h, but only presents the equations for the case of a cylindrical airbag with inextensible perimeter:

$$A_t = L_t L_0, \qquad V_t = L_0\, D_t \left(\frac{\pi D_t}{4} + L_t\right), \qquad D_t = D_0 - h, \quad L_t = \frac{\pi h}{2}, \tag{4.8}$$

  where L_t is the length of the cross-section perimeter in contact with the ground and L_0 is the extension of the airbag in the out-of-page direction (i.e. the length of the cylinder). The expression for L_t follows from the inextensible-perimeter assumption: the original circumference πD_0 is redistributed into two semicircles of diameter D_t and two flat segments of length L_t, so that $\pi D_0 = \pi D_t + 2L_t$, which gives $L_t = \pi h/2$. To expand the search for winning airbag candidates to different shapes, the following geometric assumption was introduced regarding a spherical airbag, truncated by a planar cut at the contact surface, so that the removed cap has a height equal to the stroke h:

$$A_t = \pi\, h\, (D_0 - h), \qquad V = \frac{\pi}{6} D_0^3 - \frac{\pi}{3} h^2 \left(\frac{3}{2} D_0 - h\right). \tag{4.9}$$

  which allows the code to sweep candidates across two typical airbag geometries;
- **Increased complexity in venting mechanism**: while Zhou et al. models venting as relegated only to an orifice/valve present on the surface of the airbag, a more realistic venting mechanism would include the contribution of fabric porosity. In most applications across both the automotive and the space industries, weaving patterns of airbag fabrics are engineered to control the venting rate during impact events. To include a similar venting contribution to the model, the variable φ_{vent} (*vent open-area fraction*) was introduced to indicate the fraction of fabric area treated as open pores. Hence, the total venting area (evolving throughout the simulation) is modelled as:

$$A_{vent} = A_{or} + \varphi_{\text{vent}}\, (S_{surface} - A_t) \tag{4.10}$$

  where $A_{or} = \frac{\pi}{4} d_{or}^2$ is the area of the discrete venting orifice of diameter d_{or}, $S_{surface}$ is the undeformed fabric surface area, and A_t is the contact patch pressed against the ground, whose pores are considered as non-venting and which grows as the airbag is crushed. Keeping the venting calculations true to Zhou et al.'s modelling choices, the vented mass flow through the updated definition of the total venting area A_{vent}, with pressure ratio $\lambda = P_{\text{down}}/P_{\text{up}}$, is subsonic for $\lambda \geq 0.528$,

$$\dot{m} = C_{\text{dis}}\, A_{vent}\, P_{\text{up}} \sqrt{\frac{2\gamma}{(\gamma - 1)RT} \left(\lambda^{\frac{2}{\gamma}} - \lambda^{\frac{\gamma+1}{\gamma}}\right)}, \tag{4.11}$$

  and choked otherwise,

$$\dot{m} = C_{\text{dis}}\, A_{vent}\, P_{\text{up}} \sqrt{\frac{\gamma}{RT} \left(\frac{2}{\gamma + 1}\right)^{\frac{\gamma+1}{\gamma-1}}}, \tag{4.12}$$

  where the discharge coefficient $C_{\text{dis}}(\lambda)$ accounts for frictional and viscous losses at the orifice and is adopted by in the original model in Zhou et al. after Do [87] as an empirical 6th-order polynomial fit in the pressure ratio:

$$C_{\text{dis}} = -3.8399\lambda^6 + 9.4363\lambda^5 - 7.2326\lambda^4 + 1.6972\lambda^3 - 0.2908\lambda^2 - 0.0130\lambda + 0.8426. \tag{4.13}$$

- **Fabric membrane integrity**: a completely novel addition to Zhou et al.'s model consists in a membrane strength check, thus including the possibility of airbag rupture. The fabric is treated as a thin membrane (made of a discrete number of plies n_{ply}, with a fabric-type-specific ply thickness value) under uniform internal pressure, so the tension per unit length in the fabric follows from the equilibrium of the pressurised shape, $N = k\,\Delta P\,R$, with $k = 1$ for the hoop direction of a cylinder and $k = 0.5$ for a sphere [88]. A fundamental assumption of this model is that the critical seam lies in the maximum-tension direction, i.e. the longitudinal seam of the cylinder, which carries the hoop tension, and it governs the bag's resistance to rupture. Therefore, the membrane is checked against the hoop burst pressure, obtained from the seam-reduced fabric strength:

$$P_{\text{burst}} = P_{\text{amb}} + \frac{\eta_{\text{seam}}\, N_{\text{ult}}}{SF_{\text{burst}}\, k\, R_0}, \qquad N_{\text{ult}} = 100\, n_{\text{ply}}\, N'_{\text{ult}}, \tag{4.14}$$

  where P_{amb} is the ambient pressure, η_{seam} the seam efficiency, i.e. the fraction of the fabric strength retained at the seam, SF_{burst} the safety factor on bursting, R_0 the initial radius of the bag, n_{ply} the number of fabric plies and N'_{ult} the ultimate tensile strength of a single ply in N/cm. The factor 100 converts N/cm into N/m, so that P_{burst} is obtained in Pa with R_0 in m.

The termination checks implemented in this model are also different from the ones presented in Zhou et al. Specifically, they are carried out in the following order:

1. **Bursting**: if the internal pressure exceeds P_{burst} at any time during the stroke, the simulation is stopped and the candidate design is discarded;
2. **Impact velocity at bottomed-out condition**: if the payload has used the entire stroke and its residual velocity at the impact with the terrain is too high compared to the manually set threshold, i.e. $u_{\text{land}} \geq u_{\text{res,max}}$, the combination is discarded;
3. **Rebound**: if the payload bounces back, i.e. $y_{\text{pen}}^{k+1} = y_{\text{pen}}^{k} + u_y^{k+1}\,\Delta t \leq 0$, where y_{pen} is the vertical penetration of the payload measured from the point of first contact (which would correspond to stroke h in case of vertical impact at $\theta = 0°$), u_y its vertical velocity and Δt the timestep: the payload has been pushed back above the height at which it first touched the terrain. Hence, the design is discarded as it does not allow efficient dissipation of the payload's initial kinetic energy, given back as elastic energy from the compressed gas.
4. **Settling**: if none of the previous events occurs within $t_{\max}$, the payload is considered caught by the airbag. The design is accepted only if its residual speed along the impact axis at $t_{\max}$ satisfies $|u_n(t_{\max})| \leq u_{\text{res,max}}$.

In addition to these checks, every design that bottoms out or settles must satisfy the acceleration constraint, which is the main requirement of the sizing. The peak load transmitted by the bag to the payload along the impact axis, $a_{\text{peak},n} = \max_t F_n/M$, must not exceed the limit $a_{\text{lim}} = a_{\text{allow}}/SF$. Unlike the checks above, this one does not stop the simulation, but rather it is evaluated on the whole time history once the run has ended, and a design that violates it is discarded. As it cannot be considered a stopping criterion, this check is mentioned here only for clarity, but it is clear that its role becomes significantly more predominant in the optimisation layer of the code.

Adaptive timestep

Depending on the progression of the impact event simulation, the oracle uses either one of two timestep sizes: a fine step `DT_FINE = 1e-5` s (10 μs) and a coarse step `DT_COARSE = 2e-4` s (200 μs). Specifically, the choice of either timestep size depends on what value the bag load on the payload along the contact normal `load_n = Fn / M_payload`, taken from the previous step (`load_hist[-1]`), has compared to the established acceleration limit a_{allow} (scaled by `SF`): the size switches to fine when the load goes above 3% of a_{allow}, but then it switches back to the coarse step when the load drops below 1% of a_{allow}. The compression and venting phases, where pressure and acceleration change quickly, require a much finer resolution and, thus, are resolved with steps 20× smaller. The hysteresis band between 1% and 3% stops the step from flipping back and forth near a single threshold. The adoption of the adaptive timestep affects the venting subiterations too, as each step is divided into `max_inner_iters` venting sub-steps (`dt_inner = dt / max_inner_iters`). During the DE search, the number of sub-iterations depends on the step size: 5 on the fine step and 20 on the coarse step, so `dt_inner` is 2 μs and 10 μs respectively.

With 50 on both steps in the final re-simulation of the winning candidate, it is brought down to 0.2 μs and 4 μs.

Overall, this adaptive timestep concept was designed to conduct the low-load phases before and after the stroke run with large steps, with the effect of shortening oracle calculation runtimes.

Limitations of the `simulate_airbag` *oracle*

Overall, the `simulate_airbag` oracle has significant limitations: assuming a rigid impact surface, a fixed impact axis $\mathbf{n}$, a crude approximation of the coupling of horizontal and vertical impact velocities, and the assumption that the airbag flattens to zero thickness, as well as excluding the thermal degradation of the fabric caused by rising gas temperatures during compression. However, at the scope of *preliminary* airbag sizing, embracing these simplifications results in a computationally inexpensive model, allowing it to be embedded into an optimisation loop.

Not all of these assumptions are necessarily non-conservative: in fact, hypothesising a rigid ground implies that the terrain cannot absorb kinetic energy, thus increasing peak acceleration, while assuming that the bag can flatten to zero thickness overestimates the stroke actually available. Therefore, the candidates produced by the code should be interpreted as starting points for the design, to be verified with higher-fidelity simulations or drop tests (see Steps 5-7 of the *Inner design loop: a payload-protection subsystem design workflow* in Section 3.2.2).

Algorithm D.1 shows the pseudocode of the algorithm implemented as the oracle function.

4.5.3. `differential_evolution` optimisation algorithm

Algorithm choice and design vector

The optimisation algorithm used in the wrapper is *differential evolution* (DE), a population-based algorithm proposed by Storn and Price [89]. The implementation is `scipy.optimize.differential_evolution` from the open-source Python library SciPy [90]. This algorithm was purposefully chosen as a generally robust, although computationally heavier, method based on simple comparisons among candidate designs. Given categorical variables as inputs and binary checks in the oracle core as described earlier, where a small change in design variables can radically change the oracle's response, an alternative gradient-based method would have required gradient estimation through finite differences, at the cost of one extra simulation per variable, the unpredictability of the resulting gradient estimate, and the risk of stopping the search at local minima.

As the problem's variables fall into three categories (continuous, discrete integers, and discrete categorical), the discrete categorical variables (two shapes, three fabrics, and five gas options) do not partake directly in the DE runs. Considering also the landing speed caps entering through the constraint $u_{\text{land}} \leq u_{\text{res,max}}$, which yields the minimum achievable mass as a function of the allowed landing speed, a separate optimisation is run for each of the $2 \times 3 \times 5 \times 4 = 120$ combinations. Therefore, the design vector on which the DE wrapper acts is as follows:

$$\mathbf{x} = \begin{cases} [D_0,\ L_0/D_0,\ n_{\text{ply}},\ P_0,\ \varphi_{\text{vent}}] & \text{cylinder} \\ [D_0,\ n_{\text{ply}},\ P_0,\ \varphi_{\text{vent}}] & \text{sphere} \end{cases} \tag{4.15}$$

The variables contributing to the simulation are orifice diameter, set to $d_{\text{or}} = 0$ so that all venting takes place through the fabric while keeping the possibility to implement relief valves in future simulations, and the initial gas temperature, fixed at $T_0 = 288\,\text{K}$. These two variables are kept constant as opting for delegating all venting to porous fabric is common practice in airbag design within the automotive industry, while $T_0 = 288\,\text{K}$ corresponds to the ISA reference temperature adopted in the mission description (Section 3.1.1), not considering the heating processes linked to gas generation mechanisms.

The bounds of the numerical variables are listed in Table 4.8. The lower bound of the airbag diameter is the stroke needed to stop the payload along $\mathbf{n}$ at the constant load a_{lim}, net of gravity (inspired by the airbag working principle in Section 2.3.2, Equation (2.1)):

$$D_{0,\text{min}} = \frac{u_{n0}^2}{2\,(a_{\text{lim}} - g\cos\theta)}, \tag{4.16}$$

which exists only if $a_{\text{lim}} > g\cos\theta$. This bound does not exclude any feasible design because lower strokes would result in peak accelerations above a_{lim}, while accelerations below a_{lim} require longer strokes.

Design feasibility and score

Each simulation ends when either one of the termination criteria described in Section 4.5.2 is met. A design is feasible if the payload lands, by reaching the full stroke or by settling, with a peak load $a_{\text{peak},n} = \max(F_n/M) \le a_{\text{lim}}$ and a landing speed $u_{\text{land}} \le u_{\text{res,max}}$. In the full-fidelity re-simulation of the winners, a tolerance of 0.1% above a_{lim} is accepted, as explained in Table 4.9. Each design $\mathbf{x}$ is then scored by:

$$f(\mathbf{x}) = \begin{cases} m_{\text{tot}} & \text{if feasible,} \\ 10^6 + PEN \sum_i v_i & \text{if infeasible,} \\ 10^{12} & \text{if the simulation fails or } m_{\text{tot}} \text{ is not finite.} \end{cases} \tag{4.17}$$

where m_{tot} is the mass of the fabric plus the initial gas mass needed for the airbag's inflation, $PEN = 10^4$, and v_i are the graded constraint violations, defined as:

$$\begin{cases} v_{\text{acc}} = \dfrac{\max(0,\ a_{\text{peak},n} - a_{\text{lim}})}{a_{\text{lim}}}, & \\ v_{\text{burst}} = \dfrac{\max(0,\ P_{\text{peak}} - P_{\text{burst}})}{P_{\text{burst}}}, & \\ v_{\text{reb}} = \dfrac{|u_{\text{reb},y}|}{u_{y0}}, & \text{if rebounded,} \\ v_{\text{ures}} = \dfrac{\max(0,\ u_{\text{land}} - u_{\text{res,max}})}{u_{n0}}, & \text{if landed.} \end{cases} \tag{4.18}$$

If requirements are met, each violation is zero. Alternatively, the offset 10^6 is far larger than the mass of any airbag in the design space, so every infeasible design scores worse than every feasible one. Among infeasible designs, the score varies with the total violation, which steers the search towards the feasible region while no feasible design has been found. A design whose simulation fails, or returns a non-finite mass, scores 10^{12}.

Steps of one differential evolution run

The steps of one DE run[11] are shown in Figure 4.29 and consist in the following:

1. **Initial population:** the run starts from a population of N designs placed in the design space with a Sobol sequence, which spreads the points evenly within the bounds instead of drawing them randomly. The population size N is not set directly: SciPy multiplies the `POPSIZE` setting by the number of numerical design variables n_{var} (see Equation (4.15)), and the Sobol initialisation rounds the result up to the next power of two:

$$N = 2^{\lceil \log_2(\texttt{POPSIZE} \cdot n_{\text{var}}) \rceil} \tag{4.19}$$

   For example, with `POPSIZE` = 5 this gives $5 \times 5 = 25 \rightarrow 32$ designs for the cylinder and $5 \times 4 = 20 \rightarrow 32$ for the sphere. Each design of the population can be thought of as occupying a seat, and the number of seats stays the same for the whole run;

2. **Building a challenger:** in every generation, each design $\mathbf{x}_i$ of the population receives one challenger, built over two steps

[11] SciPy internally scales each variable to the interval $[0, 1]$ between its bounds, so that a step moves every variable by a comparable fraction of its range.

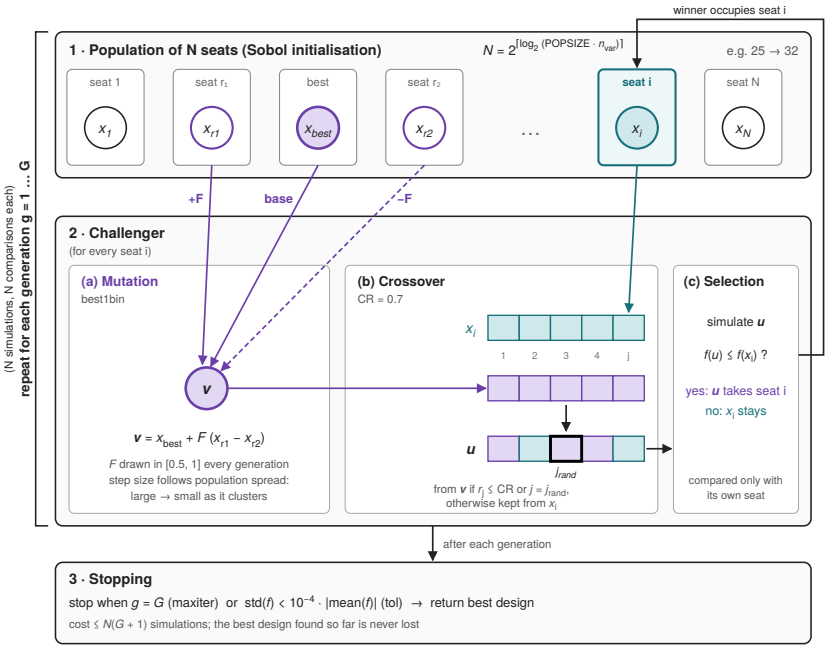


Figure 4.29: One generation of a DE run for the design in seat i. The same steps are repeated for every seat.

(a) *Mutation:* with SciPy's default method (`best1bin`), the mutant $\mathbf{v}$ is built from the best design of the current population and the difference between two other designs chosen at random:

$$\mathbf{v} = \mathbf{x}_{\text{best}} + F\ (\mathbf{x}_{r_1} - \mathbf{x}_{r_2})\,, \tag{4.20}$$

where $r_1 \neq r_2 \neq i$ are indices of two population members chosen at random, and the mutation factor F is drawn between 0.5 and 1 at every generation. Because the step is built from a difference between members, its size follows the spread of the population. Steps are large while the designs are spread over the design space and become small as the population clusters;

(b) *Crossover:* the value of some variables of the mutant $\mathbf{v}$ are then updated with the values of corresponding variables of the design $\mathbf{x}_i$ it will challenge, leading to the construction of the *challenger* $\mathbf{u}$. All variables j, except one, of the challenger $\mathbf{u}$ is taken from $\mathbf{v}$ with probability $CR = 0.7$ and kept from $\mathbf{x}_i$ otherwise, and the remaining unchanged variable j_{rand}, chosen at random, is always taken from $\mathbf{v}$ so that the trial differs from $\mathbf{x}_i$ by at least one variable. The trial is then simulated and compared only with the original design $\mathbf{x}_i$:

$$u_j = \begin{cases} v_j & \text{if } r_j \leq CR \text{ or } j = j_{\text{rand}} \\ x_{i,j} & \text{otherwise} \end{cases} \qquad \mathbf{x}_i \leftarrow \begin{cases} \mathbf{u} & \text{if } f(\mathbf{u}) \leq f(\mathbf{x}_i) \\ \mathbf{x}_i & \text{otherwise} \end{cases} \tag{4.21}$$

where r_j is a random number between 0 and 1 drawn for each variable. A value of $\mathbf{u}$ that falls outside its bounds is replaced by a random value inside them, and n_{ply} is rounded to the closest integer.

Each challenger competes only with the design in its own seat, so a generation requires N simulations and N comparisons rather than a comparison of every design with every other. Since a design leaves its seat only for a trial that scores at least as well, the best design found until then is never lost.

3. **Stopping mechanism:** a run stops after G generations set by the parameter `maxiter`, or earlier if the standard deviation of the scores in the population falls below 10^{-4} times the magnitude of their mean (`tol`). In this sense convergence means that the population has clustered, while it does not reliably indicate that the global minimum has been found. With feasible scores of the order of the airbag mass, this tolerance requires a very small spread, and about half of the runs

stop at `maxiter` (297 of the 600 runs of the full sweep), with cost of a run being at most $N(G+1)$ simulations counting the initial population.

Complete calculation procedure

The complete DE calculation procedure is shown in Figure 4.30. Overall, the parameters directly affecting the simulation are:

- `POPSIZE` (sets N through Equation (4.19)) The population size controls how widely a run searches and how quickly it clusters. A larger population is less likely to cluster around a wrong region, but it also clusters more slowly and therefore needs more generations;
- `MAXITER` (= G) The maximum number of generations sets how long a run can refine its result, but extra generations cannot move a population that has already clustered in the wrong local minimum. In this work `MAXITER` = 150, since in test runs the best score was already within 0.1% of its 200-generation value by generation 135 to 145;
- `SEEDS` Each combination of categorical variables is optimised in `#SEEDS` independent runs whose populations never interact, and the lowest score is kept. The same seed always reproduces the same result.

For each landing-speed cap, shape, fabric and gas, n_{seed} (= `SEEDS`) DE runs are performed and the lowest score is kept. The design with the lowest score over the $n_{\text{shape}} \times n_{\text{mat}} \times n_{\text{gas}}$ combinations is the winner for that specific landing-speed cap. To reduce the cost of the search, the DE runs use a compiled copy of the oracle with fewer venting sub-iterations (5 on the fine step, 20 on the coarse step). The winner is then re-simulated with the original oracle and 50 venting sub-iterations, and its masses, peak load, landing speed, pressures, violations and feasibility are reported. The full sweep consists of $n_{\text{ures}} \times n_{\text{shape}} \times n_{\text{mat}} \times n_{\text{gas}} \times n_{\text{seed}}$ DE runs, or up to $n_{\text{ures}} \times n_{\text{shape}} \times n_{\text{mat}} \times n_{\text{gas}} \times n_{\text{seed}} \times N(G+1)$ simulations.

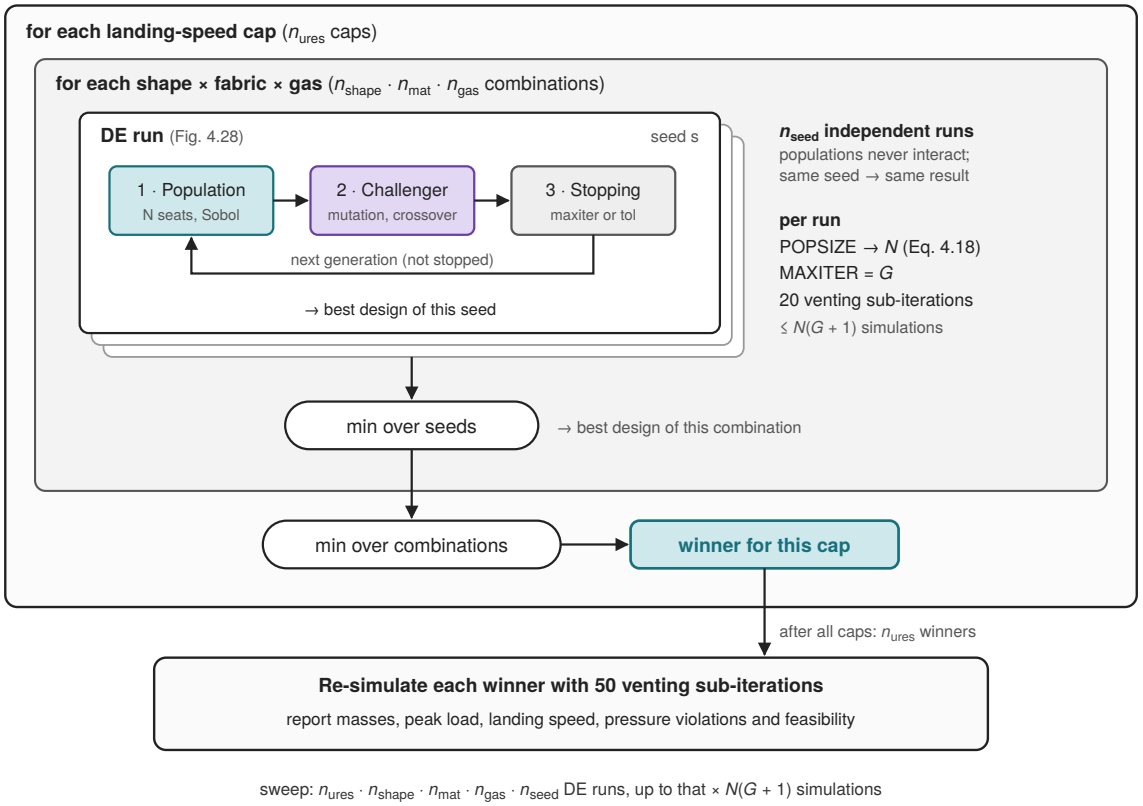


Figure 4.30: Overall optimisation procedure: nested loops over the landing-speed cap, the categorical variables and the random seed, selection of the winning design for each cap, and its re-simulation with 50 venting sub-iterations.

Limitations of differential evolution
Although robust for problem formulation adopted, the Differential Evolution algorithm has some limitations, which are important to consider when interpreting the results of Section 4.5.4. The most fundamental one is that DE does not guarantee the global minimum (since all the runs could cluster around the same local minimum), only relying on the agreement between different seeds as an indication that the result does not depend on the random initial population. Furthermore, because the optimiser reduces the mass until a constraint stops it, the winner usually lies close to a design limit.

4.5.4. Results

Table 4.9 shows the results of the differential evolution optimisation wrapper set up as described in the previous sections. Evidently, the combination of *cylinder* as a shape, *silicone-coated Nylon 6,6* as a material, and *Helium* as inflation gas wins for every residual velocity cap. This is likely motivated by the following three reasons:

- **Helium:** helium's atomic mass is about 7 to 11 times smaller than the molecular mass of the other gases. At a given P_0 and V_0, the gas mass is $m_{\text{gas},0} = P_0 V_0/(RT_0)$, and helium has $R = 2077\,\text{J/(kg K)}$, against 297 for N_2, 287 for dry air, 208 for Ar and 189 for CO_2, always giving He a notable advantage. As a side effect, the speed of sound of helium is about 2.9 times that of N_2, so it vents the same volume through a much smaller open area, which keeps φ_{vent} at very low values in all cases and points to a relatively low fabric porosity needed. The optimiser's choice of helium, however, does not account for this gas' strong permeation capacity: due to the small size of its atoms, helium escapes through a coated fabric much faster than N_2. Over the few seconds of deployment this loss might become significant and it would add to the leakage through the seams and stitch holes, which is also not accounted for in the model;
- **Nylon 6,6:** Nylon is the lightest amongst all fabrics, and its strength is high enough to win in every design with only one ply being enough to withstand membrane loads (as shown in the n_{ply} row of Table 4.9). The extra strength of aramid (1000 N/cm against 600 N/cm) is therefore never necessary in any design, while polyester's inferior characteristics both in strength and in material density are discarded every time;
- **Cylinder:** the cylinder has an innate advantage against the sphere, which is that it can use its whole height as stroke, while the sphere cannot. In the geometry model, the ground footprint of the cylinder keeps growing up to full crush, $A_t = \frac{\pi}{2} h L_0$, while the footprint of the sphere, $A_t = \pi h(D_0 - h)$, is largest at $h = D_0/2$ and shrinks back to zero after that. Since the bag force is $(P_{\text{bag}} - P_{\text{amb}})A_t$, the sphere loses its braking force over the second half of its stroke. To stop the payload without exceeding a_{lim}, the sphere therefore needs about 2 to 2.5 times the diameter of the cylinder. Since the surface grows with D_0^2 and the volume with D_0^3, its mass ends up being about ten times larger than that of the cylinder. Clearly, the bias regarding the shrinking footprint is a limitation of the geometry model instead of the behaviour of a real membrane, which should not lead to the total exclusion of the spherical shape.

Still analysing Table 4.9, it is possible to observe the expected trend of an increasing diameter D_0 as the allowed residual velocity decreases, since more deceleration power is required from the airbag as per Equation (2.1). However, $L_0/D_0 = 0.50$ in all four designs, where 0.50 is the lower limit of the allowed range (0.5 to 4.0). The fact that the optimiser landed on the lower bound of the given range means that the optimum probably lies at a lower value than it was allowed to explore. Reducing this bound further might be mathematically meaningful, but it may not be as appropriate of a choice from a geometrical perspective, which naturally depends on the medical payload container's native geometry. Similarly, the inflation pressure settles at the upper limit $P_{0,\max} = 1.400\,\text{bar}$ for 1.5 m/s and 2.0 m/s, so these two designs are also limited by a bound. However, simply raising this bound may not be realistic, since a higher inflation pressure would demand more of the inflator. Delivering more gas within the same deployment time could lead to excessive gas temperatures, and judging whether it is feasible would require a dedicated study of the inflation gases, how they are stored and the chemical reactions that take place in the inflator. A higher allowed P_0, just as well as a heavier payload, would also raise the membrane tension, possibly shifting the fabric choice towards the second-best candidate after Nylon as presented in this model, i.e. the aramid fabric. Since the aramid properties are the least certain of the three (Appendix G), such a shift would need to be confirmed with data for a specific coated aramid

fabric.

Observing the mass and performance metrics in Table 4.9, the total masses, including both the airbag fabric and the inflation gas, are promising: even in the most demanding case, a residual velocity of 0.5 m/s, the total mass stays below 400 g. This suggests that the mass added to the UAV platform may be worth the implementation of this safety device in view of certification. The same table reports the acceleration margins (with respect to the allowable) of the winning designs for each of the four residual velocity caps, where positive margins indicate that the peak acceleration of the design is below the limit. In all four cases the peak acceleration settles at about 187.5 g, i.e. on the limit a_{lim}, as expected since the optimiser reduces the mass until the acceleration constraint becomes active. The margins on the burst pressure lie between about 25 % and 33 %, in addition to the safety factor SF_{burst} already included in P_{burst}. In all four designs the peak pressure coincides with the inflation pressure P_0: as shown in Figure 4.31, the internal pressure starts to drop as soon as the impact begins, while the deceleration is still increasing, which points to a fast deflation of the bag, so that the rise in deceleration is driven by the growth of the footprint A_t rather than by the pressure.

However, the allowable acceleration of 225 g may well be too high, as it depends strongly on the sponge assumed in the sled test (Section 4.4) without which the allowable acceleration is expected to drop substantially. Overall, the airbag design is only as reliable as the characterisation of the payload and of its packaging. These results highlight the critical role of a thorough payload characterisation in finding the allowable, and the need to define packaging requirements for the operators who will prepare the payload in real mission scenarios.

For each residual velocity cap, Appendix E compares the impact transients of the lightest feasible design obtained with each of the five inflation gases (N_2, He, Ar, CO_2 and dry air), selected over both shapes and all three fabrics.

[16] In the full-fidelity re-simulation, a design is accepted as feasible up to 0.1 % above a_{lim}, since the optimiser drives $a_{\text{peak},n}$ onto the limit and the reference model lands marginally on either side of it. The winners at 1.5 m/s and 2.0 m/s are feasible only within this tolerance.

Table 4.9: Optimum airbag design for each residual speed cap $u_{\text{res,max}}$, from the full-fidelity re-simulation of the DE results.

$u_{\text{res,max}}$ [m/s]	0.5	1.0	1.5	2.0
Categorical variables				
Shape	cylinder	cylinder	cylinder	cylinder
Fabric	Nylon 6,6, silicone-coated	Nylon 6,6, silicone-coated	Nylon 6,6, silicone-coated	Nylon 6,6, silicone-coated
Gas	He	He	He	He
Design variables				
Diameter, D_0 [m]	0.774	0.671	0.534	0.522
Aspect ratio, L_0/D_0 [–]	0.50	0.50	0.50	0.50
Length, L_0 [m] (derived)	*0.387*	*0.336*	*0.267*	*0.261*
Number of plies, n_{ply} [–]	1	1	1	1
Inflation pressure, P_0 [kPa]	129.6	138.5	140.0	140.0
Vent open-area fraction, φ_{vent} [–]	0.0063	0.0067	0.0156	0.0162
Initial vent area, $A_{vent,0}$ [cm^2] (derived)	*118.3*	*94.4*	*139.7*	*138.9*
Mass				
Fabric, m_{bag} [kg]	0.339	0.255	0.162	0.154
Gas, $m_{\text{gas,0}}$ [kg]	0.040	0.027	0.014	0.013
Total, m_{tot} [kg]	**0.379**	**0.282**	**0.176**	**0.167**
Performance				
Peak acceleration, $a_{\text{peak},n}$ [g]	187.5	187.5	187.5	187.6
Margin to a_{lim} [%][16]	+0.002	+0.009	−0.017	−0.027
Landing speed, u_{land} [m/s]	0.451	0.969	1.429	1.919
Peak pressure, P_{peak} [kPa]	129.6	138.5	140.0	140.0
Burst pressure, P_{burst} [kPa]	173.6	184.8	206.1	208.7
Margin to P_{burst} [%]	25.3	25.1	32.1	32.9
Outcome	settled	bottomed out	bottomed out	bottomed out
Time to landing, t_{stop} [s]	–	0.704	0.065	0.051

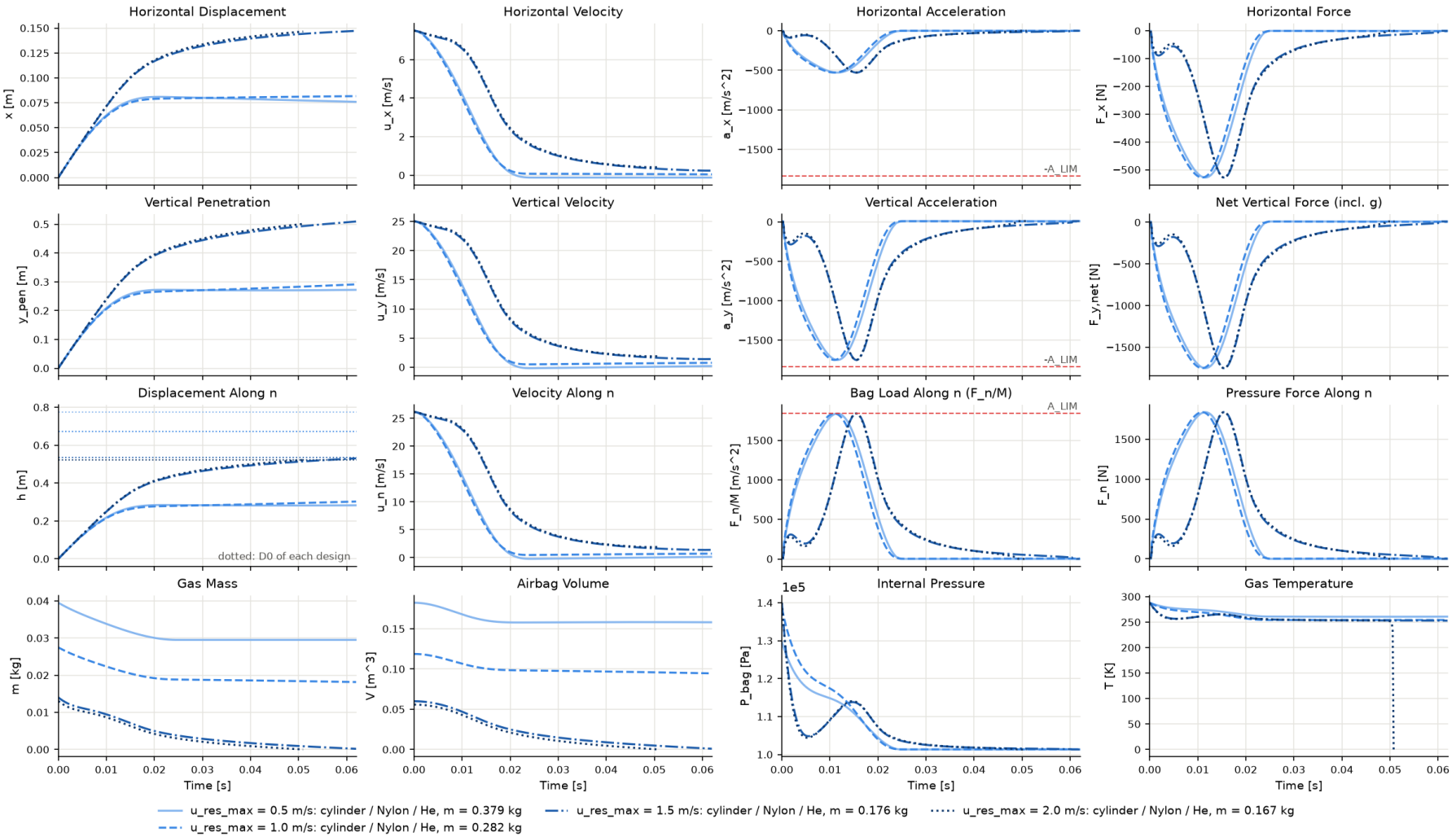


Figure 4.31: Impact transient of the optimum design for each residual-speed cap $u_{\mathrm{res,max}}$ (cylinder, Nylon 6,6, He).

4.6. System design hypothesis

Based on the information collected throughout the workflow, a hypothesis of the component design system can be formulated as a sequence of phases, from the activation of the system to the inflation of the airbag and the impact, as proposed in Figure 4.32.

One idea that emerges from this sequence, as well as from the simulation choices in the airbag design optimisation code, is to control the attitude of the payload during the fall, for example by placing its centre of gravity so that the payload falls with a predictable orientation. In this way, the airbag would not need to wrap around the entire payload, but only around the side that is guaranteed to impact, consistently with the single impact direction assumed in the optimisation code. This design choice would save material, reduce overall weight, and improve the chances of protecting the payload, provided that the payload falls with the predicted attitude.

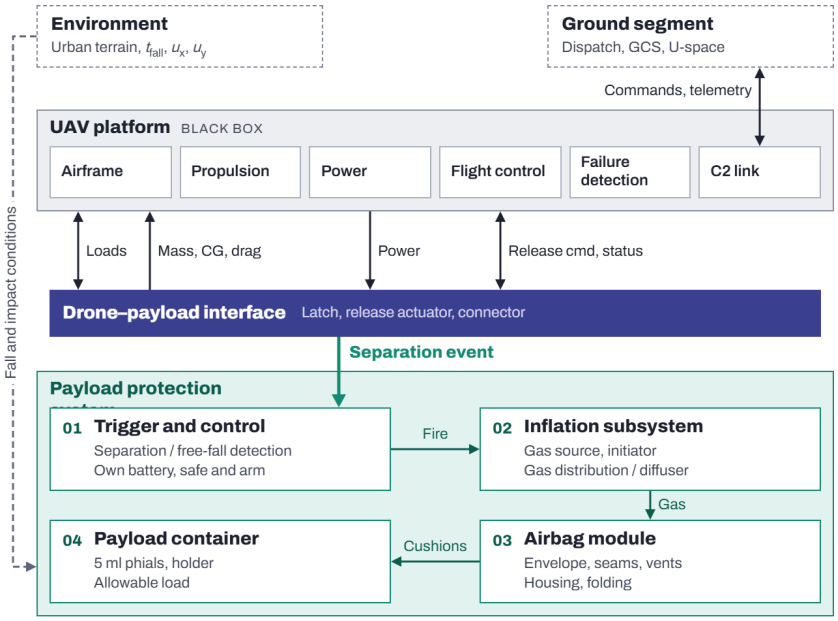


Figure 4.32: Hypothesised system architecture: UAV platform, drone-payload interface and payload protection system.

The following recommendations can then be made for the second design phase:

- **Inflation mechanism:** helium cannot be produced by the pyrotechnic reaction typically used for N_2, so it would need to be stored as pressurised gas in small cylinders. Whether this is feasible is still to be verified and, specifically, the mass of the gas containers should be included in the comparison, since it is not accounted for in the optimiser and may offset the lower gas mass of helium;
- **Time of inflation:** the time of flight to impact obtained in Step 0 ranges between 4 and 6.5 s, so the activation and the full inflation of the airbag must be completed within the shortest time of flight, i.e. about 4 s, including the time needed to detect the failure and trigger the system;
- **Folding mechanism:** the most common family of folding patterns in the literature are within the zig-zag family (e.g. Leporello, accordion or concertina folds presented in [65], [66], [69], [91]) or, alternatively, in the rolling family (including sigma-rolling, as in [58], [65]). These patterns could also be optimised for inflation time and storing method, as already suggested when proposing the inner design loop in Section 3.2.2;
- **Seaming pattern:** the stitch pattern is a fabrication variable that becomes critical if ignored, according to [92]. Seams concentrate stress because they elongate less than the fabric, and they fail either by yarn slip or by stitch failure [48]. Mu et al. [91] compare the I, H, centre and 2/3-1/3 seam layouts, with thin, medium and thick seams, while [58] studies the Y-tear configuration. The optimisation of the seaming pattern is also left for future work.

These recommendations are only a starting point: the second design phase will require new recommendations of its own, based on a dedicated literature search and on concepts beyond the scope of this workflow, such as the integration with the UAV and the deployment system as schematically illustrated in Figure 4.32

5 Conclusions

This chapter summarises the results of the thesis by returning to the research questions formulated in Section 2.4, starting from the main question and then addressing each sub-question sequentially. It concludes with recommendations on future work based on the limitations identified throughout the workflow, covering both technical developments and the collaboration between authorities, research and industry needed to enable UAV medical delivery operations.

5.1. Summary of results

How can a systematic multi-phase simulation workflow be developed to design an airbag-based protective system for medical payloads during UAV emergency-release events?
This thesis was designed as a first step to bridge the gap between the research carried out so far, which covered logistics, regulation and mission evaluation, and the long-term objective of enabling the deployment of a UAV medical delivery network over Italian territory, flanking the transportation solutions already in place. To carry the research and development effort forward, the proposed solution is rooted in safety, opposing a *mission-first* development strategy that struggles to get past pilot flights and certification (Section 2.1).

At a higher level, the thesis proposes a paradigm shift in design based on the *safety-first* principle, inspired by biomedical design, in which the payload and the UAV are treated as separate entities and safety is moved to the start of the design process (Section 3.2.1). At a more specific level, this principle is applied to the preliminary sizing of an airbag-type device, through the airbag design workflow of Section 3.2.2. Steps 0-4 of this workflow, applied in this thesis, include a mission analysis, an experimental campaign and the FE modelling of impact events. These steps are preparatory to the design of an airbag device able to protect the selected medical payload, i.e. pharmaceutical glass phials, and each of them provides the next with the data it needs: the probabilistic impact conditions over the territory of Turin, the calibrated material model of the payload, the allowable acceleration and, finally, the input for the optimisation of the airbag design.

With the safety-first approach, the protection system therefore provides the outer design loop with its inputs (airbag mass and volume, inflator and inflation time), instead of being included in the overall UAV architecture at the end of the design (Figure 3.4b). Overall, separating the payload's protection system from the UAV's allows for a modular design that could be implemented across different drones and missions while also easing certification, which is ultimately the long-term goal of *Politecnico di Torino*'s research group.

Sub-questions:

1. *What are the allowable mechanical limits (acceleration, stress, etc.) for medical products during emergency release, and how should these define the design envelope?*
   The mechanical limits were studied through the characterisation of an exemplifying payload, consisting of 5 ml borosilicate glass phials. An experimental campaign of 76 compression tests on

two batches of phials provided the basis for an approximate material model in LS-DYNA, calibrated on a tensile strength FT = 60 MPa, which reproduces the experimental breaking load of 98.7 N. The phials were then subjected to an FE sled test in LS-DYNA, hypothesising a certain payload assembly configuration. This gave an allowable peak acceleration of 225 g for an acceleration perpendicular to the phial axis (the most critical case). Together with the impact conditions obtained from the mission analysis, taken at the 98.5th percentile (u_{x0} = 7.5 m/s horizontally and u_{y0} = 25 m/s vertically), this allowable defines the design envelope of the airbag.

2. *How can the inflated airbag configuration be optimised to satisfy impact-absorption requirements while balancing competing design requirements?*
   This was achieved through the proposed design optimisation code, which takes as input the initial velocity, the payload mass and the allowable acceleration, and explores the design variables, i.e. the gas type, the fabric type, the number of fabric plies and airbag sizing parameters regarding characteristic dimensions, inflation pressure, and venting area. A differential evolution wrapper around the analytical airbag impact model returns low-mass candidates that pass the bursting, full-stroke impact velocity, and rebound checks, together with their geometry. These outputs can guide the design of the airbag towards a payload-geometry-specific solution.
3. *What recommendations facilitate transforming the inflated geometry into a compact, functional deployment system?*
   A hypothesis of a fully functional system was proposed based on an informed literature search and the findings gathered throughout the workflow. Without a specific payload geometry, the proposed system cannot go beyond the level of recommendation drawn from information collected throughout the workflow, not accounting for the design phase subsequent to this MSc thesis deliverable. The recommendations include the type of inflation mechanism (a stored-gas inflator charged with helium), the time of flight to impact (between 4 and 6.5 s), which sets the time available for the inflation, and the initial inflation pressure P_0, which ranges from 129.6 kPa to 140.0 kPa (absolute) across the optimised designs in Table 4.9. Geometry-specific tailoring is left for future work outside the scope of this thesis.

5.2. Recommendations for future work

The airbag design workflow tested in this thesis is limited by the speculative nature of the payload assembly and, thus, cannot directly lead to testing and prototyping of a fully functional device. However, it is possible to continue this research and development line in several different ways, aiming at the long-term goal of deploying a full-scale UAV-based medical delivery network on regional and national territory.

In general, technical recommendations could include a combination of the following pathways:

1. **Improve current limitations:** fix the limitations of the present work by increasing the complexity of the single workflow steps. This could involve a more detailed payload-material characterisation including standard tests and equipment, as well as extending the analysis to several types of medical payloads for a more comprehensive approach, and conducting an experimental sled testing campaign. The design optimisation code could make use of a more complex optimisation algorithm that can potentially decrease calculation times and design convergence;
2. **Proceed towards validation and prototyping:** exit the Safety loop and enter the Mission loop in Figure 3.4b, carrying the project forward based on the optimised airbag configurations obtained in Section 4.5.4 and on the system design recommendations given in Section 4.6. This would mean updating the hardware and mission design, and then re-evaluating initial conditions to re-inform the "inner" design loop. Once the capsule and UAV geometry are defined, the optimised airbag could be validated by manufacturing and drop-testing a prototype or, as a first step, by simulating the impact of the actual assembly. This validation was not carried out in the present work, since it is specific to a geometry that is not yet fixed. This step requires addressing the design of the system and its integration with the UAV, possibly through an ad-hoc solution, followed by dedicated simulation and testing, as well as informed decisions on the inflation mechanism and the folding method of the airbag;
3. **Introduce reliability-based design:** implementing a reliability-based approach to derive the

minimum-mass airbag that keeps a glass phial's survival probability above a target defined from previous SORA ground-risk assessment. This would mean combining the stochastic nature of glass strength with the impact-velocity distribution of the urban UAV missions to improve the airbag design and possibly further save on mass once probabilistic design is chosen over the worst-case, conservative approach used in the present demonstration of the workflow. The reliability-based approach should, however, consider the difficulties of certification and produce an informed design that still guarantees an acceptable level of safety;

4. **Consider alternative failure scenarios:** certification standards impose the installation of a parachute onto the UAV platform. The contribution of a parachute device could also be explored, to describe a less conservative, but possibly more likely, UAV failure scenario, in which the airbag and the parachute work in tandem. Along the above-the-river trajectory, the parachute would provide an initial deceleration, so that the impact becomes equivalent to a drop from a lower height, and could also offer some attitude control during the fall. At the same time, a slower descent increases the time of flight and therefore the wind drift, which widens the impact footprint and may increase the probability of falling onto urban areas, a critical aspect for certification. The combined system could also be formulated as a minimum-mass problem, in which the total mass of airbag and parachute is minimised for the combined falling mode.

Overall, embracing the long-term goal described above, a further recommendation is to use the work conducted until now within the feasibility studies research group at *Politecnico di Torino* to inform national guidelines on the necessary measures to actually enable this type of operation. The alignment of national authorities on operational and safety requirements, through a cooperation between the aeronautical and the medical institutions for a cross-field operation, would facilitate the implementation of a regional or national UAV network. Research groups and industries would then know which requirements to meet and could develop devices that are relevant in practice, rather than designing defensively around an aeronautical certification barrier and general safety concerns.

As a first step in this direction, the alignment of objectives could result in a collaboration between ABzero for the delivery capsules, SkyEU for network operations and UAV platforms, and the feasibility studies research group for providing expertise on safety and certification including the airbag device, to be ultimately integrated into one comprehensive system, able to safely sustain medical delivery missions and to support their scaling beyond local trials.

6

Project Management

6.1. Project organisation and stakeholders

6.1.1. Context

This thesis was conceived as a continuation of a research line initiated by *Politecnico di Torino* in 2020, during the COVID-19 emergency, and carried on until 2023/2024 with Angelini's conclusive work on the research group's UAV operation feasibility studies [16]. This research line favoured conceptual and evaluation studies ignoring a more practical, design-oriented approach. The present MSc thesis project is, therefore, the first in this research line to adopt such an approach based on the formalisation of previous findings.

The formal setup of this project consists in a TU Delft graduation agreement, with *Politecnico di Torino* as the host university, FEV Italia as the industrial partner, and the *A.O.U. Città della Salute e della Scienza* (represented by the Molinette hospital) as the advisory hospital. The official start was on 25th March 2026, with a nominal duration of eight months.

6.1.2. Supervisory team

The supervisory team is composed of five members. Prof. Bianca Giovanardi from TU Delft was the responsible supervisor of the thesis, in collaboration with four external supervisors belonging to three different Italian institutions. The host institution, *Politecnico di Torino*, provided two supervisors: Prof. Enrico Cestino, the main professor interested in this research line, and Davide Angelini, PhD student and co-supervisor of this thesis. At an industrial level, Eng. Sergio Bianco from FEV Italia advised on the FEM parts of the thesis. Finally, Dr. Daniela Cestino, representative of the *Città della Salute* and of the Molinette hospital, acted as medical advisor.

Each of the supervisors requested a different means of communication. Structured meetings were scheduled mostly with the universities, while ad-hoc meetings were organised with S. Bianco and D. Cestino, depending on their availability. The frequency of updates and meetings is summarised in Table 6.1.

Table 6.1: Supervisory team, roles, and communication channels.

Supervisor	Organisation	Role	Channel and frequency
B. Giovanardi	TU Delft	Responsible supervisor	Bi-weekly meetings
E. Cestino	Politecnico di Torino	Host supervisor	Weekly meetings, skipped when no update was needed
D. Angelini	Politecnico di Torino	Co-supervisor	Weekly meetings, together with E. Cestino
S. Bianco	FEV Italia S.r.l.	Industrial co-supervisor	Ad hoc, fitted to his availability
D. Cestino	A.O.U. Città della Salute	Medical co-supervisor	Ad hoc, for supply of the medicinal phials and exchange of medical information

6.1.3. Monitoring and communication

The variety of consultation channels adopted through the project shows a challenge in how communication was conducted throughout the thesis. Except for the only joint meeting among all the relevant parties during the pre-kick-off alignment attended by B. Giovanardi, E. Cestino, and D. Angelini, joint update meetings did not usually suit the supervisors' individual availabilities. As a result, the student was the only information link between the parties involved, and each supervisor saw a partial picture through their individual communication channel depending on the topics discussed with the student during the meetings.

6.1.4. Stakeholder priorities

The process of aligning the expectations of the different parties also posed a challenge for this thesis, as the home and host institutions initially had divergent priorities. *Politecnico di Torino* and FEV Italia were interested in a rigorous characterisation of the glass phials and, in general, of the medical payload. For TU Delft, the main concern was that the scope be feasible within the nominal duration of the thesis by establishing a clear research direction as soon as possible: the TU Delft supervisor judged the planned work to be more than sufficient from the start, and at the beginning even overshooting the workload expected for a MSc thesis.

6.1.5. Resources by party

The resources offered by the stakeholders are listed below, focusing specifically on what each of them offered and contributed to the thesis.

- **TU Delft:** supervision and access to the DelftBlue high-performance computing (HPC) cluster (including the Aerospace Engineering Faculty's partition of HPC12);
- **Politecnico di Torino:** the MTS compression testing facility, plus HPC and licences as expected in the research proposal but not fulfilled due to maintenance issues (mentioned later);
- **FEV Italia:** a workstation with the HyperWorks FE modelling suite, including pre- and post-processing, and general FE modelling support.
- **A.O.U. Città della Salute (Molinette):** the medicinal phials.

6.2. Planning: baseline versus execution

6.2.1. Baseline

The timeline hypothesised at the beginning of the MSc thesis project is shown in Table 6.2, where the different project phases and milestones presented in the first project proposal are highlighted in chronological order.

Table 6.2: Baseline planning from the research proposal.

Phase / milestone	Planned duration	Planned date (opt. / max.)
Phase 0: Mission envelope and allowables	3 weeks	19/06
Phase 1: Inflated configuration	7 weeks	07/08
Midterm review	–	31/07 - 07/08
Phase 2: Folding and deployment	10 weeks	16/10
Green light review	–	23/10 - 13/11
Final submission	–	13/11 - 11/12
Phase 3: Refinement for prototyping	*Future work*	–

6.2.2. Scope evolution and execution

As the research process progressed, the thesis work progressively evolved away from the initial planning. The project proposal at the beginning aimed at the realisation of the full airbag design, answering to four sub-questions that would focus more on the simulation and development aspect of the safety device, shifting the focus away from the preliminary steps needed to inform the design correctly. In

the research proposal following the initial literature search phase, the main research question became workflow-centred. The most substantial deviation consisted in giving higher importance to the payload characterisation steps would require much longer than the 3 to 7 weeks initially hypothesised for full, high-fidelity modelling (see Phase 0 and Phase 1 in Table 6.2). In fact, the high-fidelity modelling required 6 macro-model iterations for the compression test (as shown in Appendix C) and several more for the sled test, including explorations on the sponge material characterisation.

At the midterm review, the scope moved to its current state. Although no formal replan was made after the midterm review, it was agreed among the supervisors that the focus would be on a thorough characterisation of the payload before informing the airbag design (aligning with the host university and the industrial partner's initial expectations), without the need for a general readjustment of the timeline as it would still allow for the regular conduction of the green light meeting and the delivery of the final portfolio within the windows set in the research proposal.

6.2.3. Final scope definition

The final evolution of the research process settled in this way:

- Sub-question 1 (allowables): answered in full, through experimental and FEM characterisation;
- Sub-question 2 (inflated configuration): answered at pre-design level, using an optimisation wrapper extended to 2D (including horizontal velocity, seam efficiency and fabric transpiration);
- Sub-question 3 (folding and deployment): moved to future work, only reaching recommendation level for the completion of this thesis;
- Main question (workflow): answered.

The upstream steps were developed with high fidelity, with the closing deliverable consisting in a design recommendations chapter for a variety of geometry-specific future designs (see workflow developed in Chapter 3, shown in Figure 3.5).

6.3. Critical events and their management

6.3.1. Scope narrowing: from airbag design to workflow and characterisation

About two months into the research process, right after the literature research phase, the TU Delft supervisor raised the possibility of narrowing the scope and framed two branches: the thesis would focus either on the airbag design or on the pre-design, i.e. the workflow required for reliable airbag design. The two branches were practically mutually exclusive. The student first chose the airbag design branch considering the possibility to inform the design based on data found in literature, but the question was not formally presented to the *Politecnico di Torino* and FEV Italia supervisors, whose priority on the correct characterisation of the phials had been known since the initial definition of the thesis. In hindsight, this communication mistake from the student's side resulted in an expectation misalignment that was only formally corrected in the vicinity of the midterm meeting.

By the midterm meeting, it became clear that building high-fidelity models in the pre-design part of the thesis had become the main focus of the thesis as proper characterisation would require more iterations than planned. This decision was cemented by the realisation that the downstream sizing depends on the upstream allowables, so sizing on approximate models or scattered literature data would carry errors forward. This was the basis of the reasoning for insisting on the modelling part, and it aligned perfectly with the expectations of *Politecnico di Torino* and FEV Italia, whose position was prioritising credible allowables from the start of the project. The outcome was that the scope moved from designing the airbag, as initially planned, to the workflow before the design of the airbag. This meant taking the road the TU Delft supervisor had already outlined at the beginning of the thesis, while keeping the main research question as drafted after the initial literature search phase. Overall, the scopes aligned and all parties agreed that payload characterisation needed to be prioritised over airbag design.

6.3.2. Loss of computational resources

Until the midterm review, which took place on the 4th of August 2026, the only resources available were a local machine running Python and equipped with an LS-DYNA student licence limited to four cores and 128 000 elements. The expected access to *Politecnico di Torino*'s HPC and full LS-DYNA

licence, which was supposed to be granted about a month before the midterm meeting, was delayed by unexpected, substantial maintenance throughout the entire months of July and August, where the HPC would be fully shut down and unavailable to the entire university. This meant that a significant delay was forecast: it was hypothesised that, to unlock new computational power at *Politecnico di Torino*, the student would need to wait for the end of the maintenance period and sustain heavy queue times subsequently. The situation was resolved by discussing with B. Giovanardi in an "emergency update" meeting, organised two weeks after the news of the unexpected HPC maintenance reached the student. The student presented three possible options the TU Delft supervisor to continue with the thesis project:

1. Wait for the maintenance to take place and ask for a one-month or two-month extension, to make up for the expected waiting time;
2. Obtain computational power elsewhere;
3. Change thesis topic, either repurposing the work done until then or redefining it completely, starting anew.

In the end, computational power was unlocked thanks to the TU Delft supervisor's offer to unlock access to the DelftBlue cluster and to the Aerospace Engineering faculty's partition of HPC12, as something not agreed upon before the thesis and outside the official thesis agreement, thus not needing to change the thesis topic.

6.3.3. Split toolchain between workstations

Another constraint concerned the toolchain used for pre- and post-processing of the FEM calculations. The FEV workstation had the HyperWorks suite installed with industrial licences, but the IT policy of the company blocked the installation of the VPN (EduVPN) needed to connect to DelftBlue and launch the LS-DYNA input files there. On the contrary, the personal PC had VPN access to DelftBlue but no HyperWorks installed, and the only usable pre- and post-processor was LS-PrePost, which is more limited in functionality than the industrial software.

The impact that this limitation had on the project was that data could only flow out of the FEV PC, so the results files generated on DelftBlue or HPC12 could never be post-processed in HyperWorks. Furthermore, every large model change (repositioning of a component, remeshing, reorganisation of information in modular files, etc.) required going back to the FEV PC and editing the model there, which had often remained obsolete regarding the changes implemented on the student's machine. In several occasions, large files risked exceeding the size limits of the only transport channel guaranteed between PCs, i.e. e-mail. This caused significant loss of time and extra difficulty in keeping track of model updates, as different iterations of the same model were developed on two machines in parallel: parameters were sometimes manually edited on the personal computer and then not implemented in the original model on the FEV PC, so manual checks to implement changes uniformly across launch files in separate PCs had to be carried out often. In general, the lack of transferability of the models was particularly difficult to manage. A small mitigation that was adopted was splitting the model into modular input files (`*INCLUDE`), so that file-specific changes could become more easily trackable. Despite the fact that subdividing complex models into focused files is standard practice in industrial FEM, not being able to transfer large files (especially results files) between machines because of the e-mail transfer limitation remained an unresolved challenge throughout the thesis project.

6.3.4. Technical decisions under conflicting advice

On two occasions, the decisions of the student conflicted with the advice given by one of the supervisors:

1. **Modelling approach for the phial.** The first example concerned the modelling approach for the phial. The industrial supervisor suggested a solid-element model with a material card developed for concrete and ceramics. With this approach, however, there were persistent contact issues, the material behaviour was not adequate to the type of failure observed in the lab, and the reaction force measured on the compressing plates was zero or strongly oscillating, pointing to contact sensitivities and to a material model not formulated for glass. Therefore, the student decided to switch to shell elements and, after an independent and focused literature review, to adopt a glass-specific material model, `*MAT_GLASS` (MAT_280). This made it possible, for the first time, to obtain reaction forces that were measurable and comparable with the experimental data.

2. **Fidelity of the sled-test model.** The second occasion concerns the advice given by the TU Delft supervisor regarding the use of a reduced, rather than a full-scale, sled test FE model because of the higher computational cost needed, as well as for the risk of not finishing on time. However, the student realised that the reduced model's results could not be considered physically valid, nor presented as such with critical judgement. The decision was therefore to build the full sled test model, keeping the reduced model as a fallback plan and taking full responsibility for any delay. As an outcome, the full model was built successfully and no changes to the thesis timeline were needed.

6.4. Self-assessment

As the first research and development project conducted autonomously, the present thesis project represented a fundamental occasion for the student to self-evaluate on performance with respect to project management. On one hand, the student managed to adapt to the different communication styles and requirements of the supervisors, generally succeeding in keeping them updated despite the initial scope misalignment. Milestones were held notwithstanding a worrying forecast delay of some months and the logistic challenges: the midterm took place on schedule and the green light and submission remained within the proposed windows. Furthermore, recommended technical approaches were overturned with evidence: the shell elements and glass material model that were adopted produced the first usable comparison with experimental data.

On the other hand, shortcomings on the student's side must be considered too. When confronted with the scope misalignment, divergence in supervisor priorities, and timeline underestimation at the beginning of the thesis project, the student should have raised the problem openly without trying to deliver both the airbag work and the phial characterisation, which increased the workload and delayed convergence on scope. Furthermore, an exceeding amount of time was spent on the recommended solid-element approach before switching to an alternative method, as no criterion for arresting the trial-and-error feedback loop was established prior to the start of the thesis modelling phases.

6.5. Lessons learned and recommendations

Overall, the present thesis project was an important learning experience, and the lessons learned cover different aspects of project management. Possible recommendations for future work might be:

- Schedule joint milestone reviews (kick-off, midterm, green light) with all supervisors, making common participation a clear priority at the start of the project;
- Keep one shared progress record that all supervisors can access, instead of adopting different platforms to match all supervisor's expectations (e.g. GitHub, MS Teams, Dropbox, Google Drive etc.)
- Keep a shared decision log to track and officially confirm scope changes or other fundamental decisions. This builds on the notion of exposing conflicts to all parties as soon as possible to converge on a formal decision within acceptable time, rather than absorbing all research possibilities as extra work;
- Set stopping criteria for technical approaches, e.g. escalate/switch approach after N iterations with no usable output;
- Plan iteration buffers for modelling tasks that are new and unexplored, such as brittle failure and/or contact;
- Confirm resource availability (HPC access, licences, VPN, software on each machine, etc.) before kick-off.

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A

Extended Market Analysis

A.1. Value-chain actors

Table A.1: Actors involved in drone delivery of medical goods in Italy, grouped by value-chain stage. † Based in Piedmont.

Category	Institution	Scope
Drone manufacturers	SkyEU†	Autonomous high-speed eVTOL drones for first aid and critical logistics [29]
	FlyingBasket	Cargo drones; used for blood transport Pozzuoli–Ischia [93]
	H2C	Hydrogen drones and hydrogen supply system for Padova Sandbox [94]
	Dronus	"Radon" drone for medical delivery [95]
	Eurolink Systems	"Beluga" medical delivery drone, with Università Cusano [95]
	IDS Ingegneria Dei Sistemi	Defibrillator-carrying drone (WingBeat project) [96]
Technology partners (role not specified)	MAVTech	INDOOR technical partner; aerial surveillance and operational-support products [20]
	FEV Italia†	Industrial partner of the DIMEAS feasibility studies; numerical impact analysis
	Leonardo	VTOL drone demonstration flights in Lazio, with Telespazio and D-Flight [97]
	Caltec	Technical partner of SIS 118 for the SEUAM defibrillator drone [98]
	AP 74 UAV	Drone company involved in the Piadena pharmacy link [99]
Payload / container makers	ABzero	Smart Capsule: sensorised, UN3373-certified container for blood, organs and drugs [23]
	Carpitech	"Philotea" temperature- and atmosphere-controlled capsule [100]
	Air Abruzzo	"Flying Box" transport capsule [100]
	Telespazio	TPZ Air 100 drug box used in U-ELCOME [101]
Operators	ABzero	Drone operator for San Raffaele trials; handled ENAC authorisation for INDOOR [25], [102]
	Unmanned4you	Operated Aetos blood-transport flights with FlyingBasket [93]
	DL Droni	Piloted the Piadena pharmacy flights [99]
	Telespazio	Runs the U-ELCOME hospital-to-village drug delivery service [18]
	Croce Rossa Italiana	Own drone fleet and certified pilot school; urgent-drug delivery trials [103], [104]
Airspace, infrastructure and energy	ENAV	Air navigation service provider; coordinated Corus-XUAM [105]
	D-Flight	U-space services (ENAV group); Corus-XUAM partner [105]
	DTA	Corus-XUAM partner, Grottaglie airport test site [105]

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Category	Institution	Scope
	ADPM Drones	"Hangar": docking and recharging station for automated missions [105]
	Gruppo SAVE	Airport group leading Padova Sandbox [94]
	Heron Air	Operator of Padova airport, Sandbox partner [94]
	Milani; DBA Group	Build the hydrogen production and refuelling unit for Padova [106]
	Stesi	Supervision system, ground logistics and spare-parts analytics for Sandbox [107]
	Torino Caselle and Cuneo Levaldigi airports†	Proposed hub and support base for hydrogen drone logistics [26]
Research	PIC4SeR, Politecnico di Torino†	Autonomous flight technologies for transporting biological material (INDOOR) [22]
	DIMEAS, Politecnico di Torino (Prof. E. Cestino)†	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost–safety comparison of car, drone and eVTOL, hub siting [12], [13], [14], [15], [16], [17]
	Università di Torino†	INDOOR project partner [20]
	Scuola Superiore Sant'Anna	Parent institution of the ABzero spin-off [25]
	Università Cusano	Co-developer of the "Beluga" drone [95]
	San Raffaele technology centre	Developed the FF2020 drone delivery demonstrations [108]
End users	A.O.U. Città della Salute e della Scienza†	INDOOR test flight between hospital sites; proposed hydrogen logistics plan; pharmacy department partner of DIMEAS feasibility studies [24], [26]
	Regional Transplant Centre Piemonte–Valle d'Aosta†	INDOOR partner for organ and biological transport [20]
	Ospedale San Raffaele	In-campus delivery of drugs and biological samples [109]
	ASST Sette Laghi	Feasibility study for inter-hospital drug transport [110]
	Piadena community pharmacies	Pharmacy-to-pharmacy drone link [99]
	ASL 1 Abruzzo	Hospital-only drugs delivered to remote villages [18]
	ASL Napoli 2	Blood transport to Ischia [93]
	Ospedale Miulli	Demonstration flight of life-saving drugs [111]
	ASL Lecce	Planned Gallipoli–Casarano hospital link [112]
	Ospedale di Patti	Planned link to the Aeolian Islands [112]
	Ospedale Pediatrico Bambino Gesù	Partner in VTOL demonstration flights in Lazio [97]
	SIS 118	Emergency service promoting the SEUAM defibrillator drone [98]
Promoters and funders	Fondazione DOT†	Promoter of the INDOOR project [20]
	Centro Nazionale Trapianti	Funds the INDOOR research grant at PIC4SeR [21]
	City of Turin†	INDOOR partner; funded the LastMed call [20], [27]
	Regione Veneto	Signatory of the Padova Sandbox agreement [94]
	Regione Lazio	Partner in VTOL demonstration flights [97]
	Regione Campania	Institutional support for the Aetos trial [113]
	EU research programmes	H2020 Flying Forward 2020; U-ELCOME, co-funded by Horizon [102], [114]
	Abruzzo municipalities and Parco Sirente Velino	Host communities of the U-ELCOME service [18], [101]
Regulators and authorities	ENAC	National civil aviation authority; flight authorisations; protocol with the Red Cross [115]
	EASA	EU rules for drone operations, Regulation (EU) 2019/947 [116]
	Regione Piemonte†	Regional health authority; drone trials subject to benefit–risk analysis [30]

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Category	Institution	Scope
Alternatives	Tecnocad / Tecnomill / Bylogix†	LastMed autonomous ground rover for last-mile drug delivery [28]

A.2. Trials and planned initiatives

Table A.2: Drone delivery of medical goods in Italy: trials and planned initiatives (status as reported by the cited sources; checked 5 October 2026).

Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
Piedmont	Turin, Molinette ↔ San Giovanni Bosco feasibility studies	2020–2024	DIMEAS (Politecnico di Torino); A.O.U. Città della Salute e della Scienza (Pharmacy); FEV Italia	Life-saving drugs, blood-derived medication	Approx. 10 km corridor along the Po, approx. 15 min at 17 m/s	Desk studies (5 MSc theses, 1 conference paper); no flights performed	[12], [13], [14], [15], [16], [17]
	Turin, INDOOR	Oct 2024	Fondazione DOT; PIC4SeR (Politecnico di Torino); ABzero; A.O.U. Città della Salute e della Scienza	Biological samples (blood) in sensorised capsule	CTO → Molinette, 500 m, automatic navigation over the city and the Po	Single experimental flight	[20], [24]
	Turin, hydrogen drone logistics plan	Apr 2025	A.O.U. Città della Salute e della Scienza (Commissioner T. Schael)	Drugs, small medical loads	Torino Caselle hub, Cuneo Levaldigi support, future Parco della Salute	Proposal; no implementation found	[26], [117]
Abruzzo	Avezzano → Collelongo, U-ELCOME	Jul 2025	Telespazio (Leonardo/Thales); ASL 1 Abruzzo; municipalities of Collelongo, Montereale, Rocca di Mezzo	Hospital-only drugs for frail patients	Hospital to village pharmacy, approx. 20 min	Start of operational phase; further routes announced	[18], [19]
Apulia	Grottaglie → Manduria, Corus-XUAM	by 2022	ENAV; D-Flight; DTA	Medical material	Airport to hospital	Test (reported in 2022)	[105]
	Taranto, SEUAM	Feb 2024	SIS 118; Caltec (Campanian aerospace consortium)	Automated external defibrillator	Urban, simulated cardiac arrest	First test of experimental project	[98]
	Lecce, ASL Lecce	2025	ASL Lecce	Not specified in source	Gallipoli–Casarano hospitals	Intended	[112]
	Acquaviva delle Fonti, Ospedale Miulli	Nov 2025	Ospedale Miulli	Life-saving drugs, biomedical material	25 kg hexacopter; 25 km in 25 min	Operational demonstration	[111]
Campania	Pozzuoli → Ischia, Aetos	Feb 2025	FlyingBasket; Unmanned4you; ASL Napoli 2	Blood, biological samples	17 km in 15 min; 4 flights; 5 kg carried	Second trial phase	[93], [118]
Lazio	Rome	2022	Leonardo; Telespazio; D-Flight; Ospedale Pediatrico Bambino Gesù; Regione Lazio; Aeroporti di Roma	Not specified in source	Fixed-wing VTOL drones	Two demonstration flights	[97]
Lombardy	Piadena (Cremona)	2021	Two community pharmacies; DL Droni; AP 74 UAV	Drugs and devices between pharmacies	Approx. 6 kg electric drone, approx. 1 kg payload	Announced as continuous service; current status not verified	[99]

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Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
	Milan, Ospedale San Raffaele, H2020 Flying Forward 2020	Mar 2023	Ospedale San Raffaele; EuroUSC Italia (regulatory consultant); ABzero (operator)	Drugs, biological samples	Within the hospital campus	Demonstration campaign, each flight ENAC-authorised	[102], [109]
	Varese, ASST Sette Laghi	2025	ASST Sette Laghi	Drugs, urgent materials	Between two hospitals a few hundred metres apart	Preliminary feasibility study	[110]
Sicily	Trapani, Croce Rossa Italiana	2021	Croce Rossa Italiana	Urgent drugs	Port quay → vessel; 5 min 30 s vs 35 min by sea	Simulation exercise	[104]
	Patti → Aeolian Islands	2025	Ospedale di Patti	Not specified in source	Patti–Lipari	Under development	[112]
Veneto	Padova, Padova Sandbox	from Mar 2025	Gruppo SAVE; ENAC; Regione Veneto; H2C/H2G; Heron Air; Milani; DBA Group; Stesi	Medical goods, biological samples (hydrogen drones)	Up to 100 km, 4–5 kg payload (stated)	Ground tests of hydrogen system; first flight expected autumn 2026	[94], [119]

B

Safety Measures for the Experimental Campaign

B.1. Safety precautions

The glass phials used in this test contained saline solution. The possibility of skin or eye contact with the liquid content or glass fragments in case of phial rupture was nonetheless considered. To mitigate the risks associated with both fragment ejection and liquid release, the following precautions were adopted:

- **Precaution 1:** the phial was fully constrained within the PETG adaptor up to the height of its main body, as shown in Figure 4.7, limiting the lateral dispersion of both glass fragments and liquid content upon failure. A small clearance was left between the top of the phial and the top of the PETG adaptor, sufficient to allow the compression head to apply the load directly onto the phial without interference from the adaptor itself.
- **Precaution 2:** all personnel present in the test area wore safety glasses for eye protection for the entire duration of the test.
- **Precaution 3:** the testing machine was operated under velocity control, with the applied force continuously monitored; upon detection of the load drop associated with phial failure, the machine automatically and immediately halted crosshead motion, preventing any further loading of the fragmented test article.

No other significant hazards (electrical or otherwise) were associated with the test, as the testing machine was operated within its standard, intended use envelope.

B.2. Persons allowed in the test area

Access to the test area during the execution of the test was restricted to the following personnel:

- the author of the present thesis, Annamaria Iannuzzo;
- PhD candidate, MSc thesis co-supervisor, and test supervisor Davide Angelini;
- MSc thesis co-supervisor, Prof. Enrico Cestino.

No other personnel were permitted within the test area for the entire duration of the test.

C

Development Iterations of the Glass Phial FE Model

Table C.1: Sequence of FE modelling iterations for the axial compression test.

V	Solver	Model / elements	Material / failure	Outcome / reason for change
1	Abaqus/Standard (implicit, static)	2 ml Akineton phial, quarter model; C3D10 (glass), S4/S4R stiff platens; contact pairs	Elastic + MAXPE damage initiation, linear displacement-based evolution	Contact convergence issues; usable force–displacement curve only in the purely elastic case
2	Abaqus/Explicit	As V1; general contact; variable mass scaling	*BRITTLE CRACKING/SHEAR/FAILURE; strain- and energy-based cracking variants	Immediate element erosion, strong oscillations, no readable reaction force. Switch to LS-DYNA (limited Abaqus licences; student licence available) and to the 5 ml saline phial (safety)
3	LS-DYNA explicit	5 ml phial; 10-node tetrahedra (ELFORM 16)	*MAT_BRITTLE_DAMAGE (MAT_096), as recommended by the industrial supervisor	No functional breaking criterion; reaction force not readable
4	LS-DYNA explicit	0.1 mm hexahedra; platen 100 mm/s, 2 mm stroke, no mass scaling	MAT_096	Breaking criterion active, but reaction force null or strongly oscillating
5	LS-DYNA implicit	Hexahedral model of V4	Linear elastic; implicit breaking criteria explored	Stiffness far above MTS value; implicit criteria unsatisfactory. Return to explicit
6	LS-DYNA explicit	Shells ELFORM 2 (default), NIP = 5, mid-surface map mesh (thick shells rejected). Quarter model at 0.5 mm first, then full model at 0.1 mm	*MAT_GLASS (MAT_280); EPSCR raised from 0.005 to 0.1 after it was found to govern deletion	First measurable reaction force comparable with MTS data. Quarter model dropped: convergence issues at 0.5 mm and imposed symmetric crack pattern; full model enabled by DelftBlue HPC
	LS-DYNA explicit	As V6; wall thickness varied from 0.40 to 0.75 mm	MAT_280	Iterations to fine-tune stiffness against MTS data; 0.40 mm retained
7	LS-DYNA implicit (eigen)	Full 0.1 mm model of V6	–	Eigenmode 18 (bending-type) selected as deterministic imperfection
8	LS-DYNA explicit (final)	Full model, 0.1 mm shells (batch-A geometry, caliper-measured); mode 18 at 0.08 mm thickness perturbation via *PERTURBATION_NODE + 1° tilt about horizontal centroidal axis; $T_{load} \gtrsim 10\,T_1$	MAT_280: FT = 0.060 GPa (calibrated, Batch A), FTSCL = 1.2, FRACEN not set, EPSCR = 0.1; no *MAT_ADD_EROSION	Final calibrated model

D

Pseudocode of the Airbag Simulation Function

D.1. Airbag impact simulation (`simulate_airbag`)

Algorithm D.1 gives the core impact simulation, and Algorithm D.2 its geometry and orifice-flow helper functions.

Algorithm D.1: Core airbag impact simulation (`simulate_airbag`)

Require: design **x**: shape, D_0, L_0, P_0, n_{ply}, φ_{vent}, fabric, gas (Table 4.8)
impact M, u_{x0}, u_{y0} (y positive downward); environment; safety factors; orifice d_{or}; solver settings $t_{\max}$, n_{inner}

Ensure: mass m_{tot}; peak accelerations; flags *burst*, *bottomed*, *rebounded*, *settled*, *survived*
residual and rebound velocities; final velocity $u_{\text{final},n}$; P_{peak}, $h_{\max}$, x_{final}

Pre-processing (done once)

$A_{\text{or}} \leftarrow \pi d_{\text{or}}^2/4$
$R_0 \leftarrow D_0/2$
$P_{\text{burst}} \leftarrow$ Equation (4.14) ▷ hoop burst limit
$S \leftarrow \pi D_0 L_0 + \pi D_0^2/2$ (cyl.) or πD_0^2 (sph.) ▷ fabric surface
$m_{\text{bag}} \leftarrow n_{\text{ply}}\, w\, S$
$\theta \leftarrow \text{atan2}(u_{x0},\, u_{y0})$ ▷ impact axis **n**, kept fixed

Initial state

$t, h, y_{\text{pen}}, x, a_n \leftarrow 0$
$(u_x, u_y) \leftarrow (u_{x0}, u_{y0})$
$(A_t, V) \leftarrow$ Geometry(0)
$m_{\text{gas},0} \leftarrow P_0 V/(R T_0)$ ▷ initial gas charge, ideal gas
$m_{\text{gas}} \leftarrow m_{\text{gas},0}$
$m_{\text{tot}} \leftarrow m_{\text{bag}} + m_{\text{gas},0}$ ▷ objective: fabric + initial gas
$(P, T) \leftarrow (P_0, T_0)$
$\rho \leftarrow m_{\text{gas}}/V$

Time integration

while $t < t_{\max}$ **do**

1) Adaptive timestep

  if $a_n > 0.03\, a_{\text{allow}}$ **then**
    $\Delta t \leftarrow 10^{-5}$ s
  else if $a_n < 0.01\, a_{\text{allow}}$ **then**
    $\Delta t \leftarrow 2 \cdot 10^{-4}$ s

end if ▷ in between: keep Δt

2) Dynamics (2-DoF, semi-implicit Euler)

$F_n \leftarrow \max\left((P - P_{\text{amb}})\, A_t,\ 0\right)$ ▷ contact pressure = gauge pressure
$a_x \leftarrow -F_n \sin\theta / M$
$a_y \leftarrow -F_n \cos\theta / M + g$
$a_n \leftarrow F_n / M$
$u_x \leftarrow u_x + a_x \Delta t$
$u_y \leftarrow u_y + a_y \Delta t$
$u_n \leftarrow u_x \sin\theta + u_y \cos\theta$
$h \leftarrow \text{clip}(h + u_n \Delta t,\ 0,\ D_0)$ ▷ stroke along **n**
$y_{\text{pen}} \leftarrow y_{\text{pen}} + u_y \Delta t$ ▷ vertical penetration
$x \leftarrow x + u_x \Delta t$

3) Deformed geometry and vent area

$(A_t, V) \leftarrow \textsc{Geometry}(h)$ ▷ Algorithm D.2
$A_{\text{vent}} \leftarrow A_{\text{or}} + \varphi_{\text{vent}}\ \max(S - A_t,\ 0)$ ▷ ground patch does not vent

4) Gas state: adiabatic compression, then venting

$\rho' \leftarrow m_{\text{gas}} / V$
$P \leftarrow P\,(\rho'/\rho)^{\gamma}$ ▷ adiabatic compression
$T \leftarrow T\,(\rho'/\rho)^{\gamma-1}$
$\rho \leftarrow \rho'$
for $j = 1$ **to** n_{inner} **do** ▷ sub-step $\delta t = \Delta t / n_{\text{inner}}$
  if $P \le P_{\text{amb}}$ **then break**
  $\delta m \leftarrow \min\left(\textsc{OrificeFlow}(P, P_{\text{amb}}, T, A_{\text{vent}})\, \delta t,\ m_{\text{gas}}\right)$
  $m_{\text{gas}} \leftarrow m_{\text{gas}} - \delta m$
  $\rho' \leftarrow m_{\text{gas}} / V$
  $P \leftarrow m_{\text{gas}} R T / V$
  $T \leftarrow T\,(\rho'/\rho)^{\gamma-1}$
  $\rho \leftarrow \rho'$
end for
$P \leftarrow \max(P,\ P_{\text{amb}})$
store $t + \Delta t$, h, y_{pen}, x, $\mathbf{u}$, $\mathbf{a}$, P, V, m_{gas} in history

5) Stopping criteria (first one triggered wins)

if $P > P_{\text{burst}}$ **then** ▷ membrane tension exceeds seam allowable
  burst $\leftarrow$ true
  $t_{\text{stop}} \leftarrow t + \Delta t$
  break
else if $h \ge D_0$ **then** ▷ full stroke used
  bottomed $\leftarrow$ true
  $(u_{\text{res},n}, u_{\text{res},x}, u_{\text{res},y}) \leftarrow (u_n, u_x, u_y)$
  $t_{\text{stop}} \leftarrow t + \Delta t$
  break
else if $y_{\text{pen}} \le 0$ **then** ▷ payload back above contact height
  rebounded $\leftarrow$ true
  $u_{\text{reb},y} \leftarrow u_y$
  $t_{\text{stop}} \leftarrow t + \Delta t$
  break
end if
$t \leftarrow t + \Delta t$
end while ▷ $t \ge t_{\max}$: no flag set

Post-processing

$a_{\text{peak},n} \leftarrow \max a_n$ ▷ maxima over the stored history
$a_{\text{peak},y} \leftarrow \max |a_y|$

$a_{\text{res,peak}} \leftarrow \max \sqrt{a_x^2 + a_y^2}$
$P_{\text{peak}} \leftarrow \max P$
$h_{\max} \leftarrow \max h$
$x_{\text{final}} \leftarrow x$
$\mathit{survived} \leftarrow \neg(\mathit{burst} \vee \mathit{rebounded})$
return outputs

Algorithm D.2: Helper functions of `simulate_airbag`

function Geometry(h) ▷ contact area and volume at stroke h
  if cylinder **then**
    $(A_t, V) \leftarrow$ Equation (4.8) ▷ stadium section, inextensible perimeter
  else
    $(A_t, V) \leftarrow$ Equation (4.9) ▷ sphere truncated by a cap of height h
  end if
  return A_t, V
end function

function OrificeFlow($P_{\text{up}}, P_{\text{down}}, T, A$) ▷ vented mass flow rate
  if $P_{\text{up}} \le P_{\text{down}}$ **then return** 0
  $\lambda \leftarrow P_{\text{down}}/P_{\text{up}}$
  $C_{\text{dis}} \leftarrow C_{\text{dis}}(\lambda)$ ▷ 6th-order polynomial fit
  if $\lambda \ge 0.528$ **then**
    $\dot{m} \leftarrow$ Equation (4.11) ▷ subsonic
  else
    $\dot{m} \leftarrow$ Equation (4.12) ▷ choked
  end if
  return $\dot{m}$
end function

E

Optimisation Results per Inflation Gas

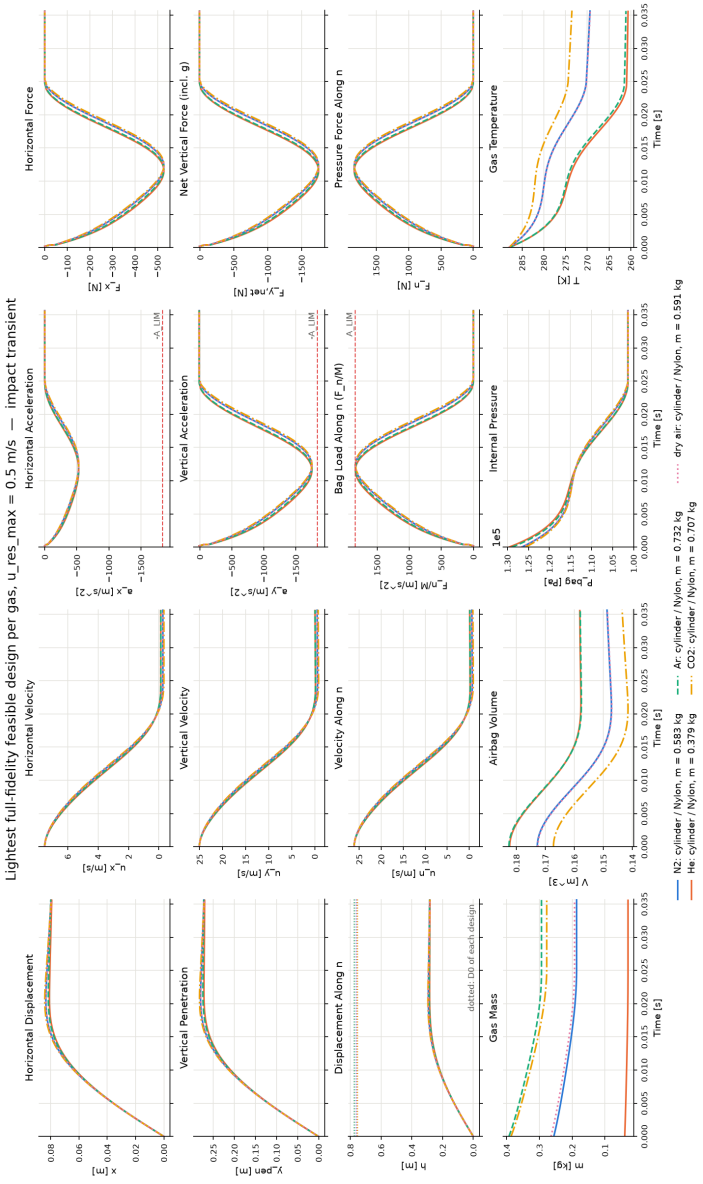

Figure E.1: Impact transient of the lightest feasible design per inflation gas, $u_{\text{res,max}} = 0.5\,\text{m/s}$.

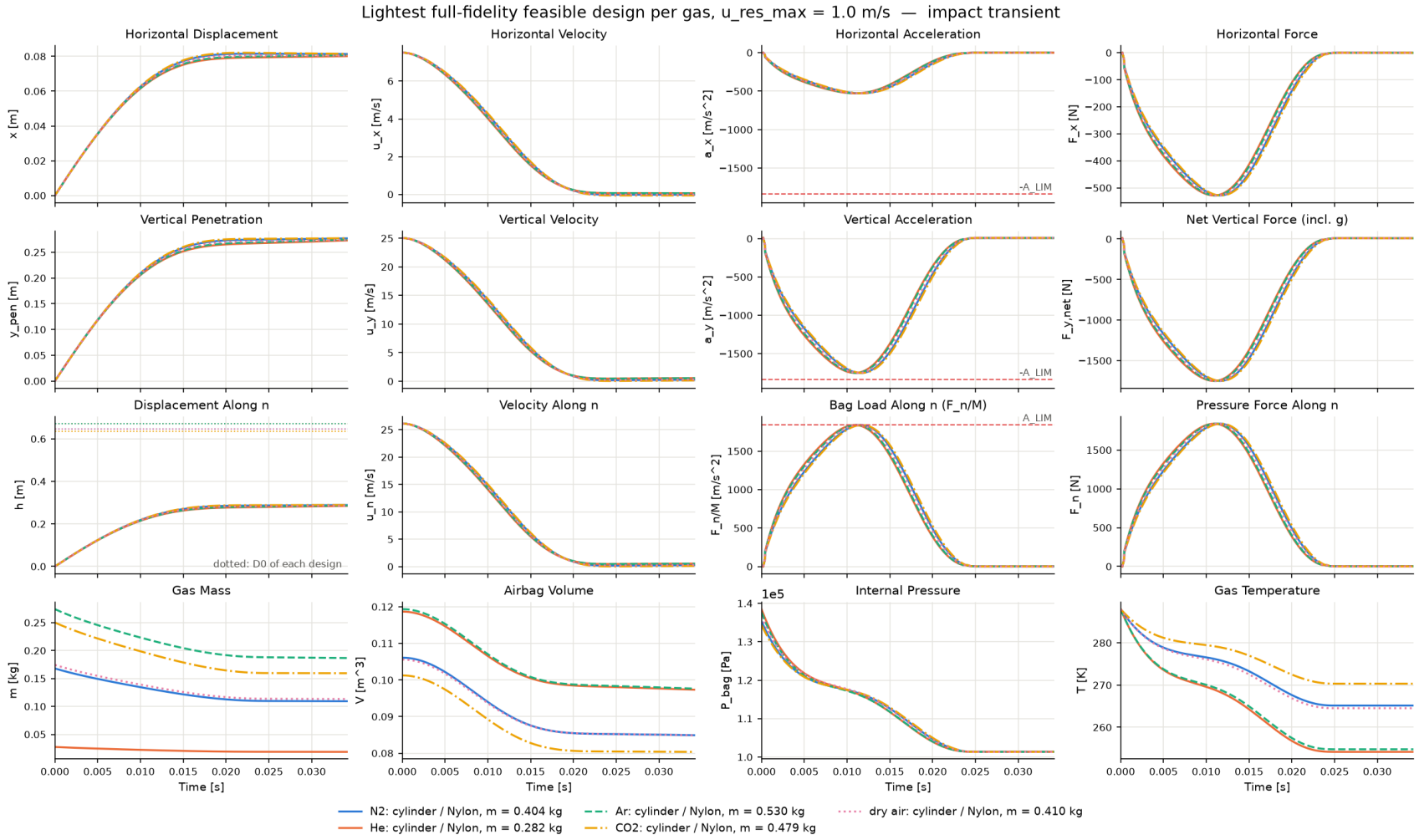

Figure E.2: Impact transient of the lightest feasible design per inflation gas, $u_{\mathrm{res,max}} = 1.0\,\mathrm{m/s}$.

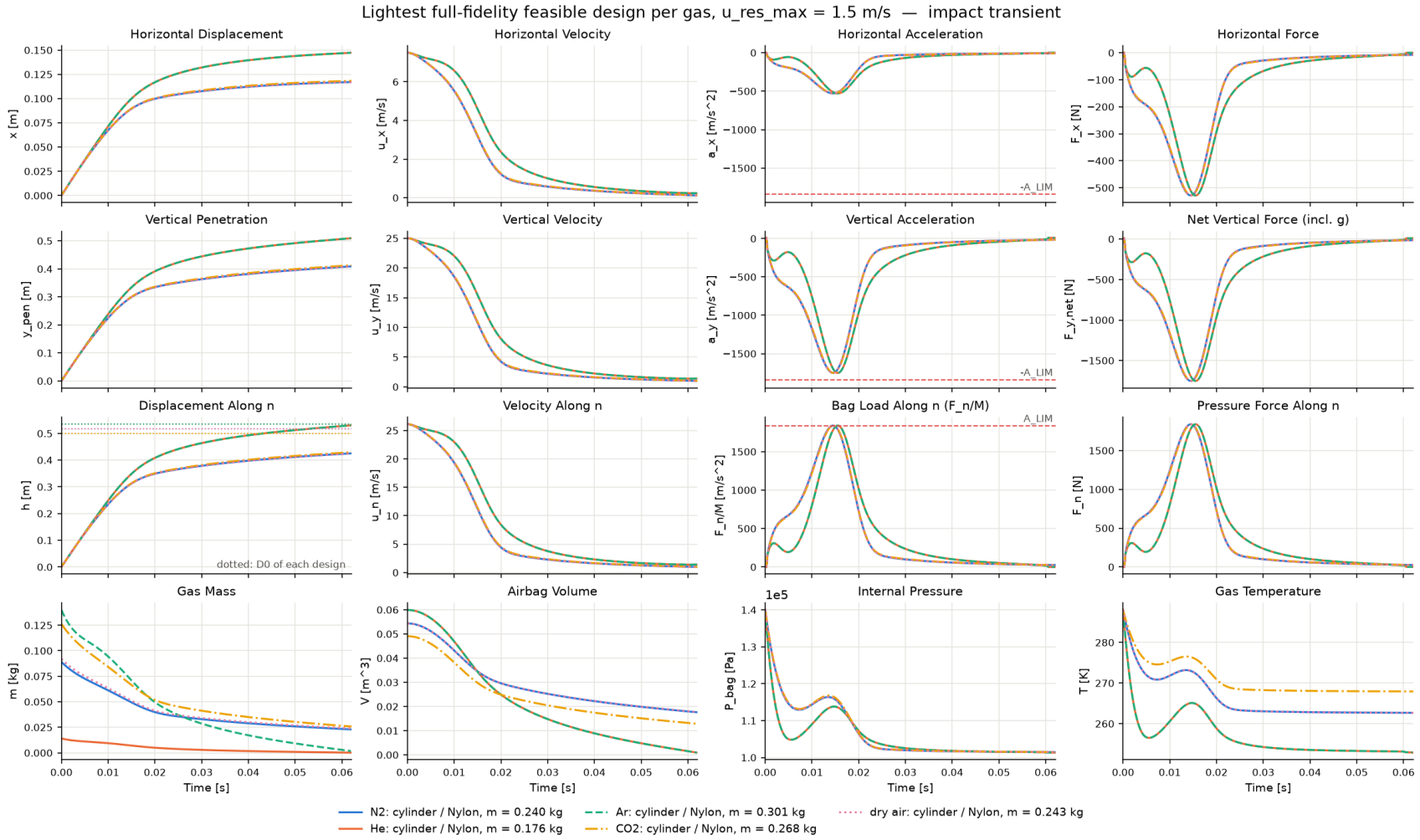

Figure E.3: Impact transient of the lightest feasible design per inflation gas, $u_{\mathrm{res,max}} = 1.5\,\mathrm{m/s}$.

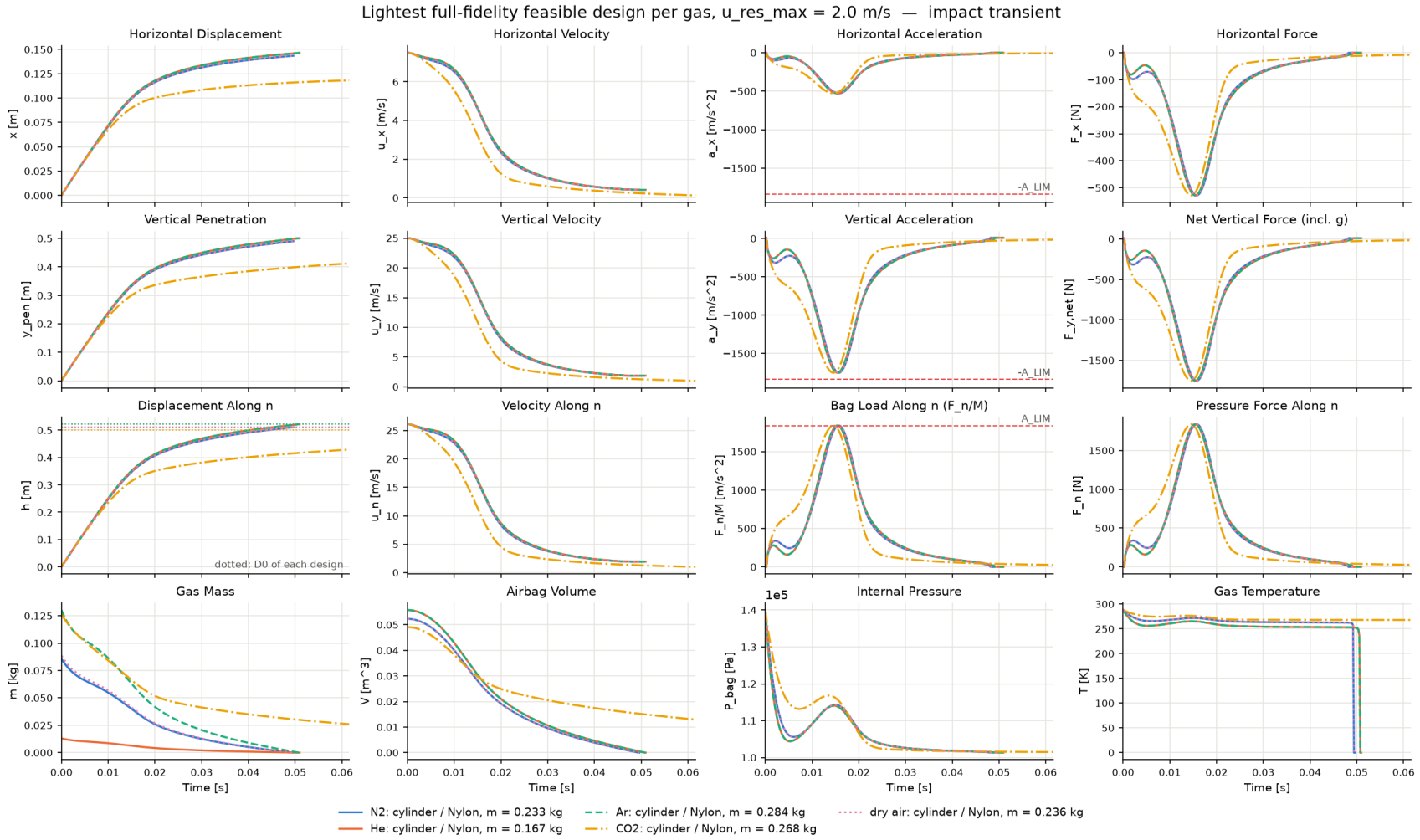

Figure E.4: Impact transient of the lightest feasible design per inflation gas, $u_{\mathrm{res,max}} = 2.0\,\mathrm{m/s}$.

F

Packing Instruction P650: Relevant Provisions

Packing Instruction P650 sets the packaging requirements for Biological Substance, Category B (UN3373). It is harmonised across transport modes, appearing in the ADR for road transport [31] and in the ICAO Technical Instructions for air transport [32]. Table F.1 summarises the provisions that bear on the design of a payload protection system. Paragraph numbers refer to the ADR edition, where P650 is found in Section 4.1.4.1. Provisions on package marking, solid substances, refrigerants, overpacks, filling instructions, mixed packing with other dangerous goods, decontamination and animal material are omitted.

Table F.1: Provisions of Packing Instruction P650 relevant to the design of a payload protection system, paraphrased from [31].

Par.	Requirement
(1)	The packaging shall withstand the shocks and loadings *normally encountered during carriage*, including transhipment and manual or mechanical handling.
(2)	The packaging shall consist of a primary receptacle, a secondary packaging and an outer packaging. Either the secondary or the outer packaging shall be rigid.
(3)	Under normal conditions of carriage, primary receptacles shall not break, be punctured or leak into the secondary packaging. The secondary packaging shall be secured in the outer packaging with suitable cushioning material.
(6)	The completed package shall withstand a 1.2 m drop in any orientation without leakage from the primary receptacle(s). Capability may be demonstrated by testing, assessment or experience.
(7)(a–b)	For liquid substances, the primary receptacle(s) and the secondary packaging shall be leakproof.
(7)(c)	Multiple fragile primary receptacles in a single secondary packaging shall be individually wrapped or separated to prevent contact between them.
(7)(d)	Absorbent material, sufficient to absorb the entire contents of the primary receptacle(s), shall be placed between the primary receptacle(s) and the secondary packaging.
(7)(e)	The primary receptacle or the secondary packaging shall withstand an internal pressure of 95 kPa without leakage.
(11)	Substances packed and marked in accordance with P650 are not subject to any other requirement of the ADR.

G

Fabric Property Data

The fabric properties used in the optimisation (Table 4.8) are rounded, representative values rather than measurements of a specific product. Table G.1 compares them with the published data they are based on.

Table G.1: Adopted fabric properties and the published data they are based on.

Fabric	Adopted values	Published data	Source
Nylon 6,6, silicone-coated	600 N/cm, 180 g/m^2	Airbag fabrics woven from 470 dtex yarn: 166 g/m^2 to 239 g/m^2, about 600 N/cm to 785 N/cm; silicone-coated example: 198 g/m^2, 674 N/cm	[84, Tabs. 1-2]
Polyester, silicone-coated	550 N/cm, 190 g/m^2	Indicated as suitable for silicone-coated airbag fabrics: at least 550 N/cm, total weight 180 g/m^2 to 300 g/m^2; example: 201 g/m^2, about 600 N/cm	[85, paras. 0026, 0027, 0041, Tab. 2]
Aramid (Kevlar 29), elastomer-coated	$\approx$ 1000 N/cm, $\approx$ 260 g/m^2	Two-ply laminates of about 92 g/m^2: 690 N/cm to 735 N/cm at room temperature	[86, Tab. IX]

The polyester and nylon values fall within the published ranges. The aramid entry is extrapolated from data for lighter laminates to a heavier coated fabric, and is the least certain of the three.

Acknowledgements - *Ringraziamenti*

ciao